

ANKARA ÜNİVERSİTESİ
MÜHENDİSLİK FAKÜLTESİ
BİLGİSAYAR MÜHENDİSLİĞİ BÖLÜMÜ



BLM 4522 Ağ Tabanlı Paralel Dağıtım Sistemleri Dersi

Final Proje Ödevi

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GitHub: <https://github.com/SaberZafari1/paralelDag-t-mOdevi>

Proje1: <https://youtu.be/o0JO0JHvL8E>

Proje2: <https://youtu.be/75O08AY4D8M>

proje3: <https://youtu.be/Y9Z-YY3MUNA>

Proje4: <https://youtu.be/cZTk1pgHQNs>

Proje5: <https://youtu.be/izMkOYWY0uA>

Proje6: https://youtu.be/HjsF_3wfeOg

Proje7: <https://youtu.be/LBnwmK5j2F0>

Projelerde kullanılan veri tabanı: AdventureWorks2019, Microsoft tarafından sağlanan örnek bir ilişkisel veri tabanıdır ve SQL Server üzerinde eğitim, test ve performans analizleri yapmak isteyen kullanıcılar için geliştirilmiştir. Bu veri tabanı, bir bisiklet üretim ve satış şirketinin ticari süreçlerini simüle eder. İçeriğinde ürün yönetimi, satış siparişleri, müşteriler, çalışanlar, üretim, satın alma ve envanter gibi pek çok iş alanını kapsayan tablolar, görünümüler ve saklı yordamlar bulunur. Özellikle veri analizi, sorgu optimizasyonu, ETL süreçleri, raporlama ve güvenlik testleri gibi birçok konunun pratik edilmesine olanak tanır. AdventureWorks2019 hem bireysel hem de kurumsal müşteri yapısını içermesi sayesinde karmaşık veri ilişkilerini ve gerçek hayattaki iş senaryolarını test etmek için ideal bir modeldir.

1. Veri tabanı Performans Optimizasyonu ve İzleme

Veri tabanı İzleme:

SQL Server Profiler, yavaş çalışan veya sistem kaynaklarını fazla tüketebilecek sorguları analiz etmek için kullanılmıştır.

- "Standard" şablonu kullanıldı. Events Selection için: SQL:BatchCompleted, Showplan XML, RPC:Completed, Performance statistics olayları seçildi.

The screenshot displays the Microsoft SQL Server Management Studio interface. The main window shows a SQL query being executed against the AdventureWorks2019 database. The query is as follows:

```
USE AdventureWorks2019;
SELECT p.Color, SUM(sod.LineTotal) AS TotalSales
FROM Sales.SalesOrderDetail sod
JOIN Production.Product p ON sod.ProductID = p.ProductID
GROUP BY p.Color
ORDER BY TotalSales DESC;
```

The query results are displayed in a table with two columns: Color and TotalSales. The results are as follows:

Color	TotalSales
Black	38247018.631195
Red	21623269.540854
Silver	19777339.945262
Yellow	18669505.218895
Blue	9602850.958400
NULL	1099318.907552
Multi	649849.180346
SilverBlack	147483.909000
White	29745.127584

The right-hand pane shows the Properties window for the current connection. The connection name is SAZ-PC (SAZ-PC\saber). The connection details include the connection elapsed time (00:00:00.116), connection encryption (Encrypted), connection finish time (4/22/2025 10:03:49 AM), connection rows returned (9), connection start time (4/22/2025 10:03:49 AM), connection state (Open), display name (SAZ-PC), login name (SAZ-PC\saber), server name (SAZ-PC), server version (16.0.1000), session tracing ID (71), and SPID (71). The TDS protocol version is 0x74000004.

The bottom status bar indicates that the query was executed successfully, returning 9 rows in 00:00:00.116 seconds.

SQL Server Profiler - [opt (SAZ-PC)]

File Edit View Replay Tools Window Help

EventClass	TextData	ApplicationName	NTUserName	LoginName	CPU	Reads	Writes	Duration	ClientProcessID	SPID	StartTime	EndTime
Trace Start											2025-04-22 10:03:36...	
ExistingConnection	-- network protocol: LPC set quoted...	SQLServerCEIP	SQLTELE...	NT SER...					19180	65	2025-04-22 09:58:56...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	66	2025-04-22 10:00:05...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	67	2025-04-22 10:00:13...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	68	2025-04-22 10:01:02...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	69	2025-04-22 10:00:19...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	71	2025-04-22 10:00:17...	
ExistingConnection	-- network protocol: LPC set quoted...	Microsoft SQ...	saber	SAZ-PC...					15516	72	2025-04-22 10:02:22...	
SQL:BatchStarting	USE Adventureworks2019; SELECT p.Co...	Microsoft SQ...	saber	SAZ-PC...	78	1301	0	100	15516	71	2025-04-22 10:03:49...	2025-04-22 10:03:49...
SQL:BatchCompleted	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...	NT SER...					19180	65	2025-04-22 10:03:57...	
SQL:BatchCompleted	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...	NT SER...	0	0	0	0	19180	65	2025-04-22 10:03:57...	2025-04-22 10:03:57...
SQL:BatchStarting	SELECT target_data FROM sy...	SQLServerCEIP	SQLTELE...	NT SER...					19180	65	2025-04-22 10:03:57...	
SQL:BatchCompleted	SELECT target_data FROM sy...	SQLServerCEIP	SQLTELE...	NT SER...	31	0	0	63	19180	65	2025-04-22 10:03:57...	2025-04-22 10:03:57...
SQL:BatchStarting	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...	NT SER...					19180	65	2025-04-22 10:03:57...	
SQL:BatchCompleted	SET DEADLOCK_PRIORITY -10	SQLServerCEIP	SQLTELE...	NT SER...	0	0	0	0	19180	65	2025-04-22 10:03:57...	2025-04-22 10:03:57...
SQL:BatchStarting	if exists(select * from sys.server_e...	SQLServerCEIP	SQLTELE...	NT SER...					19180	65	2025-04-22 10:03:57...	
SQL:BatchCompleted	if exists(select * from sys.server_e...	SQLServerCEIP	SQLTELE...	NT SER...	0	674	0	20	19180	65	2025-04-22 10:03:57...	2025-04-22 10:03:57...

USE Adventureworks2019;
SELECT p.Color, SUM(sod.LineTotal) AS Totalsales
FROM Sales.Salesorderdetail sod
JOIN Production.Product p ON sod.ProductID = p.ProductID
GROUP BY p.Color
ORDER BY Totalsales DESC;

Trace is running. Ln 10, Col 1 Rows: 18

Connections: 1

10:04 AM 4/22/2025

SQL Server Profiler - [opt (SAZ-PC)]

File Edit View Replay Tools Window Help

EventClass	TextData	ApplicationName	NTUserName	LoginName	CPU	Reads	Writes	Duration	ClientProcessID	SPID	StartTime	EndTime
SQL:BatchCompleted	SELECT SCHEMA_NAME(tt.schema_id) AS ...	Microsoft SQ...	saber	SAZ-PC...	0	7	0	0	15516	64	2025-04-22 10:11:57...	2025-04-22 10:11:57...
SQL:BatchStarting	SELECT satypes.name AS [Schema], aty...	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:11:57...	
SQL:BatchCompleted	SELECT satypes.name AS [Schema], aty...	Microsoft SQ...	saber	SAZ-PC...	0	14	0	0	15516	64	2025-04-22 10:11:57...	2025-04-22 10:11:57...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...					15516	64	2025-04-22 10:12:28...	
SQL:BatchCompleted	use [Adventureworks2019]	Microsoft SQ...	saber	SAZ-PC...	0	0	0	0	15516	64	2025-04-22 10:12:28...	2025-04-22 10:12:28...
SQL:BatchStarting	USE Adventureworks2019; SELECT Sale...	Microsoft SQ...	saber	SAZ-PC...	15	1453	6	54	15516	52	2025-04-22 10:12:31...	2025-04-22 10:12:31...

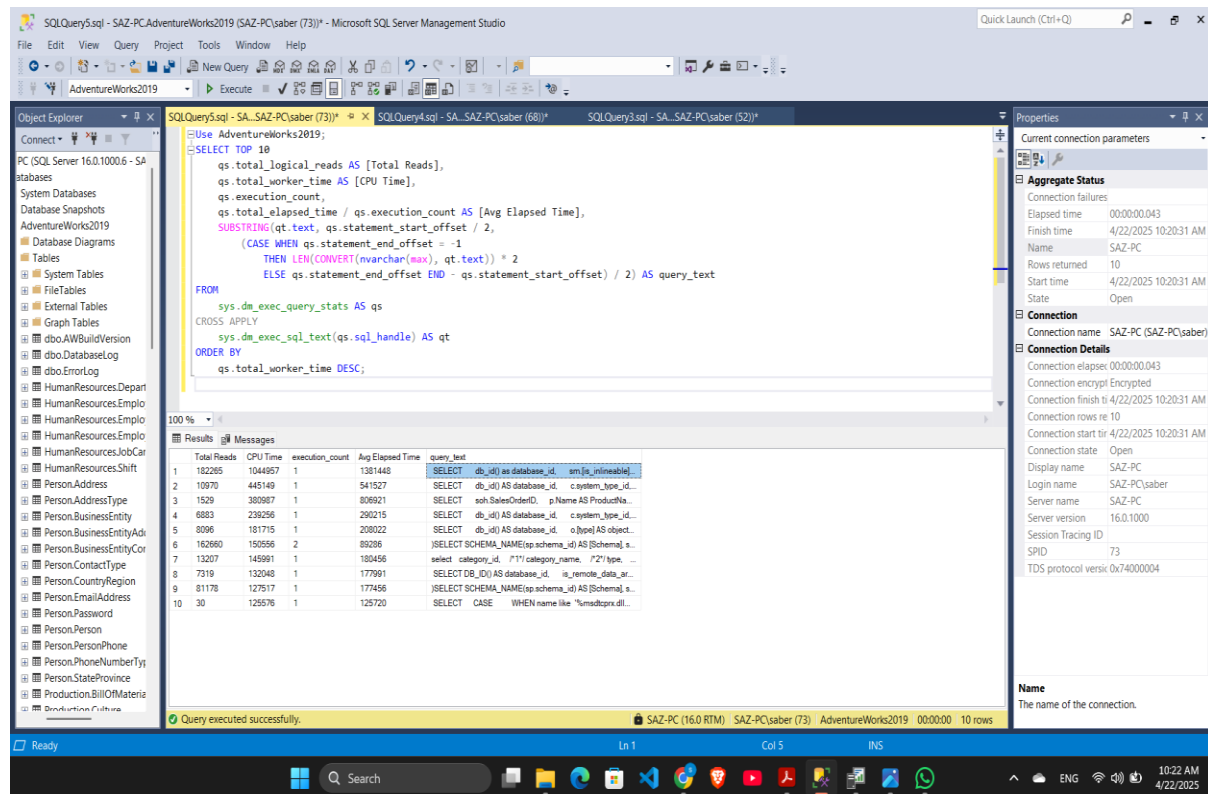
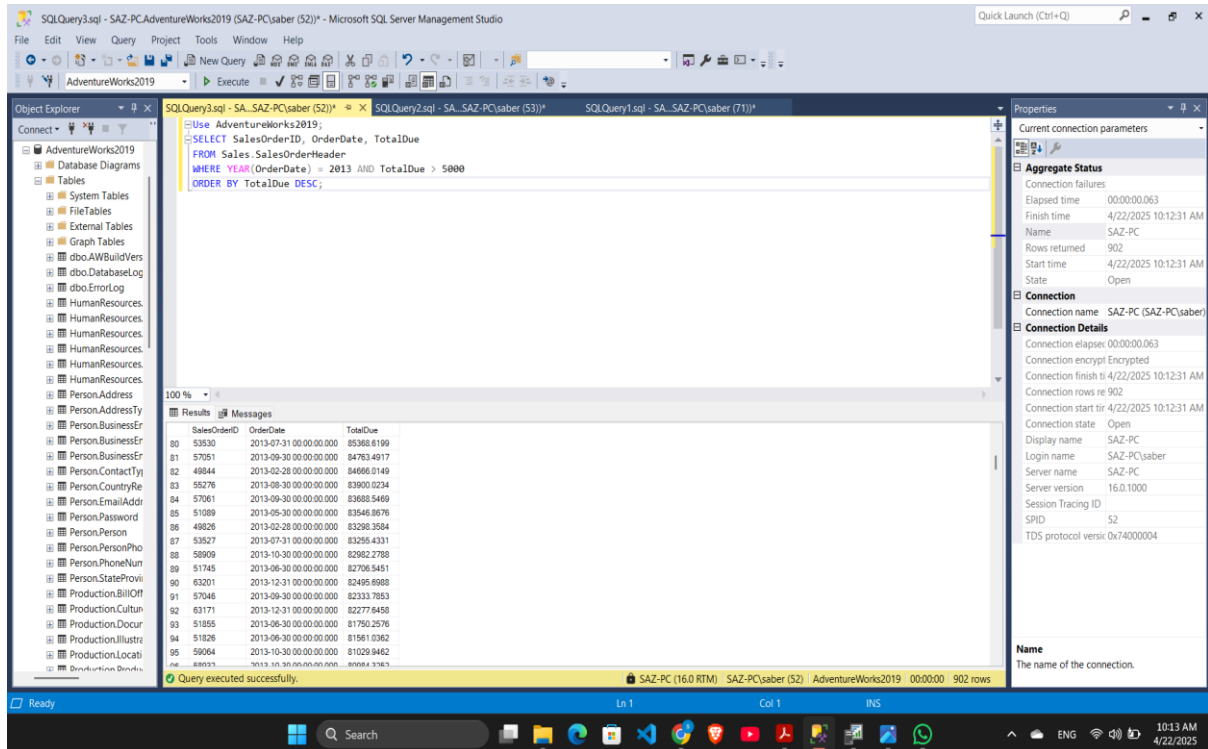
USE Adventureworks2019;
SELECT salesorderID, OrderDate, TotalDue
FROM Sales.SalesorderHeader
WHERE YEAR(OrderDate) = 2013 AND TotalDue > 5000
ORDER BY TotalDue DESC;

Trace is running. Ln 156, Col 1 Rows: 156

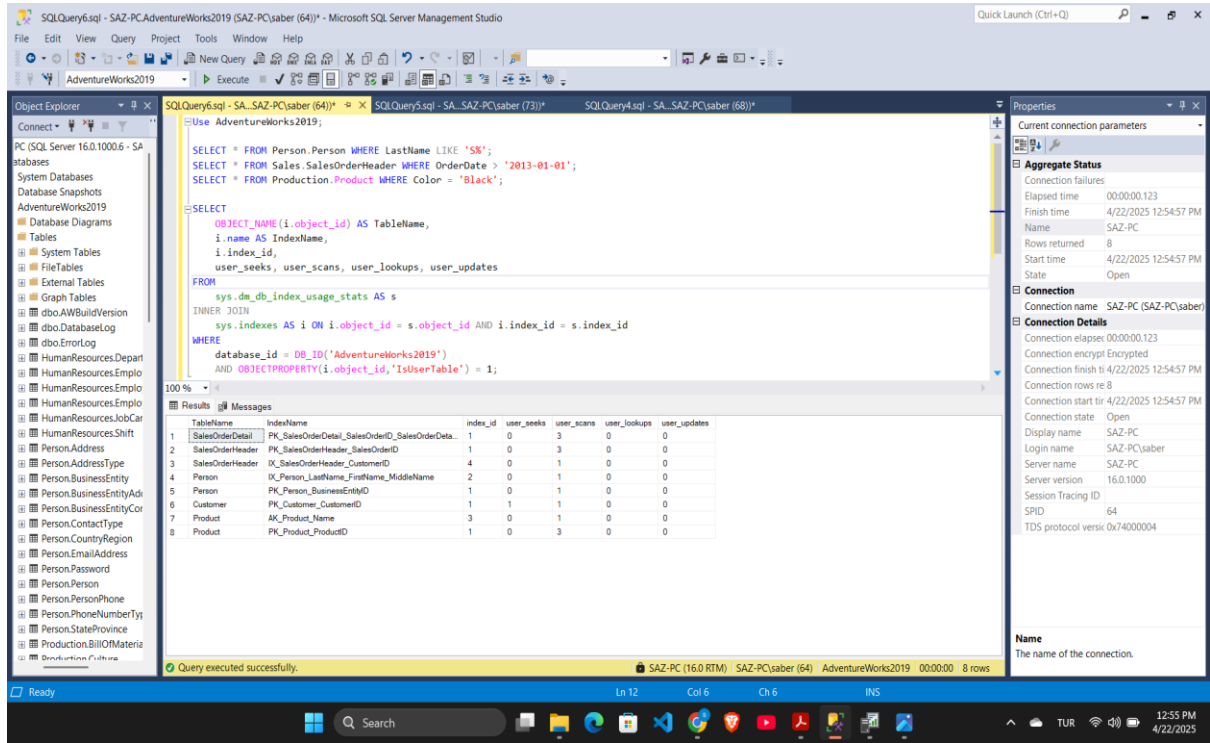
Connections: 1

10:12 AM 4/22/2025

SQL Server DMV'leri ile sistem kaynaklarını en çok tok tüketen sorgular tespit edilmiştir.



Kullanılmayan indekslerin tespiti:



SQLQuery6.sql - SAZ-PC\AdventureWorks2019 (SAZ-PC\saber (64)) - Microsoft SQL Server Management Studio

Object Explorer: PC (SQL Server 16.0.10006) - SA stables, System Databases, AdventureWorks2019, Database Diagrams, Tables, System Tables, FileTables, External Tables, Graph Tables, dbo.AWBuildVersion, dbo.DatabaseLog, dbo.ErrorLog, HumanResources.Department, HumanResources.Employee, HumanResources.JobCandidate, HumanResources.Shift, Person.Address, Person.AddressType, Person.BusinessEntity, Person.BusinessEntityAddress, Person.ContactType, Person.CountryRegion, Person.EmailAddress, Person.Password, Person.Person, Person.PersonPhone, Person.PhoneNumberType, Person.StateProvince, Production.BillOfMaterials, Production.Product, Production.ProductInventory, Production.ProductModel, Production.ProductPhoto, Production.ProductSubcategory, Production.ProductVendor, Production.TransactionHistory, Production.TransactionHistoryArchive, Production.TransactionHistoryRateCard, Production.TransactionHistoryRateCardArchive, Production.TransactionHistoryRateCardArchive2019, Production.TransactionHistoryRateCardArchive2020, Production.TransactionHistoryRateCardArchive2021, Production.TransactionHistoryRateCardArchive2022, Production.TransactionHistoryRateCardArchive2023, Production.TransactionHistoryRateCardArchive2024, Production.TransactionHistoryRateCardArchive2025.

SQLQuery6.sql - SAZ-PC\saber (64):

```
SELECT * FROM Person.Person WHERE LastName LIKE 'SK';
SELECT * FROM Sales.SalesOrderHeader WHERE OrderDate > '2013-01-01';
SELECT * FROM Production.Product WHERE Color = 'Black';

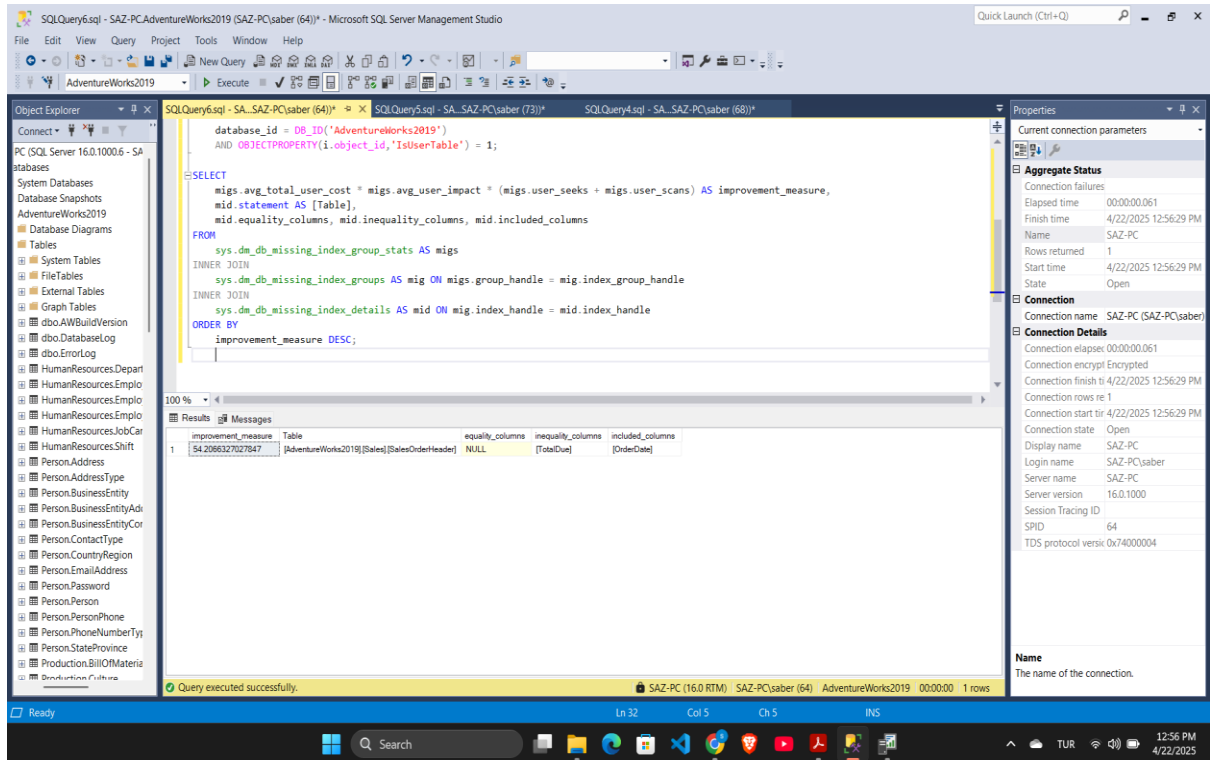
--SELECT
--OBJECT_NAME(i.object_id) AS TableName,
--i.name AS IndexName,
--i.index_id,
--user_seeks, user_scans, user_lookups, user_updates
--FROM
--sys.dm_db_index_usage_stats AS s
--INNER JOIN
--sys.indexes AS i ON i.object_id = s.object_id AND i.index_id = s.index_id
--WHERE
--database_id = DB_ID('AdventureWorks2019')
--AND OBJECTPROPERTY(i.object_id, 'IsUserTable') = 1;
```

Results:

TableName	IndexName	index_id	user_seeks	user_scans	user_lookups	user_updates
SalesOrderDetail	PK_SalesOrderDetail_SalesOrderID_SalesOrderDetailID	1	0	3	0	0
SalesOrderHeader	PK_SalesOrderHeader_SalesOrderID	1	0	3	0	0
SalesOrderHeader	IX_SalesOrderHeader_CustomerID	4	0	1	0	0
Person	IX_Person_LastName_FirstName_MiddleName	2	0	1	0	0
Person	PK_Person_BusinessEntityID	1	0	1	0	0
Customer	PK_Customer_CustomerID	1	1	1	0	0
Product	AK_Product_Name	3	0	1	0	0
Product	PK_Product_ProductID	1	0	3	0	0

Query executed successfully. SAZ-PC (16.0 RTM) SAZ-PC\saber (64) AdventureWorks2019 00:00:00 8 rows

Eksik indekslerin tespiti:



SQLQuery6.sql - SAZ-PC\AdventureWorks2019 (SAZ-PC\saber (64)) - Microsoft SQL Server Management Studio

Object Explorer: PC (SQL Server 16.0.10006) - SA stables, System Databases, AdventureWorks2019, Database Diagrams, Tables, System Tables, FileTables, External Tables, Graph Tables, dbo.AWBuildVersion, dbo.DatabaseLog, dbo.ErrorLog, HumanResources.Department, HumanResources.Employee, HumanResources.JobCandidate, HumanResources.Shift, Person.Address, Person.AddressType, Person.BusinessEntity, Person.BusinessEntityAddress, Person.ContactType, Person.CountryRegion, Person.EmailAddress, Person.Password, Person.Person, Person.PersonPhone, Person.PhoneNumberType, Person.StateProvince, Production.BillOfMaterials, Production.Product, Production.ProductInventory, Production.ProductModel, Production.ProductPhoto, Production.ProductSubcategory, Production.ProductVendor, Production.TransactionHistory, Production.TransactionHistoryArchive, Production.TransactionHistoryRateCard, Production.TransactionHistoryRateCardArchive, Production.TransactionHistoryRateCardArchive2019, Production.TransactionHistoryRateCardArchive2020, Production.TransactionHistoryRateCardArchive2021, Production.TransactionHistoryRateCardArchive2022, Production.TransactionHistoryRateCardArchive2023, Production.TransactionHistoryRateCardArchive2024, Production.TransactionHistoryRateCardArchive2025.

SQLQuery6.sql - SAZ-PC\saber (64):

```
database_id = DB_ID('AdventureWorks2019')
AND OBJECTPROPERTY(i.object_id, 'IsUserTable') = 1;

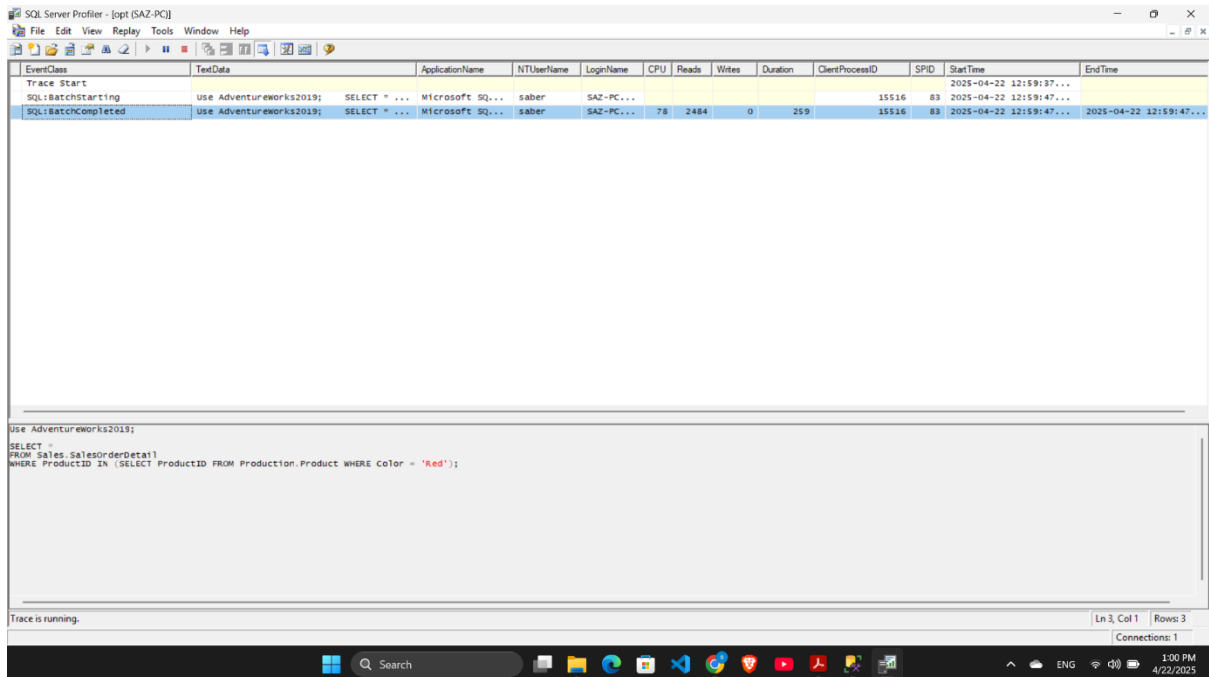
--SELECT
--migs.avg_total_user_cost * migs.avg_user_impact * (migs.user_seeks + migs.user_scans) AS improvement_measure,
--mid.statement AS [Table],
--mid.equality_columns, mid.inequality_columns, mid.included_columns
--FROM
--sys.dm_db_missing_index_group_stats AS migs
--INNER JOIN
--sys.dm_db_missing_index_groups AS mig ON migs.group_handle = mig.index_group_handle
--INNER JOIN
--sys.dm_db_missing_index_details AS mid ON mig.index_handle = mid.index_handle
--ORDER BY
--improvement_measure DESC;
```

Results:

improvement_measure	Table	equality_columns	inequality_columns	included_columns
54.2066327027847	AdventureWorks2019(Sales)[SalesOrderHeader]	NULL	[TotalDue]	[OrderDate]

Query executed successfully. SAZ-PC (16.0 RTM) SAZ-PC\saber (64) AdventureWorks2019 00:00:00 1 rows

Düşük performanslı sorgu:



İyileştirilmiş versiyonu:

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The central pane shows the results of a query executed on the AdventureWorks2019 database. The query is a SELECT statement with a JOIN and a WHERE clause. The results are displayed in a table with 15 columns and 13,065 rows. The status bar at the bottom indicates that the query was executed successfully.

Query:

```
USE AdventureWorks2019;  
  
SELECT sod.*  
FROM Sales.SalesOrderDetail sod  
JOIN Production.Product p ON sod.ProductID = p.ProductID  
WHERE p.Color = 'Red';
```

Results Table:

SalesOrderID	SalesOrderDetailID	CarrierTrackingNumber	OrderQty	ProductID	SpecialOfferID	UnitPrice	UnitPriceDiscount	LineTotal	rowguid	ModifiedDate
43660	13	6431-4D57-43	1	762	1	419.4589	0.00	419.458900	419A1302-4C7A-4044-97B2-66D9014C002E	2011-05-31 00:00:00.000
43660	14	6431-4D57-43	1	758	1	874.794	0.00	874.794000	5D0B2B03-1D4C-4C34-9696-C14C58E7301C	2011-05-31 00:00:00.000
43662	30	2E53-4802-45	3	764	1	419.4589	0.00	1258.376700	2DFD4D76-E020-4880-A045-64630C02F73	2011-05-31 00:00:00.000
43662	32	2E53-4802-45	2	730	1	183.9382	0.00	367.876400	5ACE6C85-7341-4D88-9425-56A87AE512E7	2011-05-31 00:00:00.000
43662	33	2E53-4802-45	4	754	1	874.794	0.00	3499.176000	DF2A7F2-37F2-4016-BEDF-4D8B8C388631	2011-05-31 00:00:00.000
43662	34	2E53-4802-45	3	725	1	183.9382	0.00	551.814800	92897EE9-7246-49E6-A421-C243B531AF72	2011-05-31 00:00:00.000
43662	35	2E53-4802-45	5	762	1	419.4589	0.00	2097.294500	1CE18FAA-047D-41B1-8CF6-6191C092A83C	2011-05-31 00:00:00.000
43662	38	2E53-4802-45	1	753	1	2146.962	0.00	2146.962000	4B1C5B1F-4B1E-47D0-95D5-451965A5A47	2011-05-31 00:00:00.000
43662	39	2E53-4802-45	1	756	1	874.794	0.00	874.794000	28A4CC58-8B32-47C0-4EFA-0190C29C0D83	2011-05-31 00:00:00.000
43662	40	2E53-4802-45	3	763	1	419.4589	0.00	1258.376700	D9C51A5A-4D12-4CEB-AED4-942E78BAF142	2011-05-31 00:00:00.000
43662	41	2E53-4802-45	1	732	1	356.898	0.00	356.898000	48420C59-4F8D-4817-A1DF-613DCE40E188	2011-05-31 00:00:00.000
43662	42	2E53-4802-45	6	758	1	874.794	0.00	5248.764000	2C4F704D-C794-4813-9C8B-258819FA20FA	2011-05-31 00:00:00.000
43662	43	2E53-4802-45	1	729	1	183.9382	0.00	183.938200	E53F432B-3E71-4F41-9EE8-0B05A18D98F9	2011-05-31 00:00:00.000
43662	45	2E53-4802-45	1	749	1	2146.962	0.00	2146.962000	0F38E442-512C-4EB4-85C6-27ABF81E2E2	2011-05-31 00:00:00.000
43662	46	2E53-4802-45	3	760	1	419.4589	0.00	1258.376700	260A3FDA-6977-4C9C-8678-EFE84ED56489	2011-05-31 00:00:00.000

The screenshot displays the Microsoft SQL Server Enterprise Manager interface, showing the execution plan for the same query as the previous screenshot. The query is a SELECT statement with a JOIN and a WHERE clause. The execution plan shows the query cost (relative to the batch) as 100%. The plan includes a Hash Match (Inner Join) operator, a Clustered Index Scan operator, and a Compute Scalar operator. The status bar at the bottom indicates that the query was executed successfully.

Query:

```
USE AdventureWorks2019;  
  
SELECT sod.*  
FROM Sales.SalesOrderDetail sod  
JOIN Production.Product p ON sod.ProductID = p.ProductID  
WHERE p.Color = 'Red';
```

Execution Plan:

Query 1: Query cost (relative to the batch): 100%

SELECT sod.* FROM Sales.SalesOrderDetail sod JOIN Production.Product p ON sod.ProductID = p.ProductID WHERE p.Color = 'Red'

Missing Index (Impact 98.6639): CREATE NONCLUSTERED INDEX [<Name of Missing Index, sysname>] ON [Sales].[SalesOrderDetail] [...]

The execution plan shows the following operators and their costs:

- Hash Match (Inner Join): Cost: 35 %
- Clustered Index Scan (Clustered Index Scan on [Product].[ProductID]): Cost: 1 %
- Compute Scalar: Cost: 1 %
- Clustered Index Scan (Clustered Index Scan on [SalesOrderDetail].[ProductID]): Cost: 63 %

EventClass	TextData	ApplicationName	NTUserName	LoginName	CPU	Reads	Writes	Duration	ClientProcessID	SPID	StartTime	EndTime
Trace Start												
SQL:BatchStarting	Use Adventureworks2019; SELECT * ...	Microsoft SQ...	saber	SAZ-PC...					15516	83	2025-04-22 12:59:37...	
SQL:BatchCompleted	Use Adventureworks2019; SELECT * ...	Microsoft SQ...	saber	SAZ-PC...	78	2484	0	259	15516	83	2025-04-22 12:59:47...	2025-04-22 12:59:47...
SQL:BatchStarting	SELECT dtb.name AS [Name], CAST(0 AS...	Microsoft SQ...	saber	SAZ-PC...					15516	72	2025-04-22 13:01:00...	
SQL:BatchCompleted	SELECT dtb.name AS [Name], CAST(0 AS...	Microsoft SQ...	saber	SAZ-PC...	16	92	0	25	15516	72	2025-04-22 13:01:00...	2025-04-22 13:01:00...
SQL:BatchStarting	Use Adventureworks2019; SELECT so...	Microsoft SQ...	saber	SAZ-PC...					15516	83	2025-04-22 13:01:02...	
SQL:BatchCompleted	Use Adventureworks2019; SELECT so...	Microsoft SQ...	saber	SAZ-PC...	125	2484	0	249	15516	83	2025-04-22 13:01:02...	2025-04-22 13:01:02...


```

Use Adventureworks2019;
SELECT sod...
FROM Sales.SalesOrderDetail sod
JOIN Production.Product p ON sod.ProductID = p.ProductID
WHERE p.Color = 'Red';

```

Trace is running.

Ln 7, Col 1 Rows: 7

Connections: 1

Veri Yöneticisi Roller:

Kullanıcı Oluşturma ve Yetkilendirme:

SQLQuery1.sql - SAZ-PC\AdventureWorks2019 (SAZ-PC\saber (SZ)) - Microsoft SQL Server Management Studio

File Edit View Query Project Tools Window Help

AdventureWorks2019

Execute

Object Explorer

Connect

SAZ-PC (SQL Server 16.0.1000)

Databases

System Databases

Database Snapshots

AdventureWorks2019

FirstDatabase

Database Diagrams

Tables

Views

External Resources

Synonyms

Programmability

Query Store

Regressed Queries

Overall Resource C

Top Resource Con

Queries With Force

Queries With High

Query Wait Statist

Tracked Queries

Service Broker

Storage

Security

Server Objects

Replication

Always On High Availability

Management

Integration Services Catalog

SQL Server Agent (Agent X)

XE Event Profiler

SQLQuery1.sql - SAZ-PC\saber (SZ) *

```

CREATE LOGIN ReportUser WITH PASSWORD = '123456';
USE AdventureWorks2019;
CREATE USER ReportUser FOR LOGIN ReportUser;
EXEC sp_addrolemember 'db_datareader', 'ReportUser';

SELECT
    dp.name AS PrincipalName,
    dp.type_desc AS PrincipalType,
    o.name AS ObjectName,
    p.permission_name,
    p.state_desc
FROM
    sys.database_permissions p
JOIN
    sys.database_principals dp ON p.grantee_principal_id = dp.principal_id
LEFT JOIN
    sys.objects o ON p.major_id = o.object_id
WHERE
    dp.name = 'ReportUser';

```

Results

PrincipalName	PrincipalType	ObjectName	permission_name	state_desc
ReportUser	SQL_USER	NULL	CONNECT	GRANT

Messages

1 Query executed successfully.

SAZ-PC (16.0 RTM) SAZ-PC\saber (SZ) AdventureWorks2019 00:00:00 1 rows

Properties

Current connection parameters

Aggregate Status

Connection failures

Elapsed time 00:00:00.191

Finish time 4/22/2025 2:08:05 PM

Name SAZ-PC

Rows returned 1

Start time 4/22/2025 2:08:05 PM

State Open

Connection

Connection name SAZ-PC (SAZ-PC\saber)

Connection Details

Connection elapsed: 00:00:00.191

Connection encrypt Encrypted

Connection finish ti 4/22/2025 2:08:05 PM

Connection rows re 1

Connection start tir 4/22/2025 2:08:05 PM

Connection state Open

Display name SAZ-PC

Login name SAZ-PC\saber

Server name SAZ-PC

Server version 16.0.1000

Session Tracing ID

SPID 52

TDS protocol versik 0x74000004

Name

The name of the connection.

Ready

Ln 1 Col 1 INS

Search

TUR

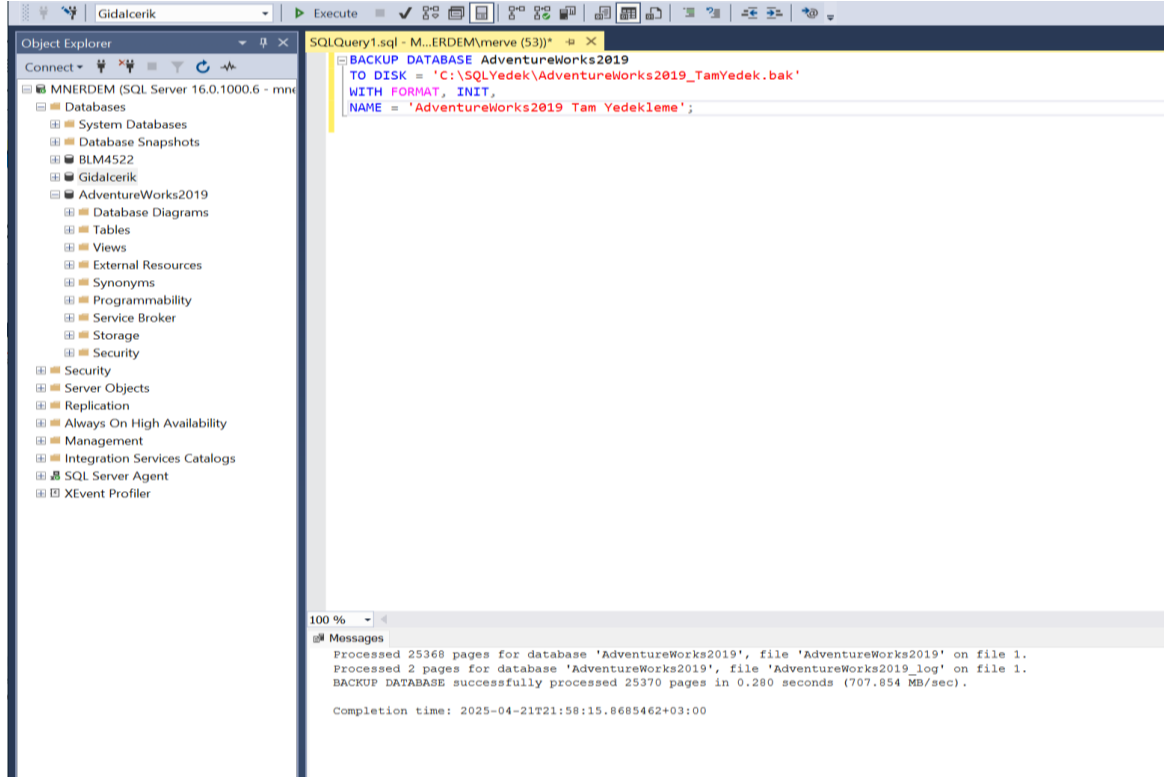
2:08 PM

4/22/2025

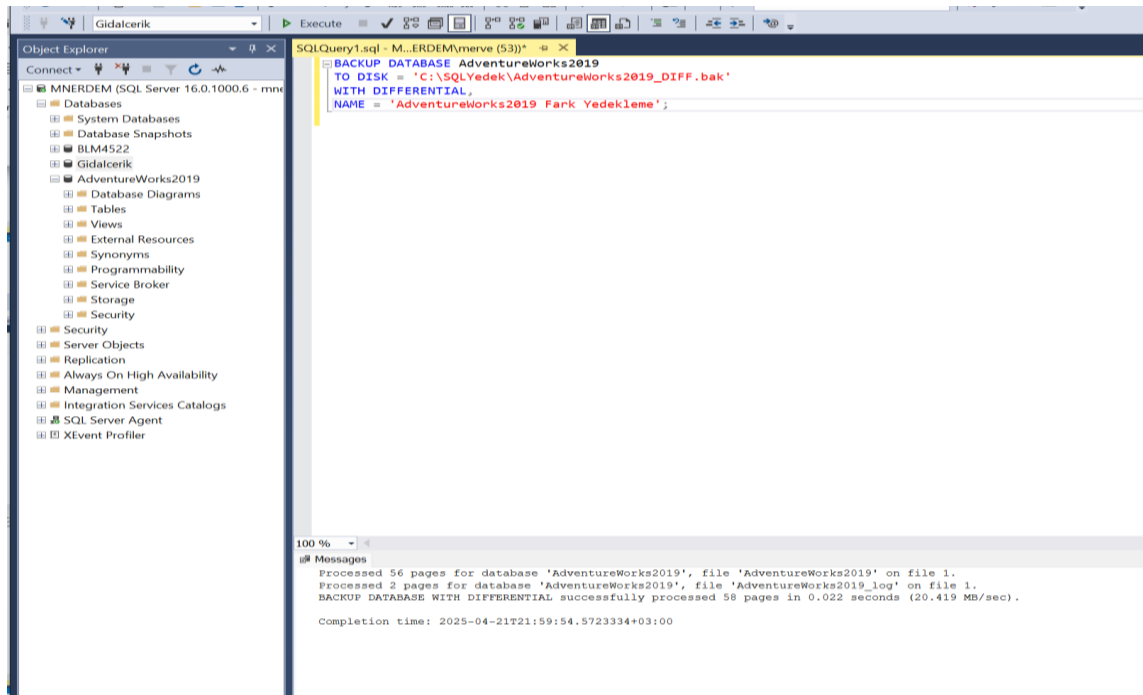
2. Veri tabanı Yedekleme ve Felaketten Kurtarma Planı

Tam, Artık, Fark Yedeklemeleri:

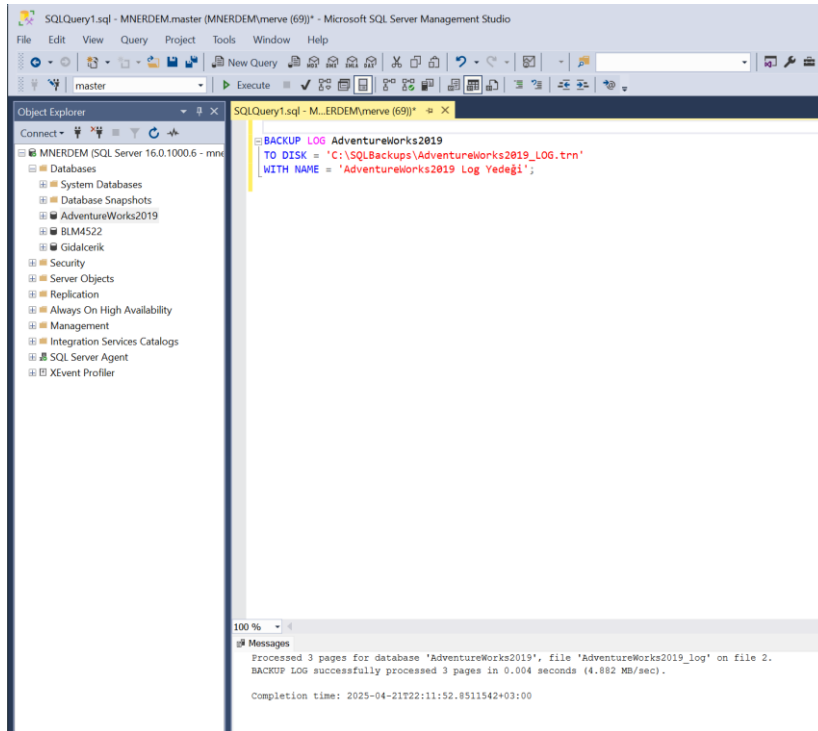
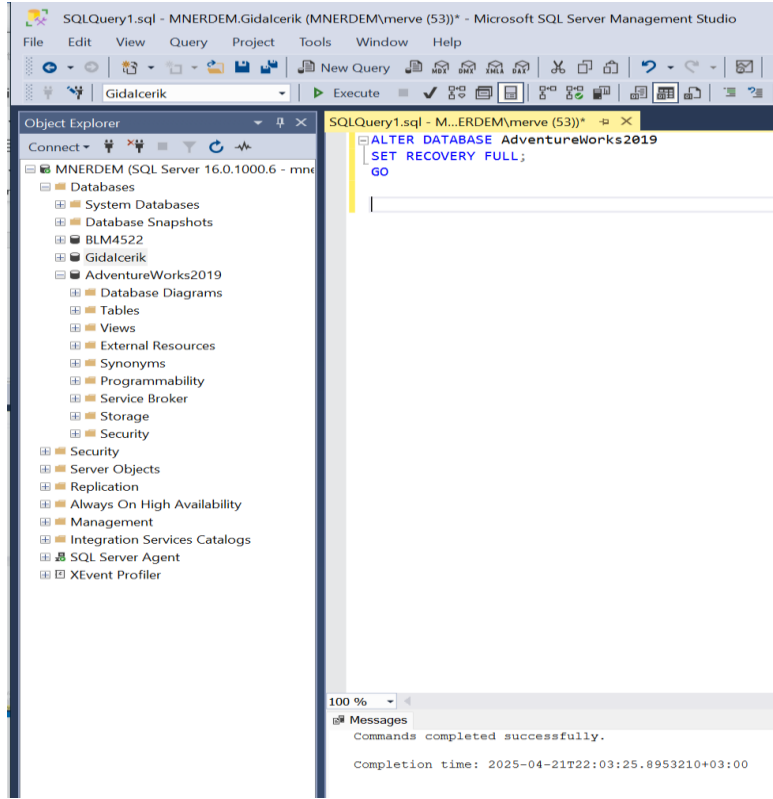
Tam Yedekleme



Fark Yedekleme:



Artırmalı (Transaction Log) Yedekleme



Zamanlayıcılarla Yedekleme:

File Edit View Project Tools Window Help

New Job Schedule

Name: MaintenancePlan2 Jobs in Schedule

Schedule type: Recurring ☒ Enabled

One-time occurrence

Date: 24.04.2025 Time: 22:05:58

Frequency

Occurs: Daily

Recurs every: 1 day(s)

Daily frequency

☐ Occurs once at: 22:00:00

☒ Occurs every: 1 hour(s) Starting at: 00:00:00

Ending at: 23:59:59

Duration

Start date: 24.04.2025 ☐ End date: 24.04.2025

☒ No end date:

Summary

Description: Occurs every day every 1 hour(s) between 00:00:00 and 23:59:59. Schedule will be used starting on 24.04.2025.

OK Cancel Help

File Edit View Project Tools Window Help

AdventureWorks2019 Execute

Maintenance Plan Wizard

Select Maintenance Task Order

In which order should these tasks be performed?

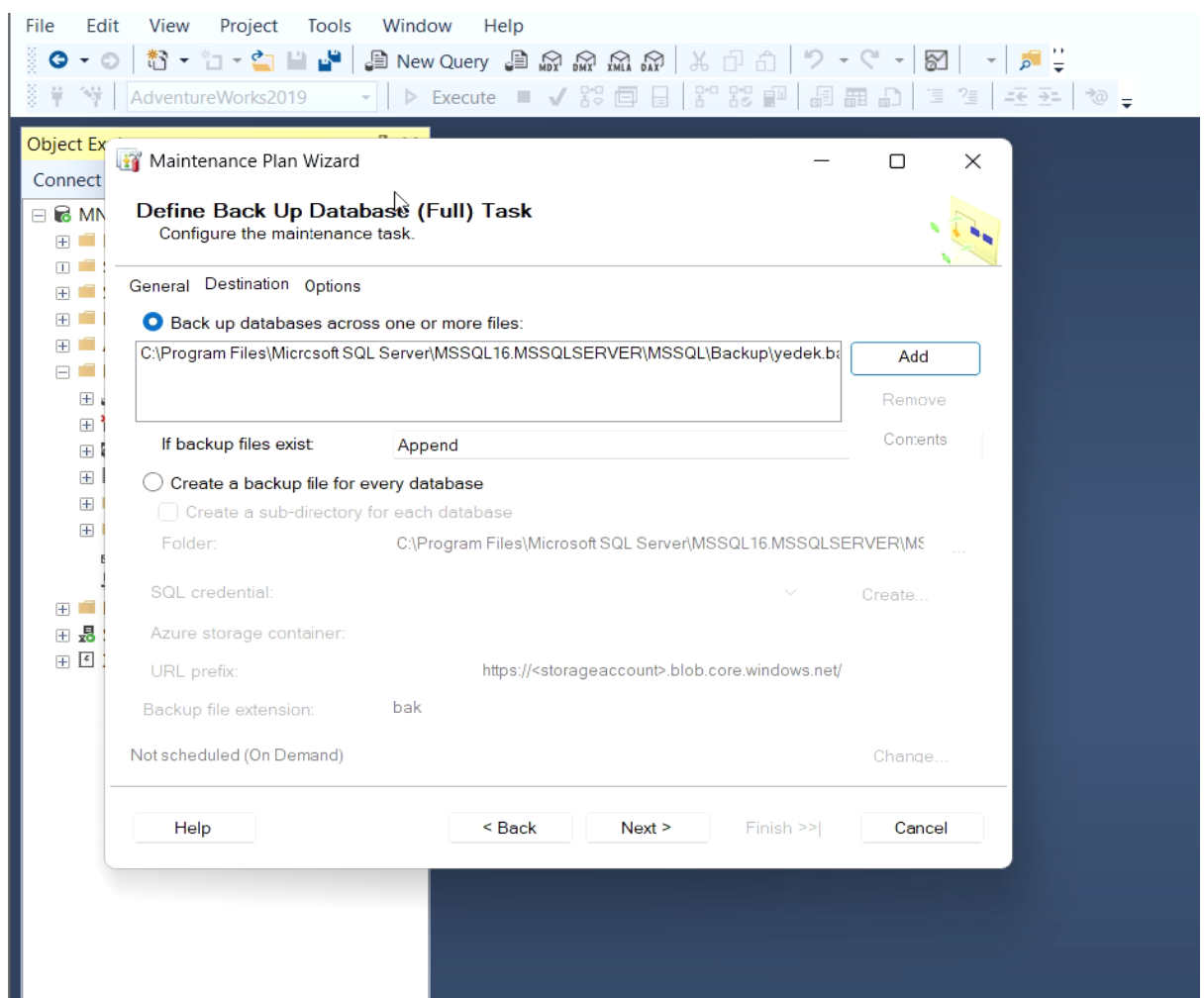
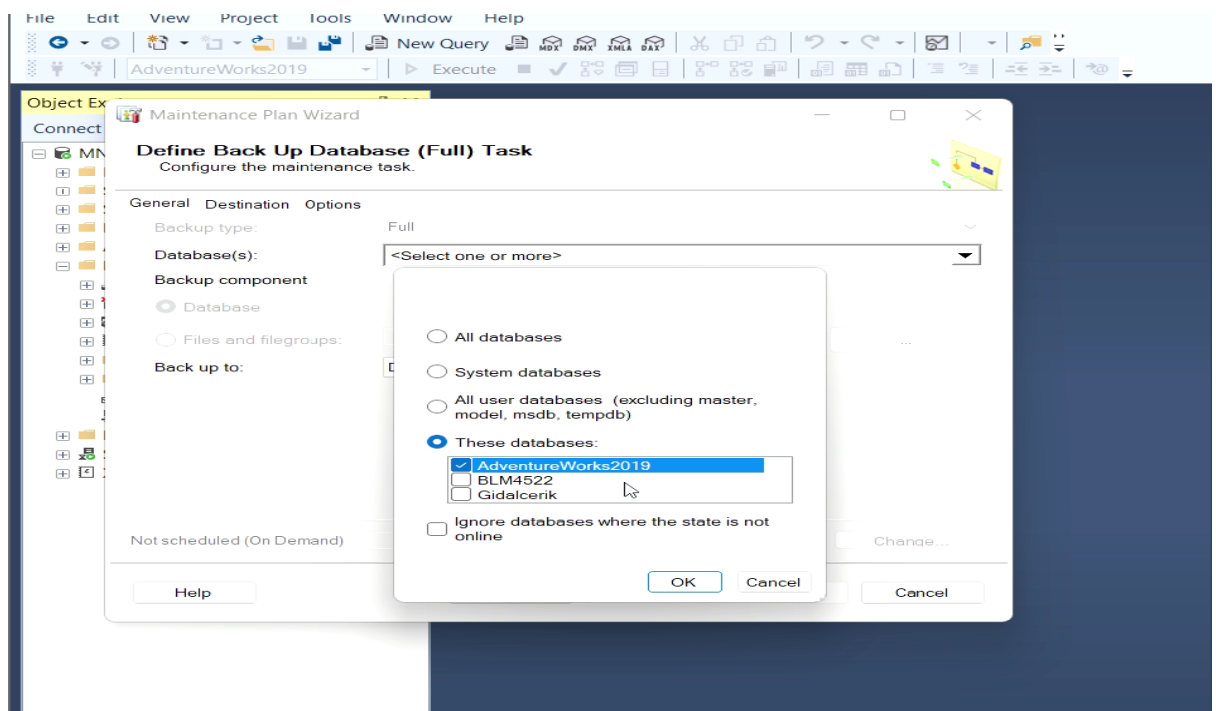
Select the order for the tasks to execute:

Back Up Database (Full)

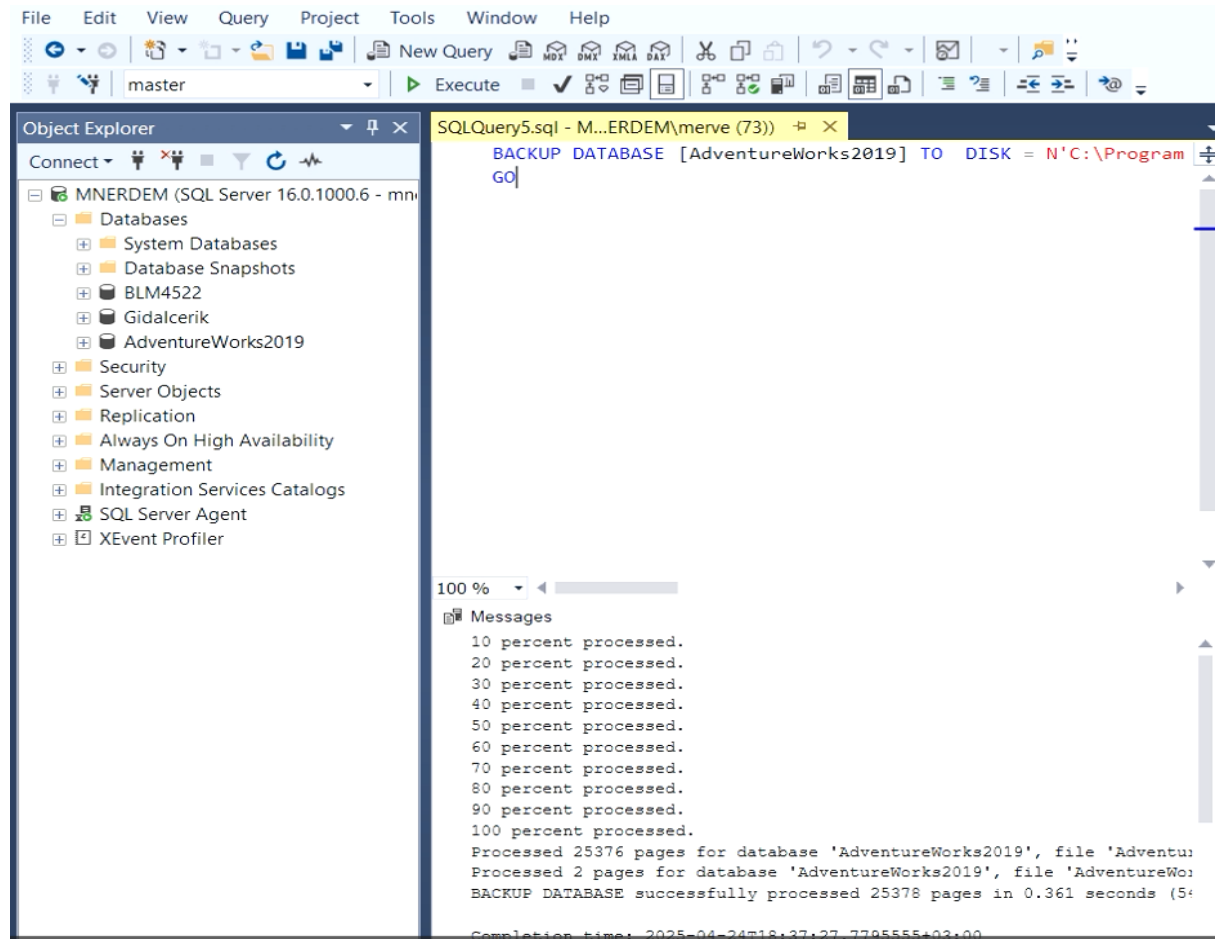
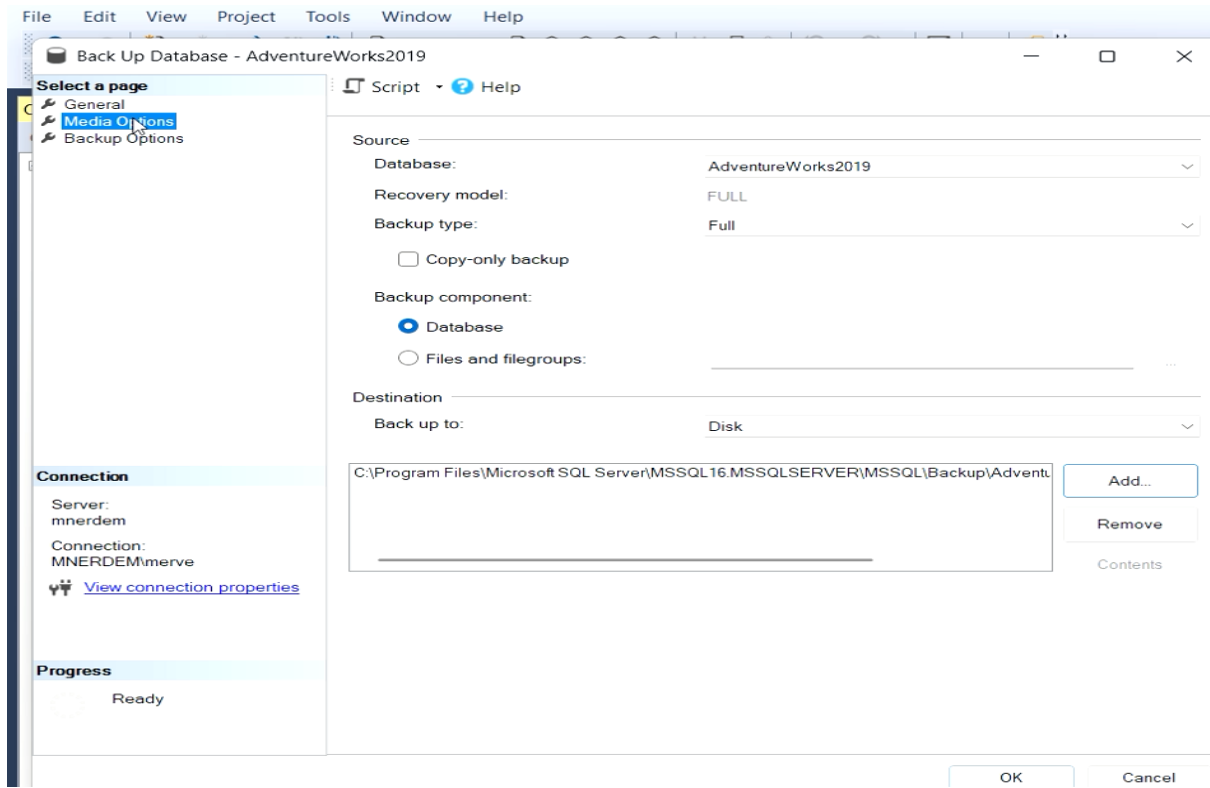
Move Up... Move Down...

The Back Up Database (Full) task allows you to specify the source databases, destination files or tapes, and overwrite options for a full backup.

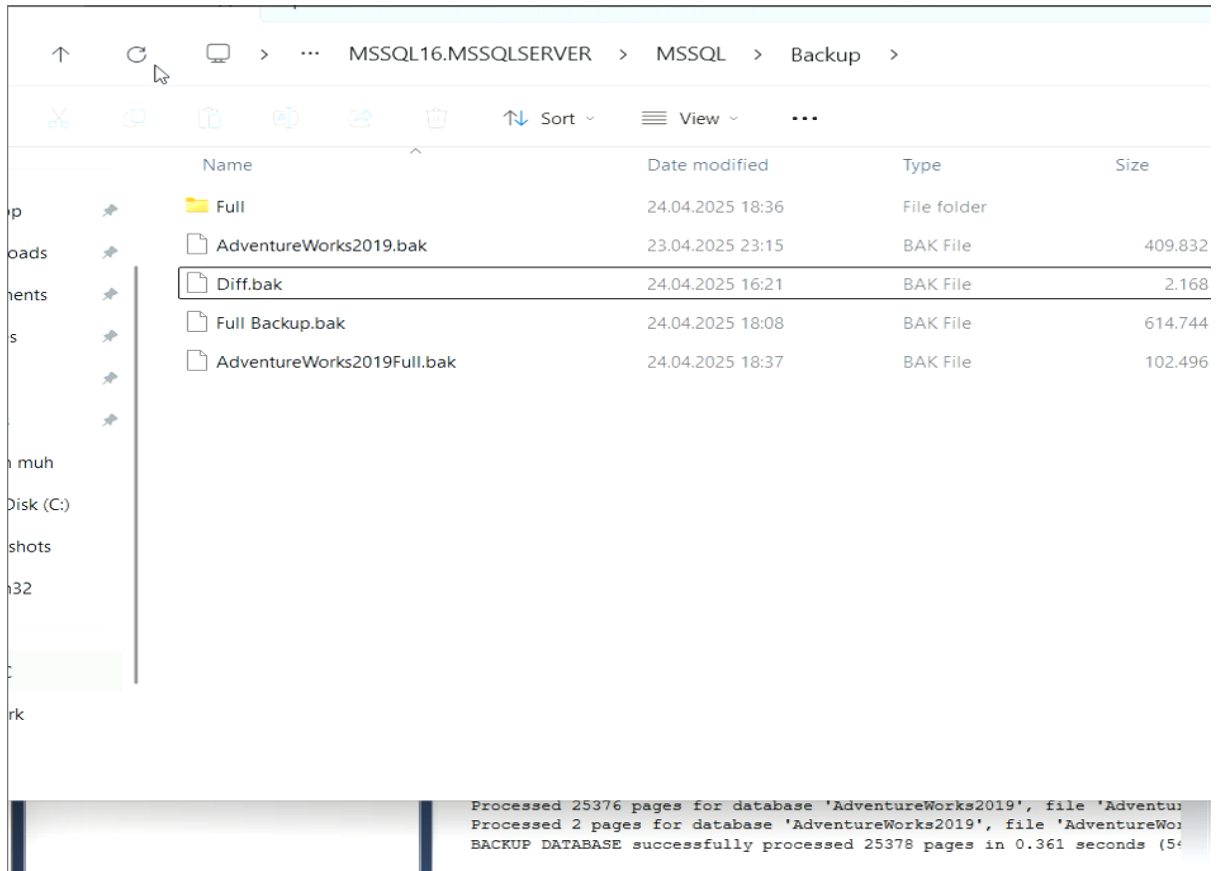
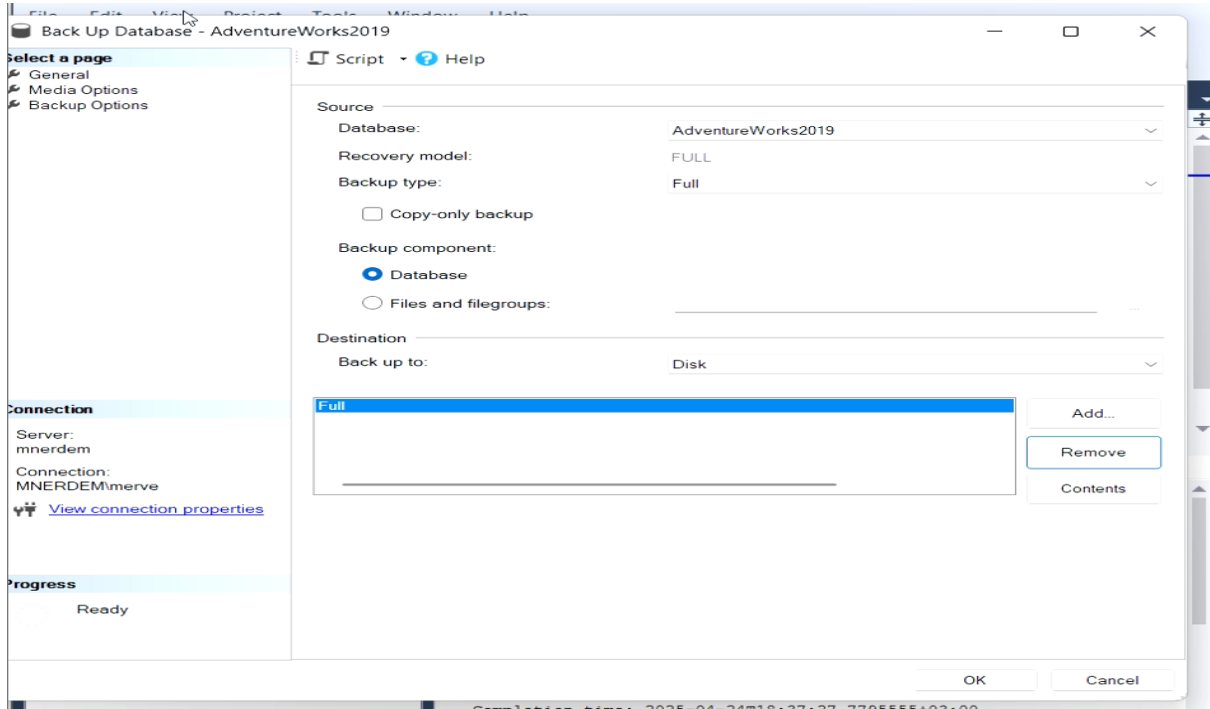
Help < Back Next > Finish Cancel



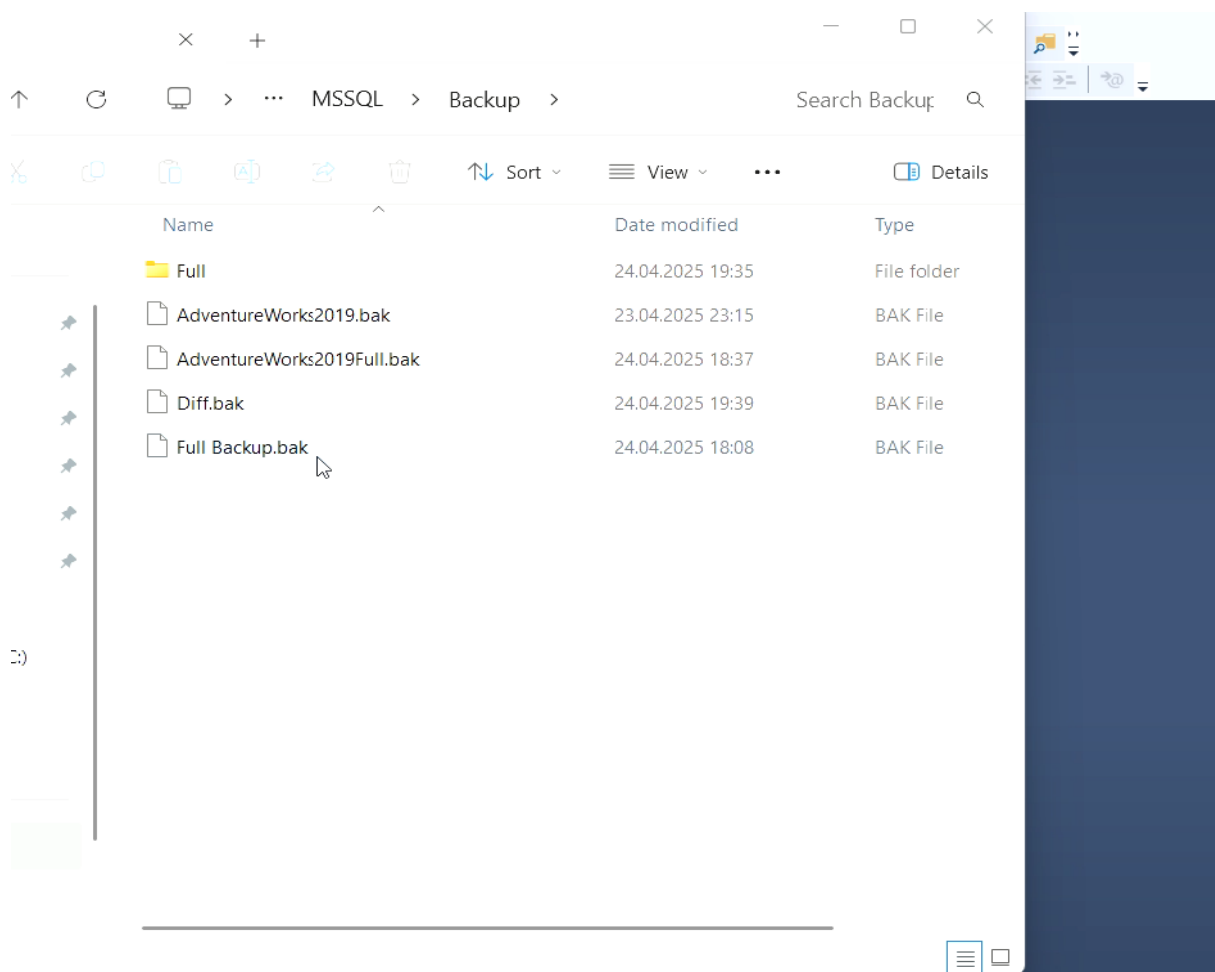
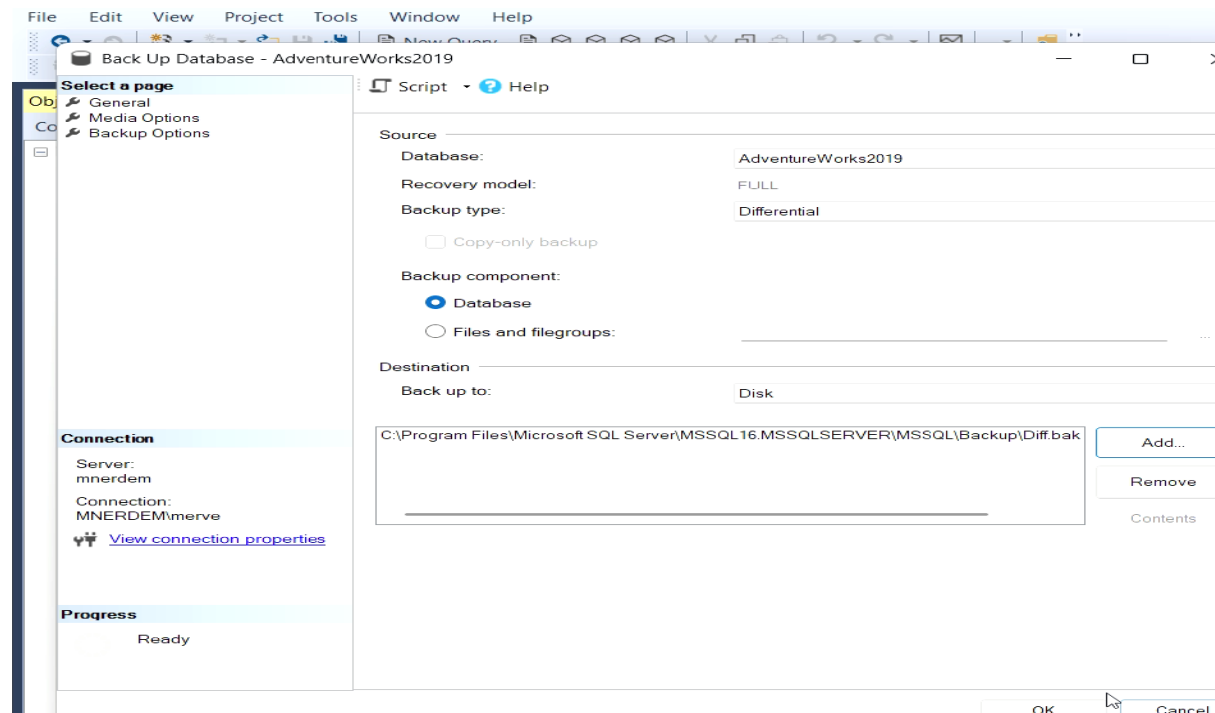
Felaketten Kurtarma Senaryoları: Full



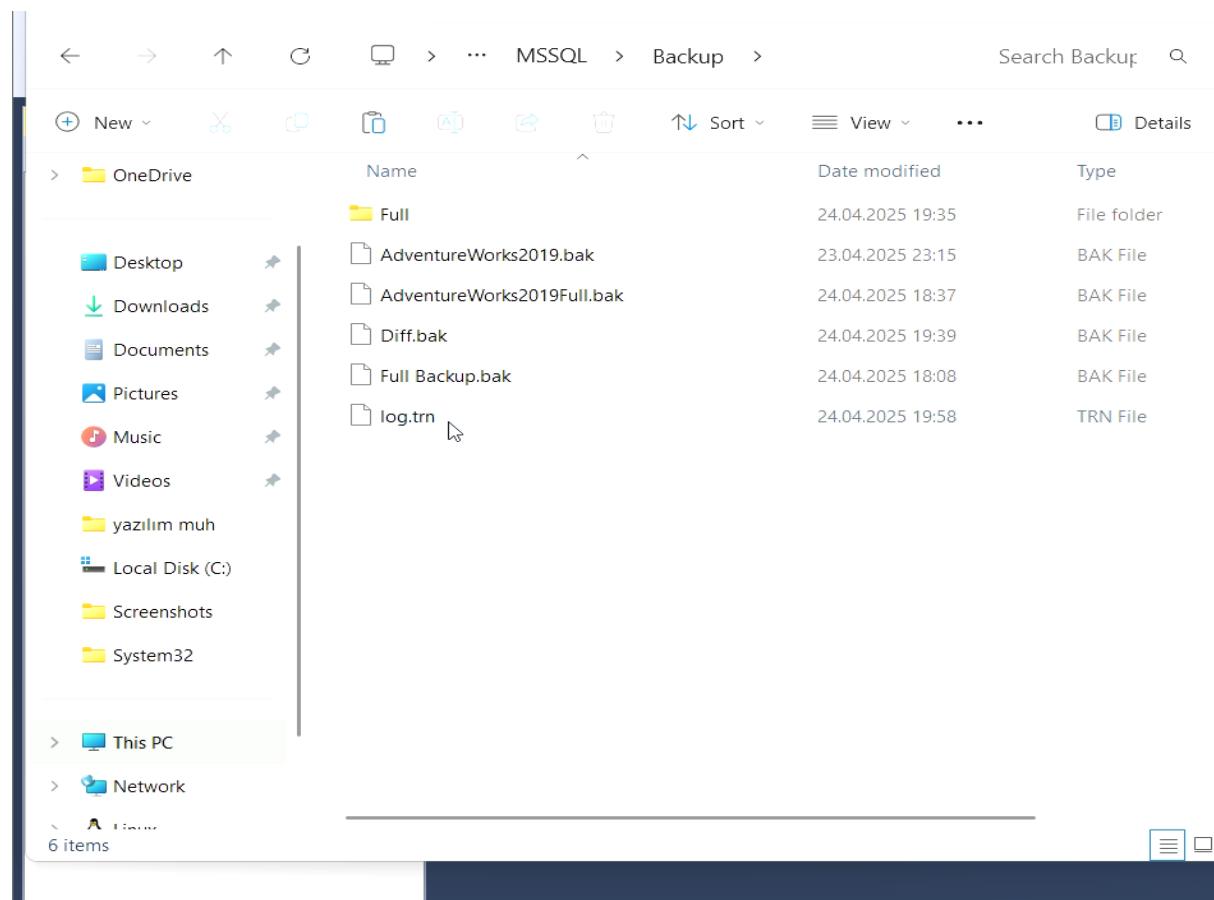
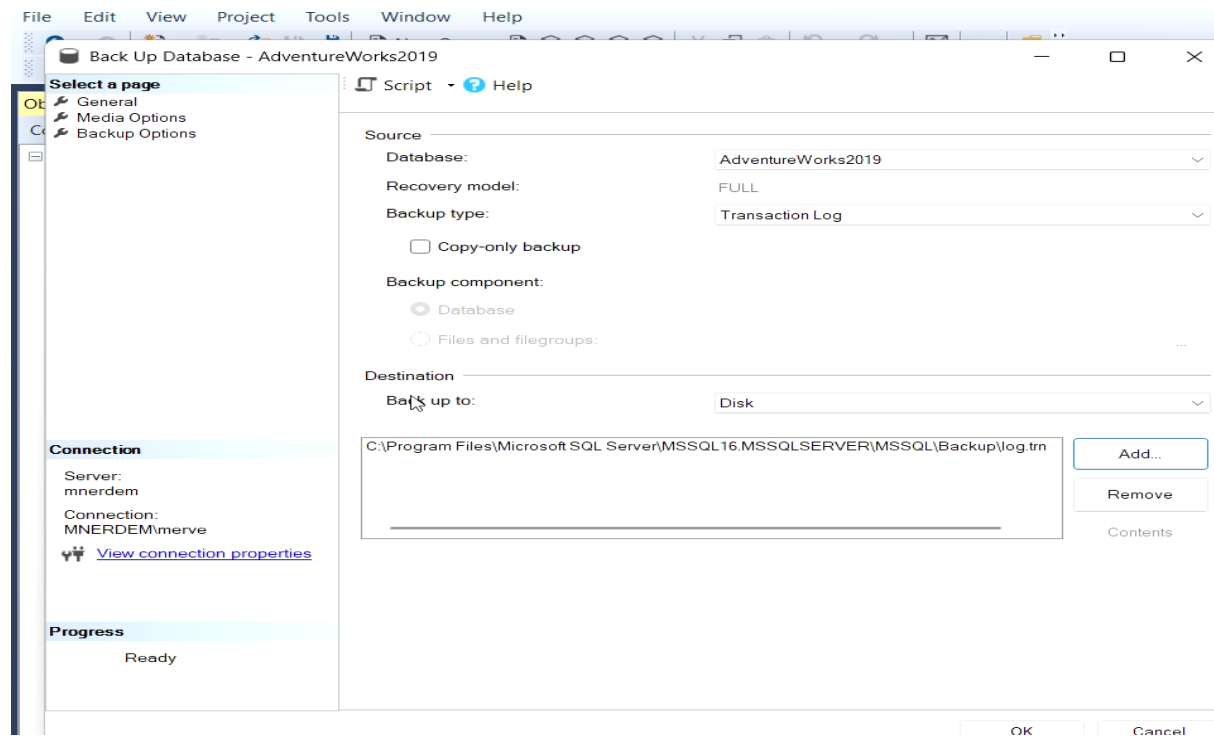
ya da ikinci seçenek olarak



Differential



Transaction Log



Test Yedekleme Senaryoları:

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the server hierarchy for 'MNERDEM (SQL Serv...)' with folders for Databases, Security, Server Objects, Replication, Always On High Availability, Management, Integration Services, SQL Server Agent, and XEvent Profiler. The main pane shows a SQL query window titled 'SQLQuery1.sql - M...ERDEM\nerve (62))' containing three 'RESTORE VERIFYONLY' commands. The Messages pane at the bottom shows the execution results, indicating that the backup set on file 1 is valid for all three files. The completion time is 2025-04-25T01:21:15.7501657+03:00. A yellow status bar at the bottom confirms 'Query executed successfully.'

```
RESTORE VERIFYONLY
FROM DISK = 'C:\SQLYedek\Adventureworks2019_TamYedek2.bak';

RESTORE VERIFYONLY
FROM DISK = 'C:\SQLYedek\Adventureworks2019_DIFF2.bak';

RESTORE VERIFYONLY
FROM DISK = 'C:\SQLYedek\Adventureworks2019_LOG2.trn';
```

100 %

Messages

The backup set on file 1 is valid.
The backup set on file 1 is valid.
The backup set on file 1 is valid.

Completion time: 2025-04-25T01:21:15.7501657+03:00

100 %

Query executed successfully.

3. Veritabanı Güvenliği ve Erişim Kontrolü

Erişim Yönetimi:

File Edit View Project Tools Window Help

Login - New

Select a page

- General
- Server Roles
- User Mapping
- Securables
- Status

Connection

Server: mnerdem

Connection: MNERDEM\merve

[View connection properties](#)

Progress

Error occurred

Script Help

Login name: limit2 Search...

☐ Windows authentication

☐ Microsoft Entra ID authentication

☒ SQL Server authentication

Password:

Confirm password:

☐ Specify old password

Old password:

☐ Enforce password policy

☐ Enforce password expiration

☐ User must change password at next login

☐ Mapped to certificate

☐ Mapped to asymmetric key

☐ Map to Credential

Mapped Credentials

Credential	Prov
------------	------

Add

Remove

Default database: master

Default language: <default>

Ara

OK Cancel

File Edit View Project Tools Window Help

Login - New

Select a page

- General
- Server Roles
- User Mapping
- Securables
- Status

Connection

Server: mnerdem

Connection: MNERDEM\merve

[View connection properties](#)

Progress

Ready

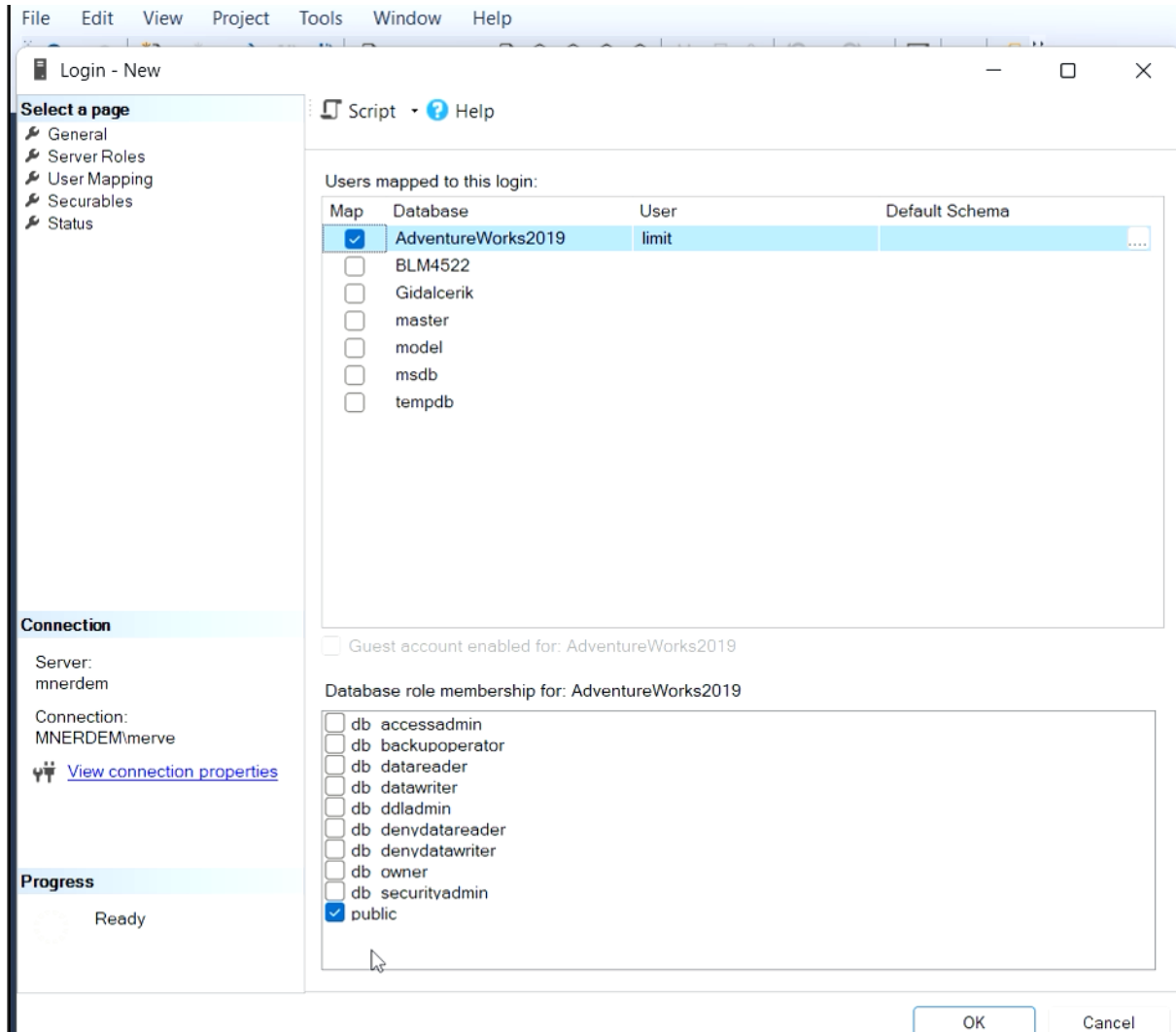
Script Help

Server role is used to grant server-wide security privileges to a user.

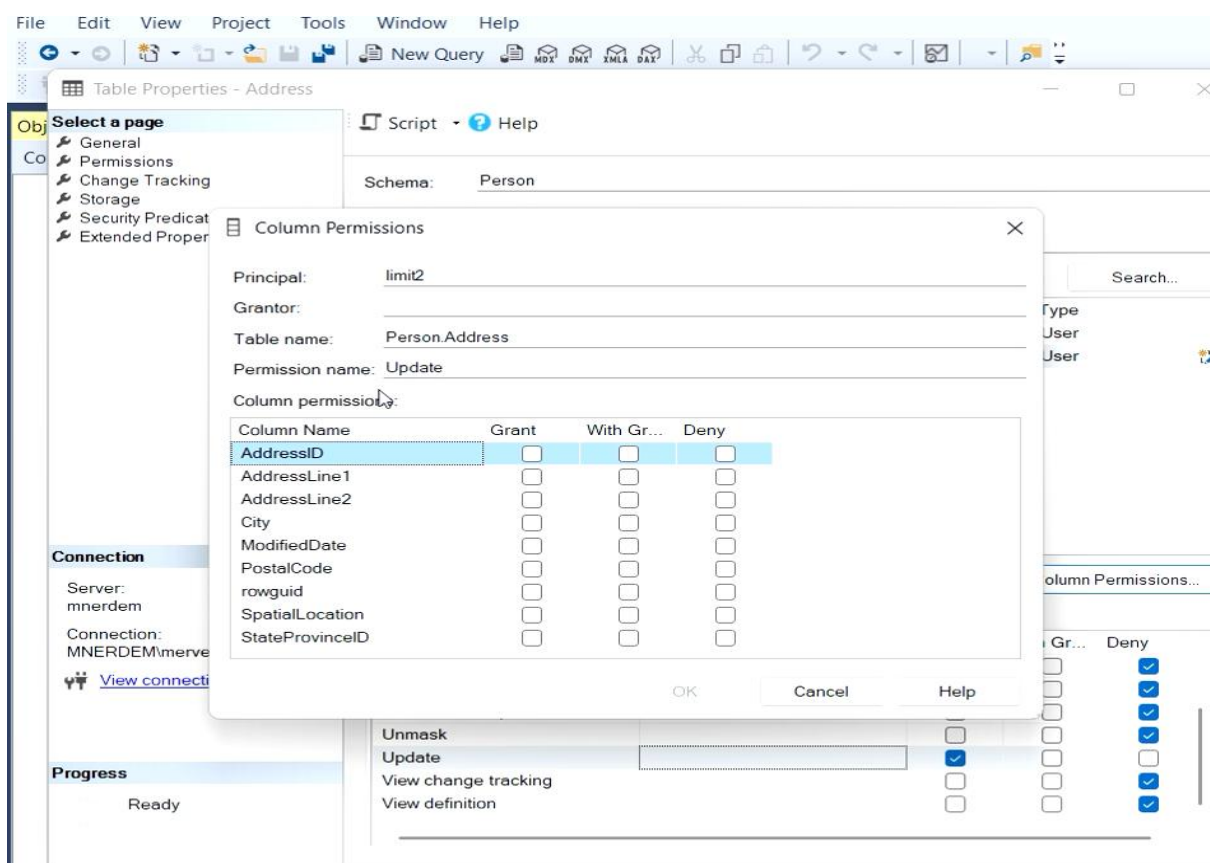
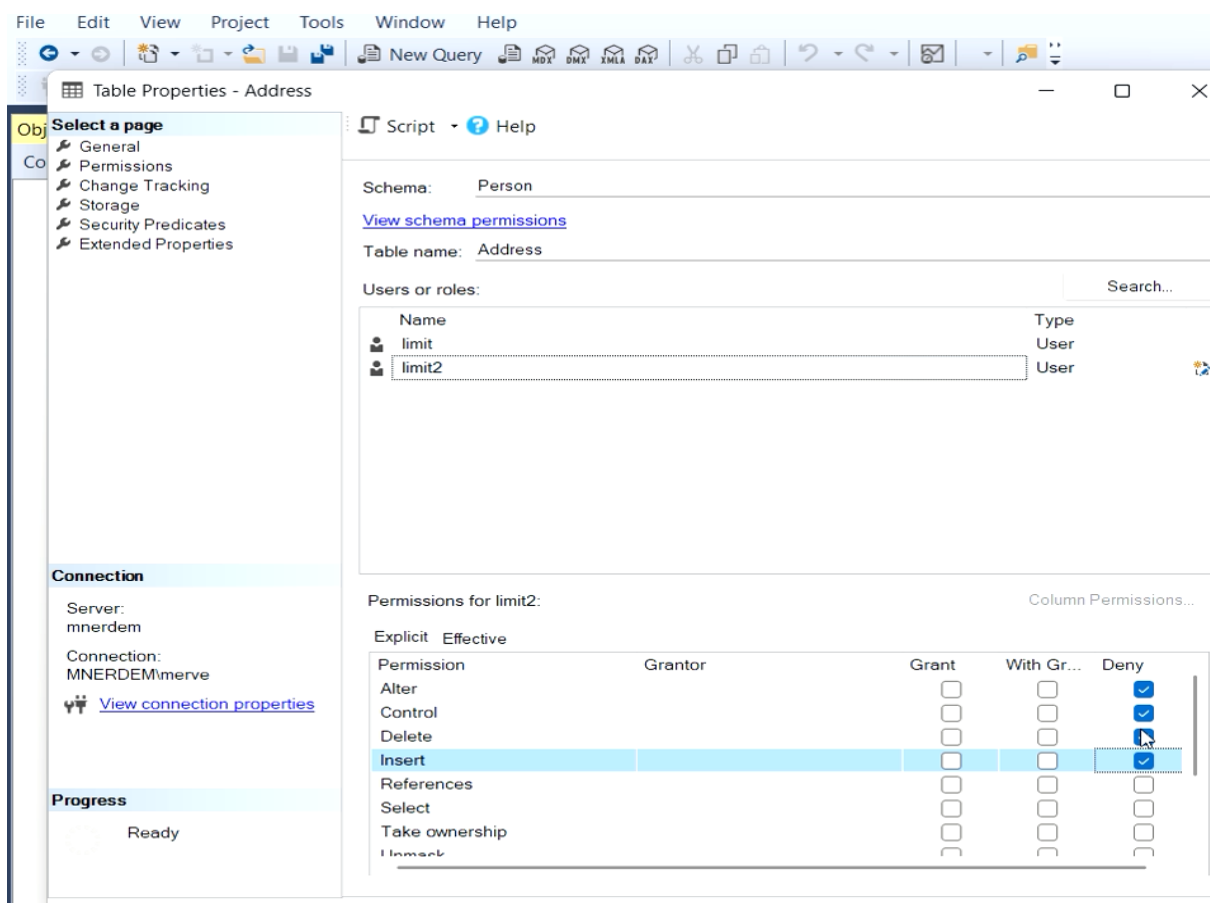
Server roles:

- ☐ ##MS DatabaseConnector##
- ☐ ##MS DatabaseManaqer##
- ☐ ##MS DefinitionReader##
- ☐ ##MS LoginManaqer##
- ☐ ##MS PerformanceDefinitionReader##
- ☐ ##MS SecurityDefinitionReader##
- ☐ ##MS ServerPerformanceStateReader##
- ☐ ##MS ServerSecurityStateReader##
- ☐ ##MS ServerStateManaqer##
- ☐ ##MS ServerStateReader##
- ☐ bulkadmin
- ☐ dbcreator
- ☐ diskadmin
- ☐ processadmin
- ☒ public
- ☐ securityadmin
- ☐ serveradmin
- ☐ setupadmin
- ☐ sysadmin

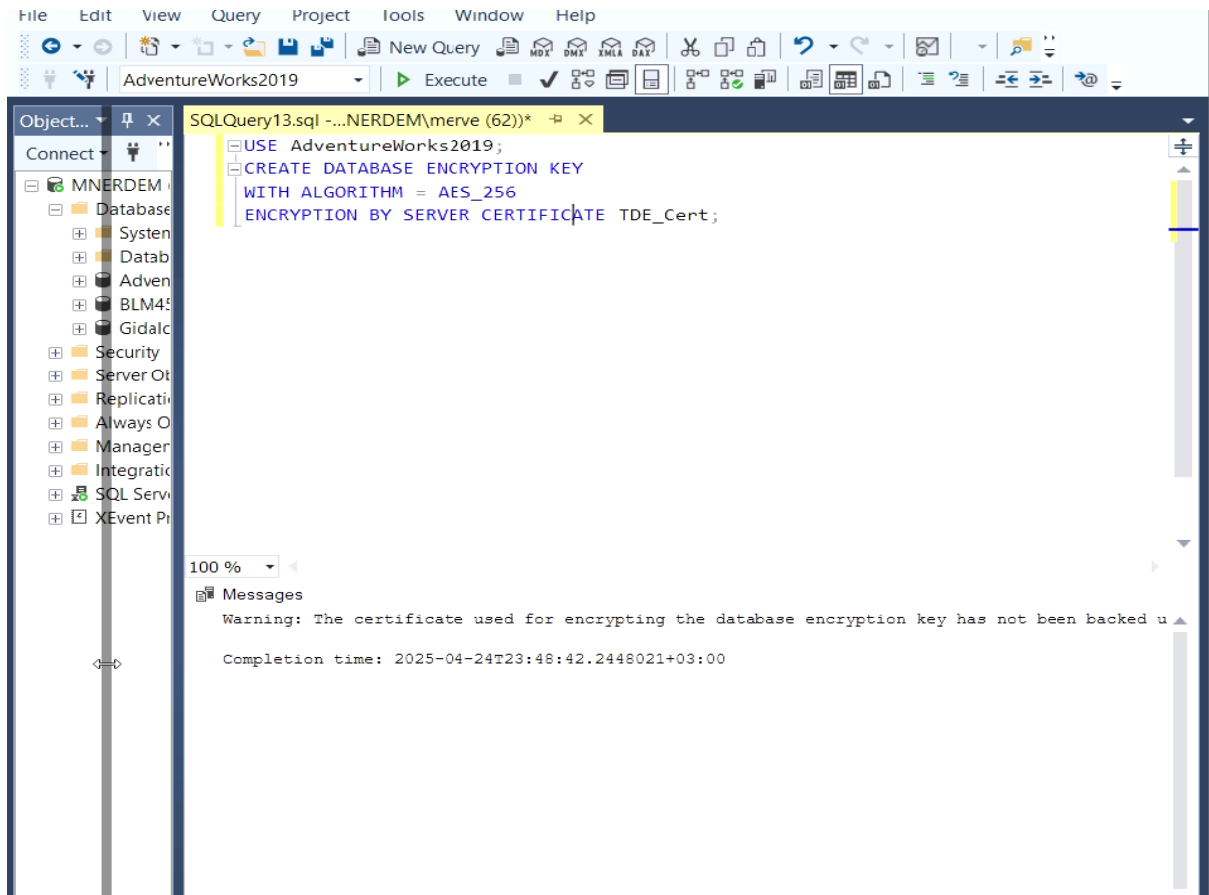
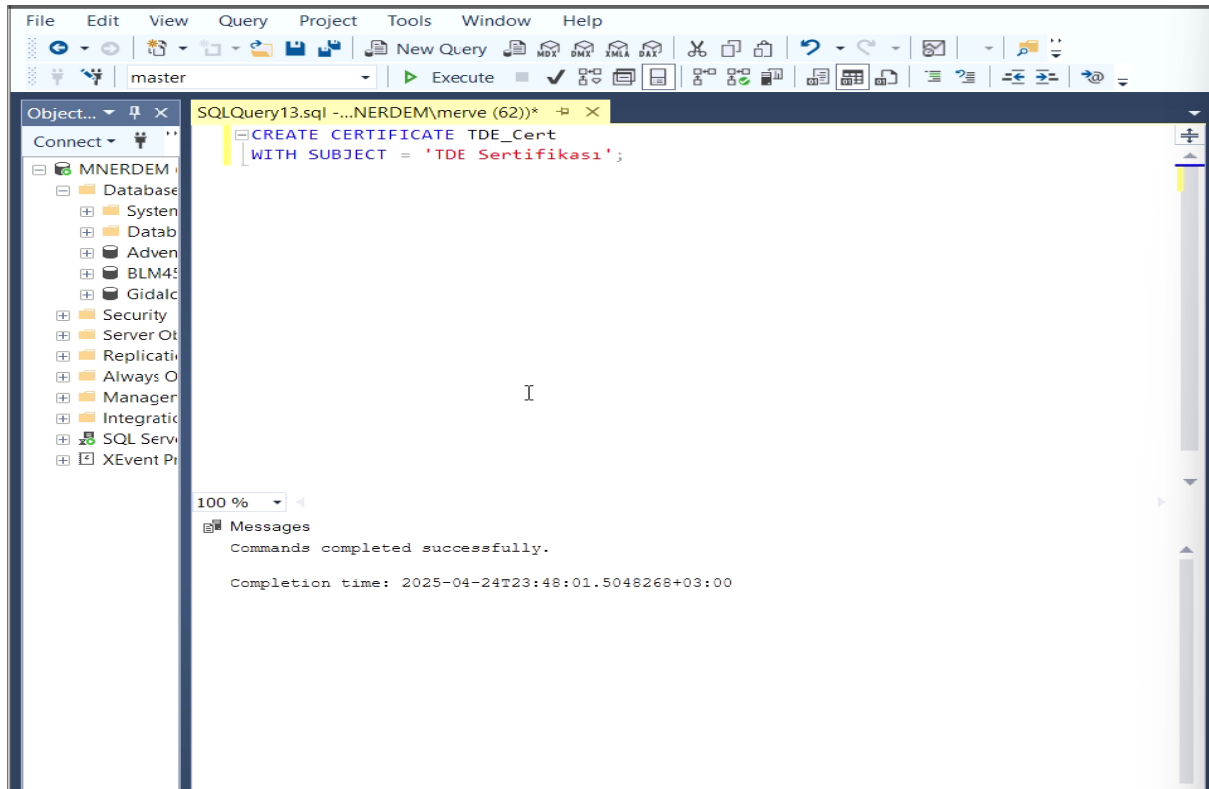
OK Cancel

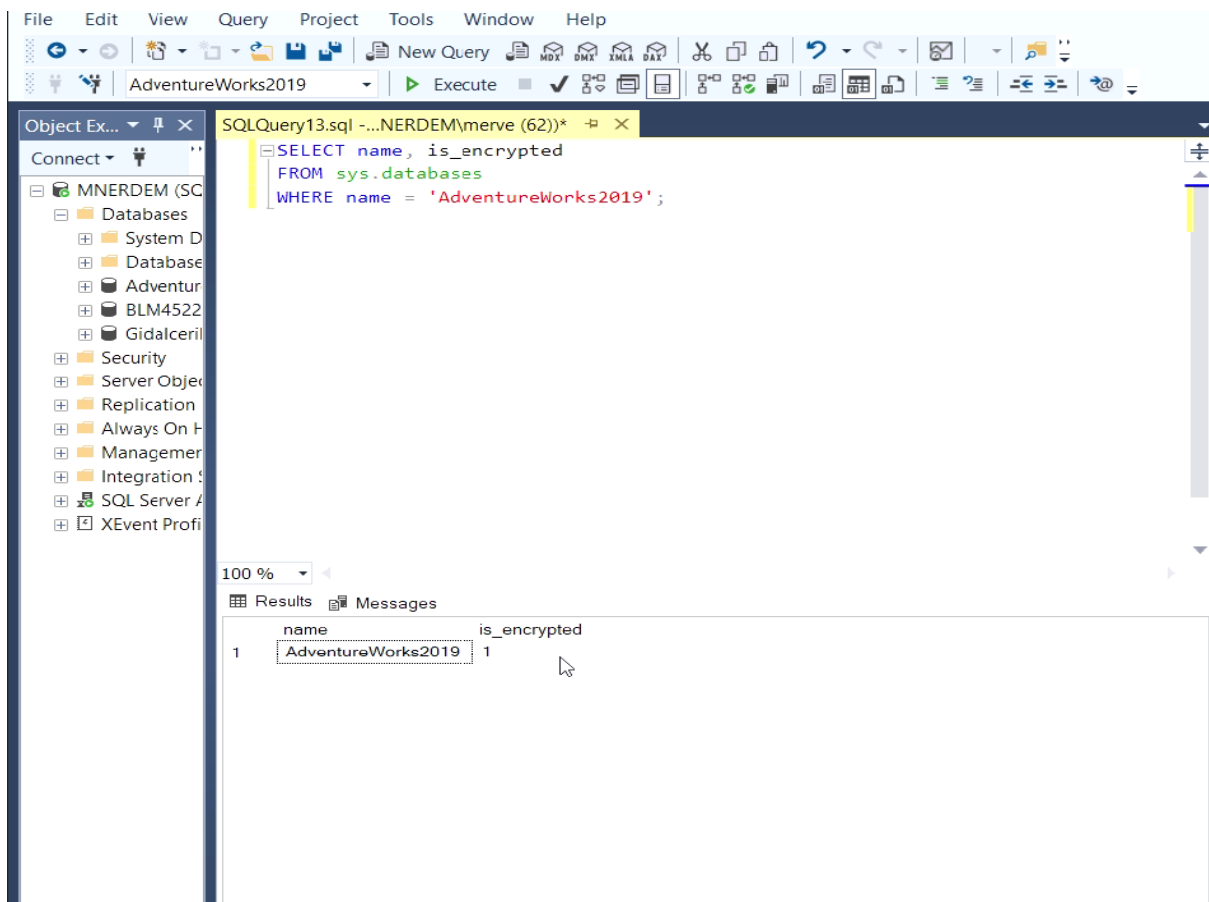
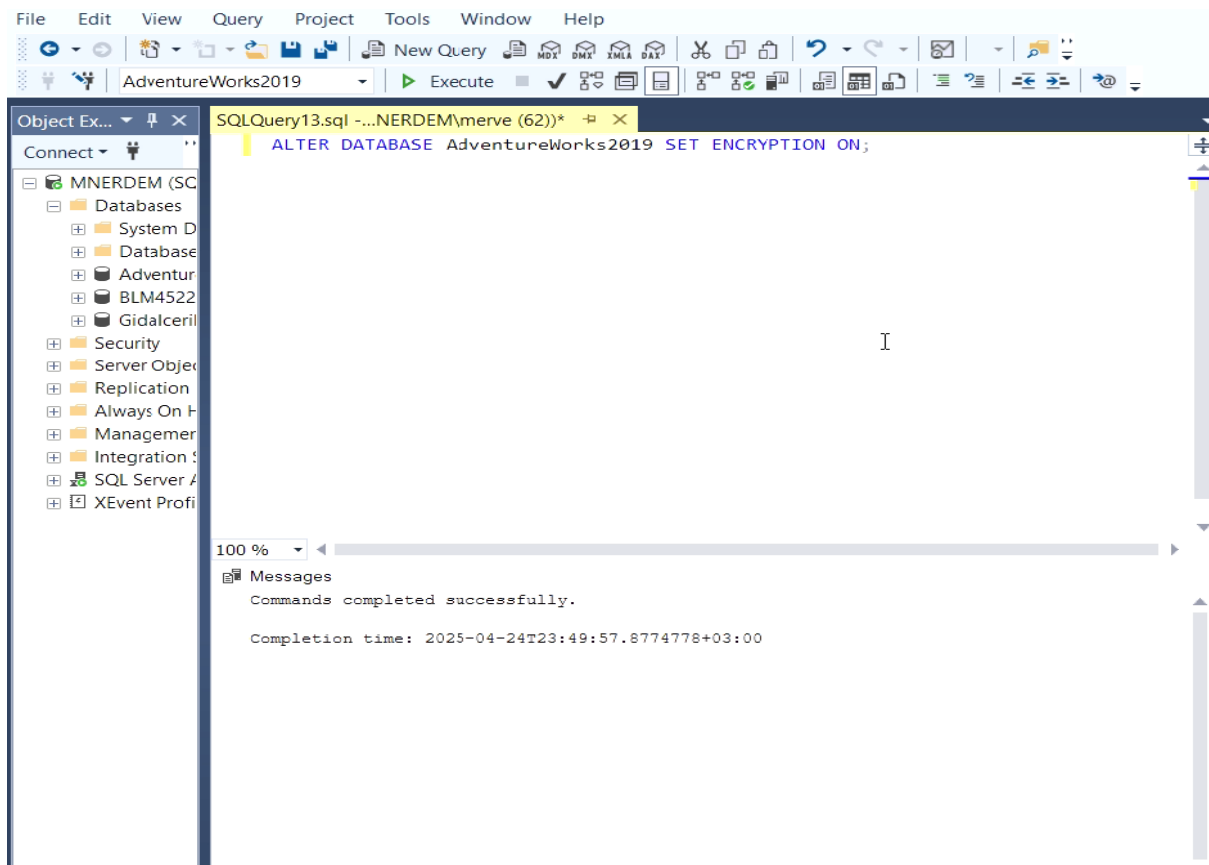


Tablolar içinde ayrıca düzenlemeler ile kısıtlamalar yapabiliriz.

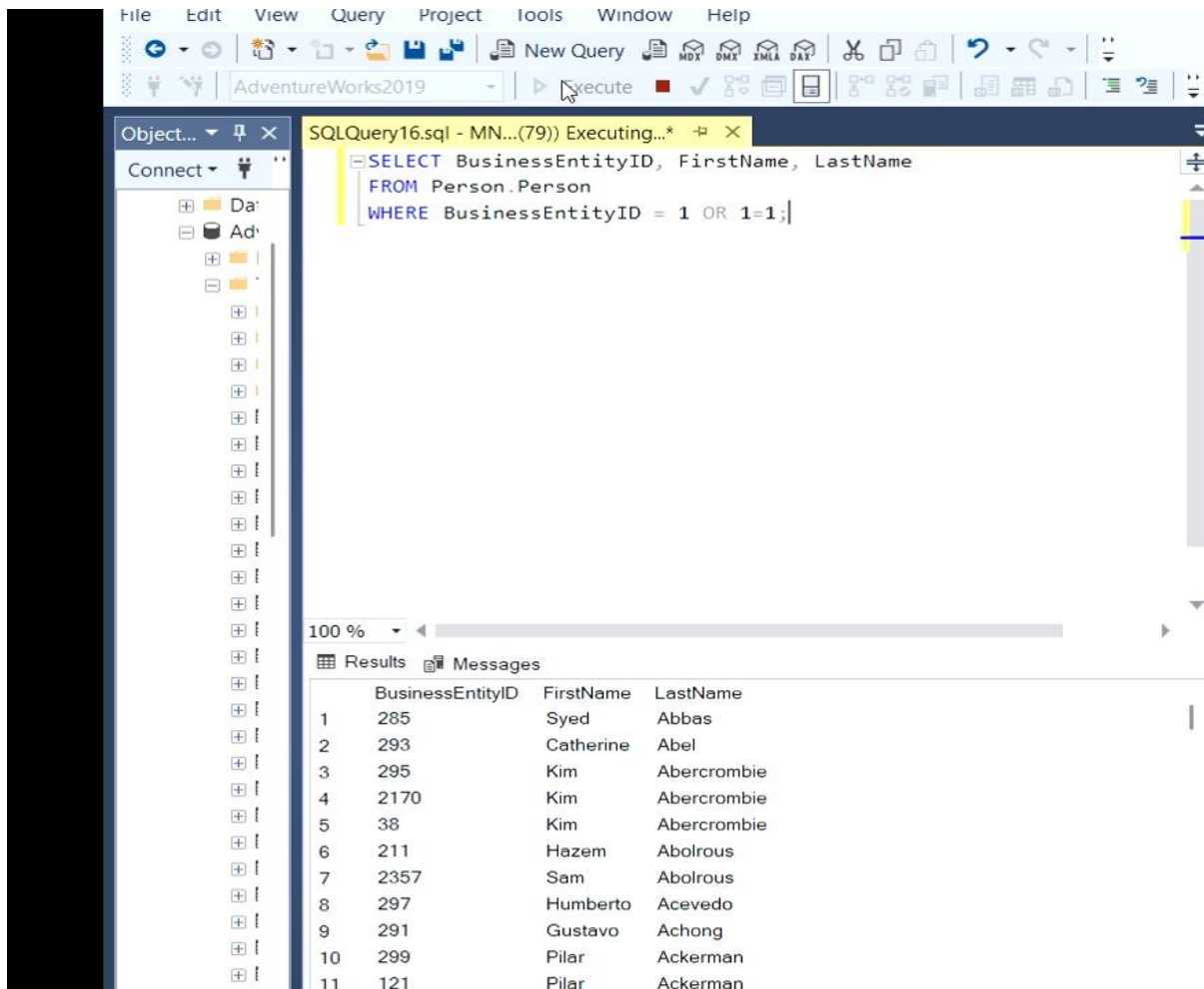
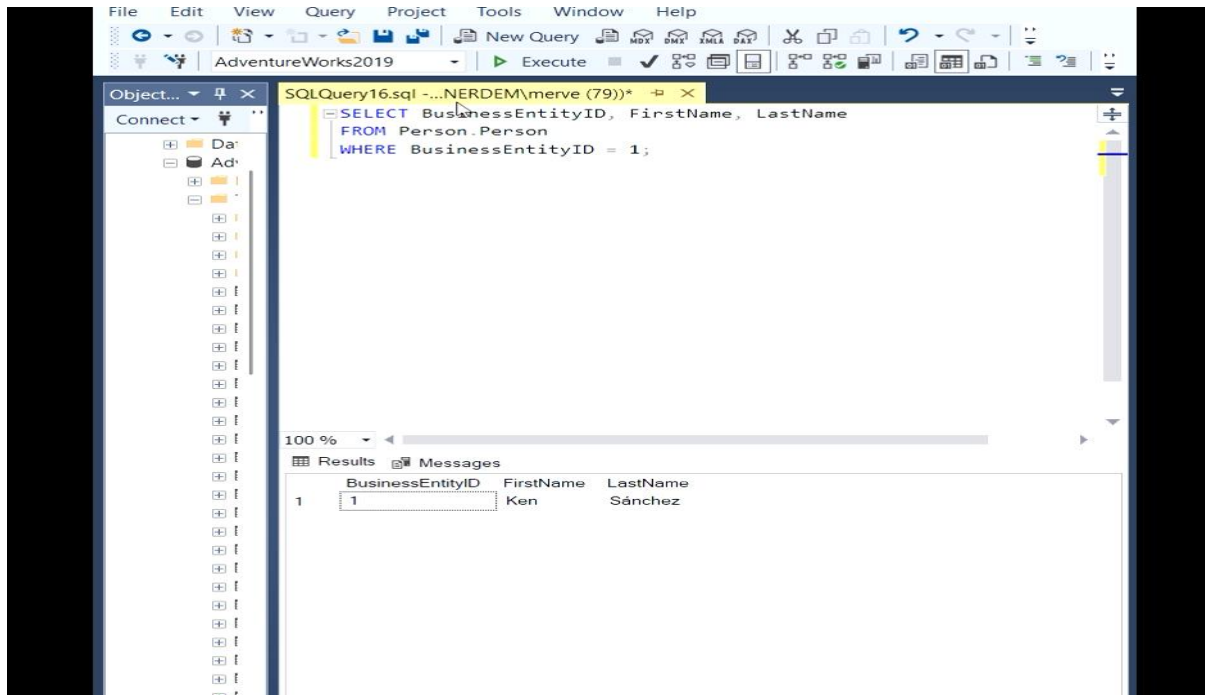


Veri Şifreleme:

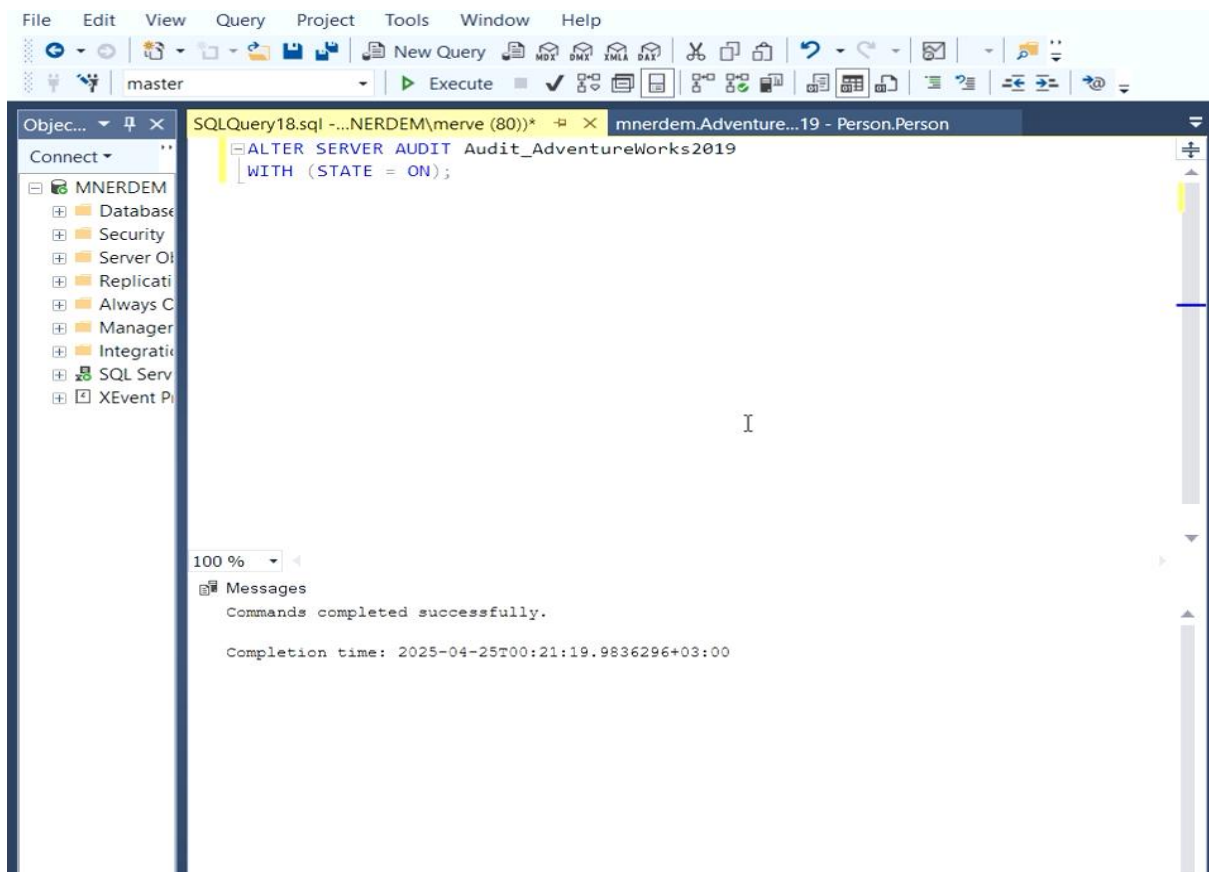
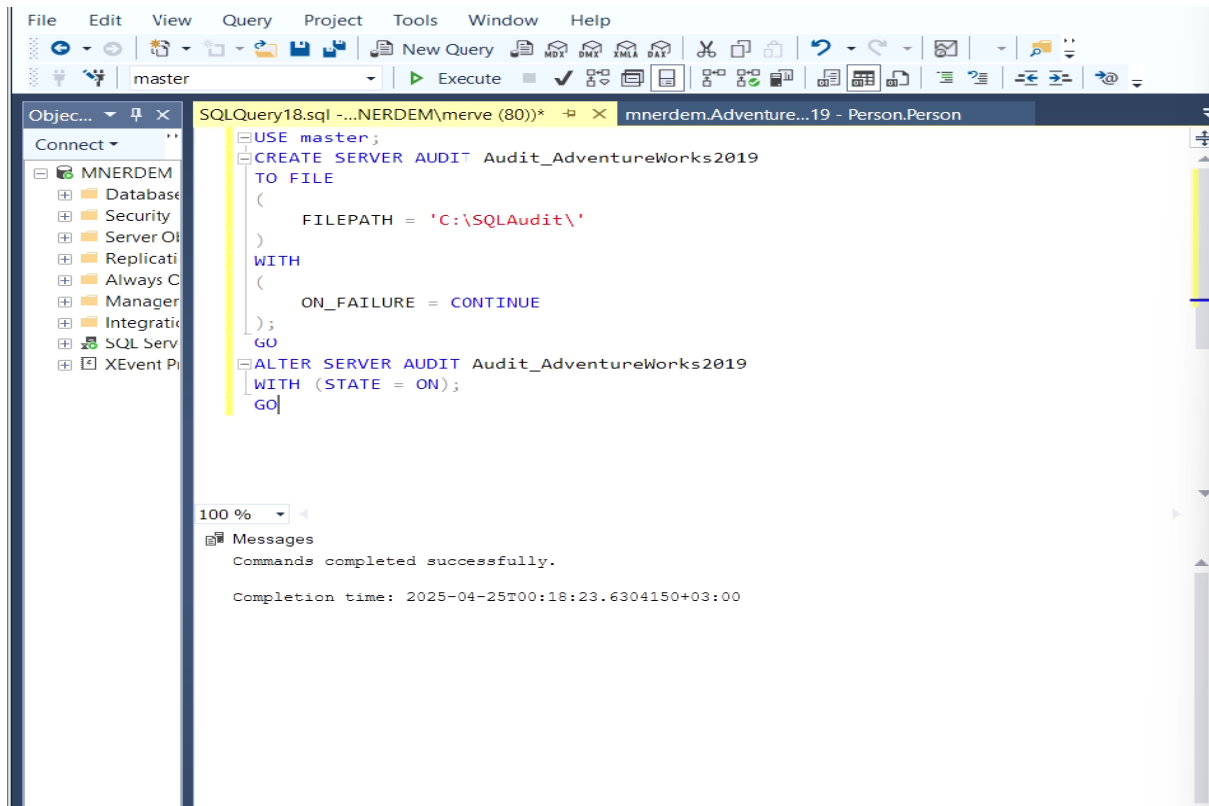


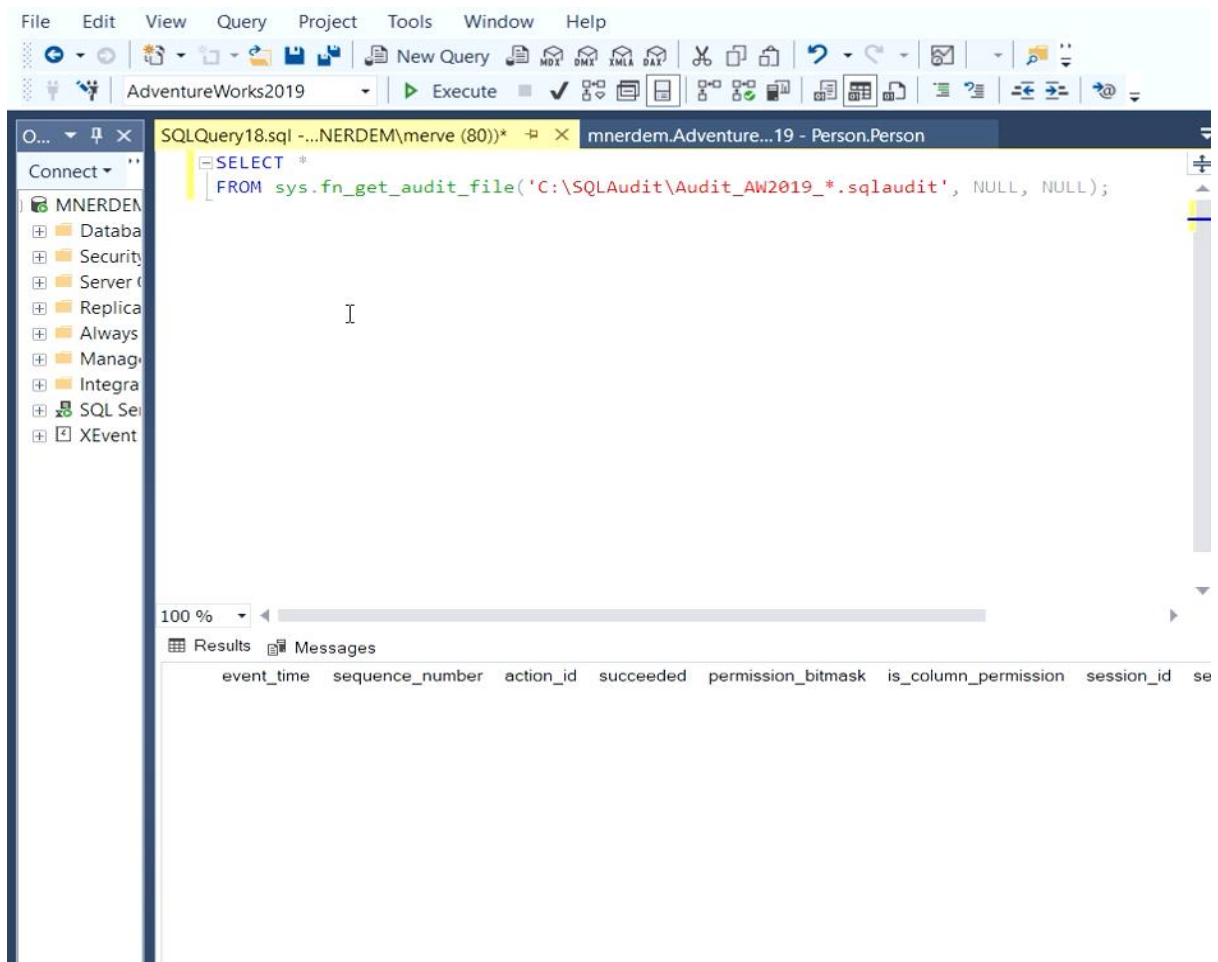
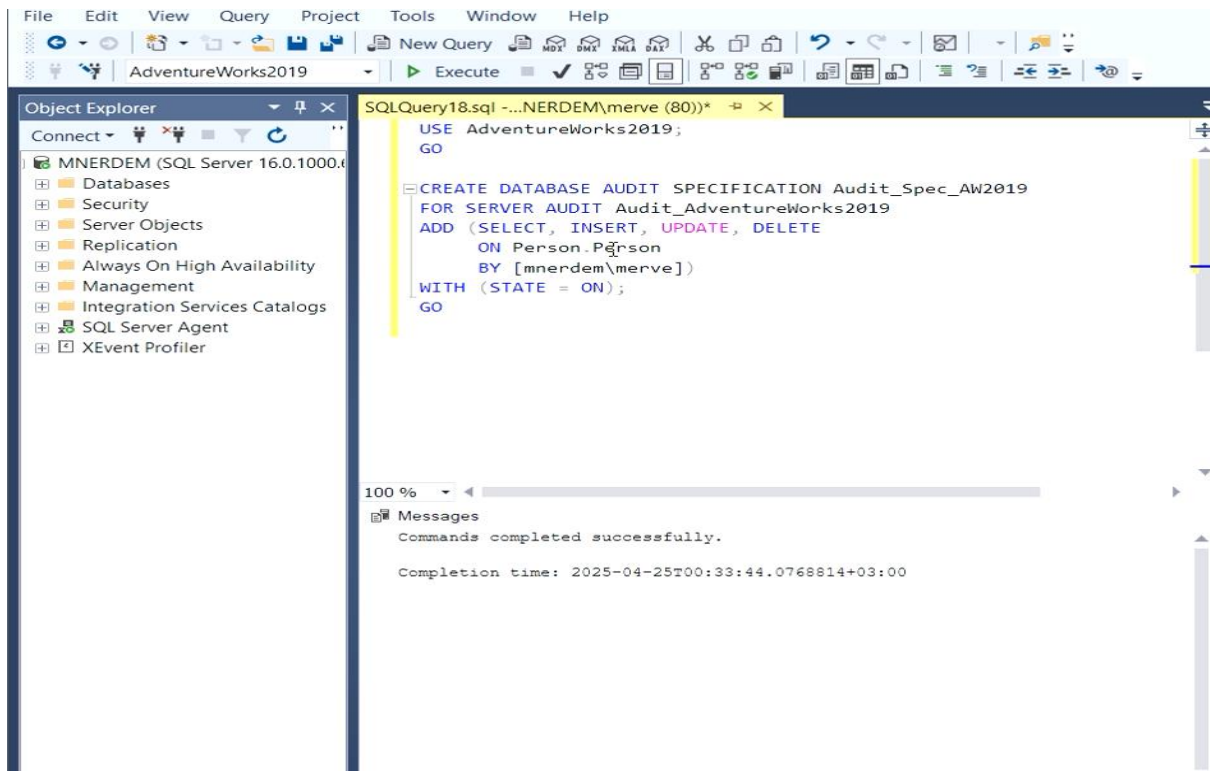


SQL Injection Testleri:



Audit Logları:

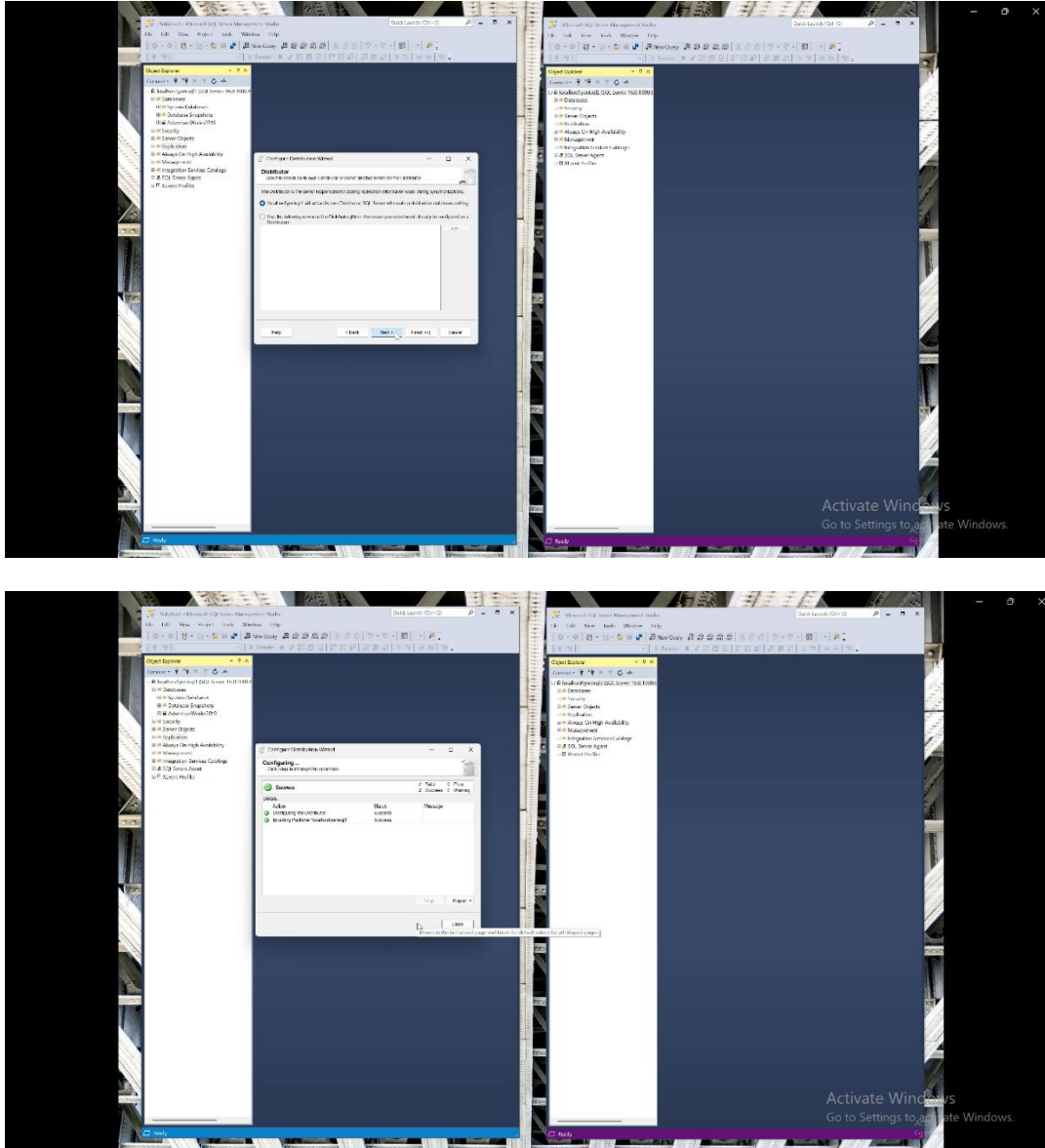




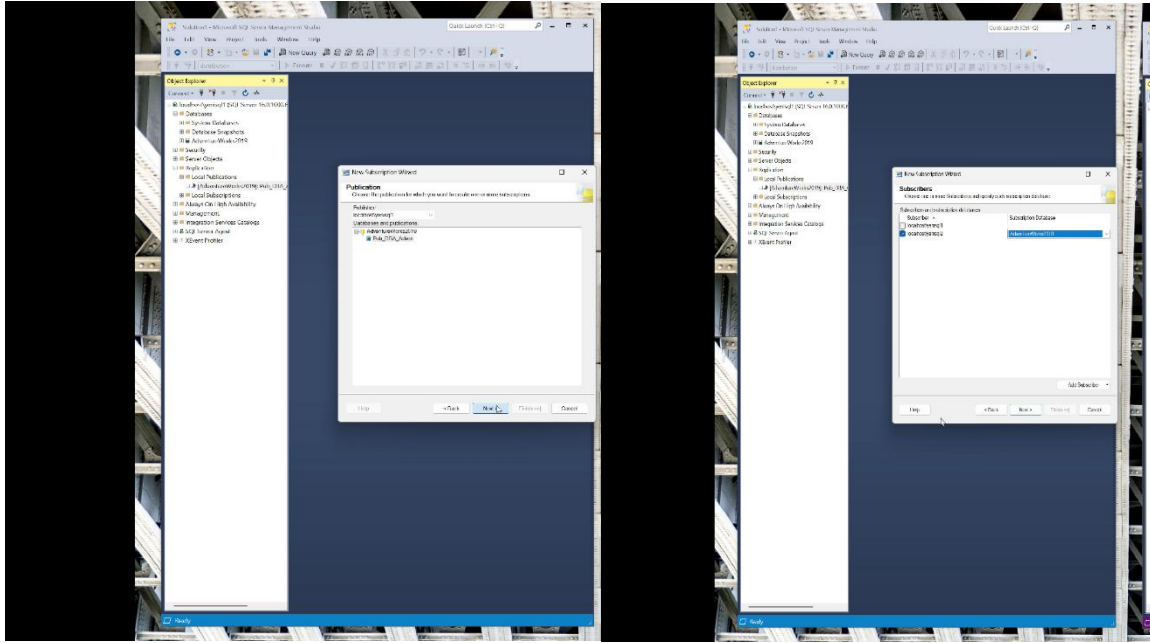
4. Veri tabanı Yük Dengeleme ve Dağıtık Veri tabanı Yapıları

Veri tabanı Replikasyonu:

SQL Server'da veri replikasyonu kurulumu, SQL Server Management Studio'da Replication sekmesinden başlatılır ve Configure Distribution Wizard aracı ile devam edilir. Bu sihirbazda, replikasyonun merkezi bileşeni olan Distributor seçilir ve bir dağıtım veritabanı oluşturulur. Ardından, hangi veritabanının yayınlanacağı belirlenir ve replikasyona dahil edilecek tablolar seçilir.

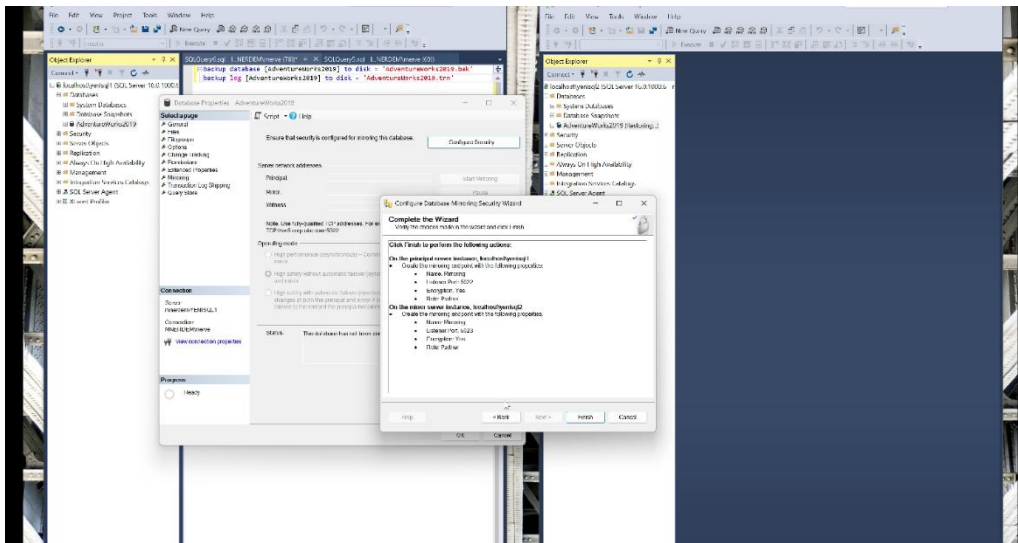


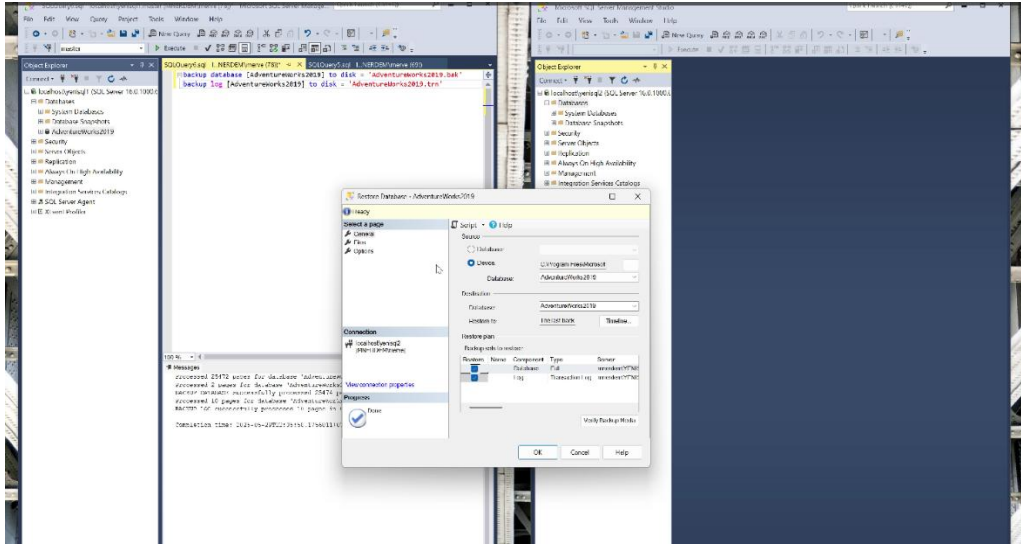
New Subscription Wizard ile belirli bir yayın seçilmekte ve bu yayına ait abonelik oluşturulmaktadır. AdventureWorks2019 veritabanı üzerinde "Pub_DBA_Adven" adında bir yayın seçildi.



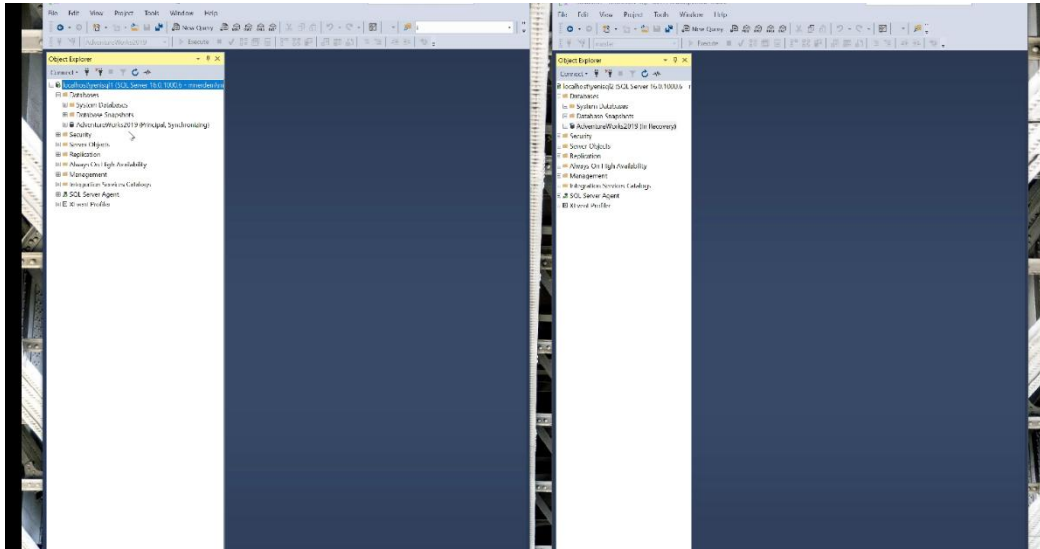
Yük Dengeleme:

Öncelikle Database Properties penceresini açtık ve sol menüden Mirroring sekmesine geçtik. Bu sekmede, Configure Security sihirbazını başlatarak Principal ve Mirror sunucuları tanımladık. Daha sonra, Service Accounts ve Endpoints yapılandırılarak iletişim için gerekli portlar belirlendi. Son olarak Start Mirroring butonuna tıklayarak mirroring oturumu başlatıldı ve veritabanı eşitlemesi sağlandı.





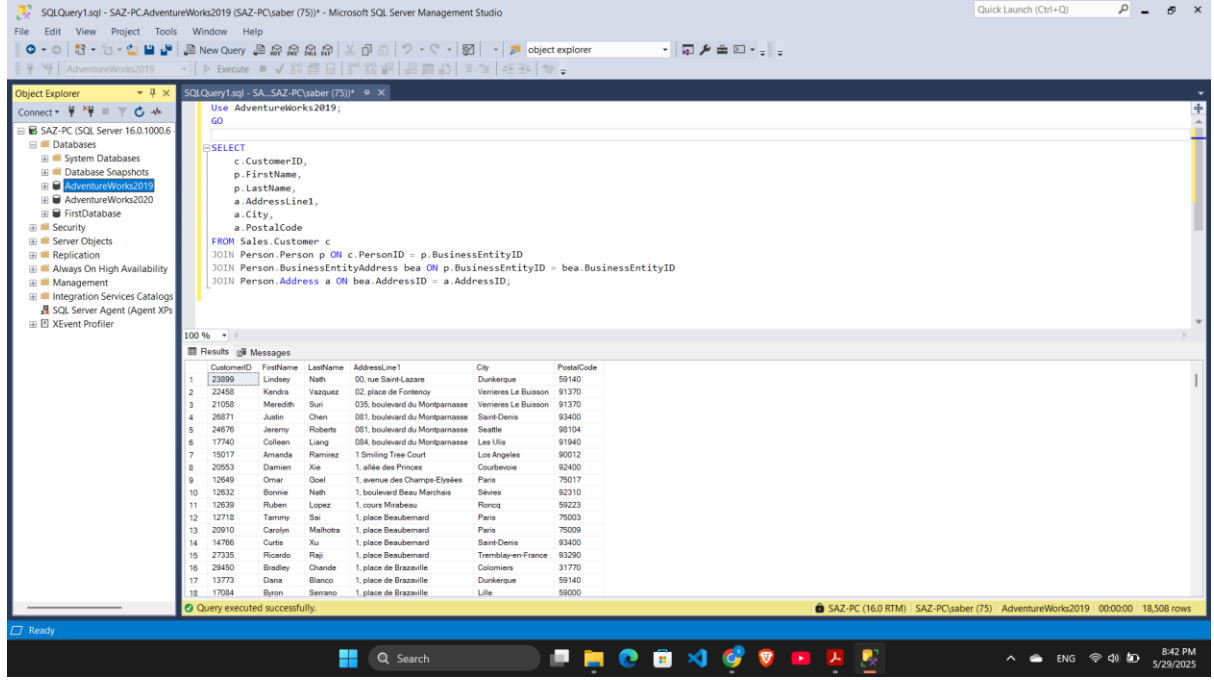
Sol tarafta principal sunucu aktif ve veri eşitleniyor, sağ tarafta ise recovery durumunda yedek sunucu hazır beklemektedir.



5. Veri Temizleme ve ETL Süreçleri Tasarımı

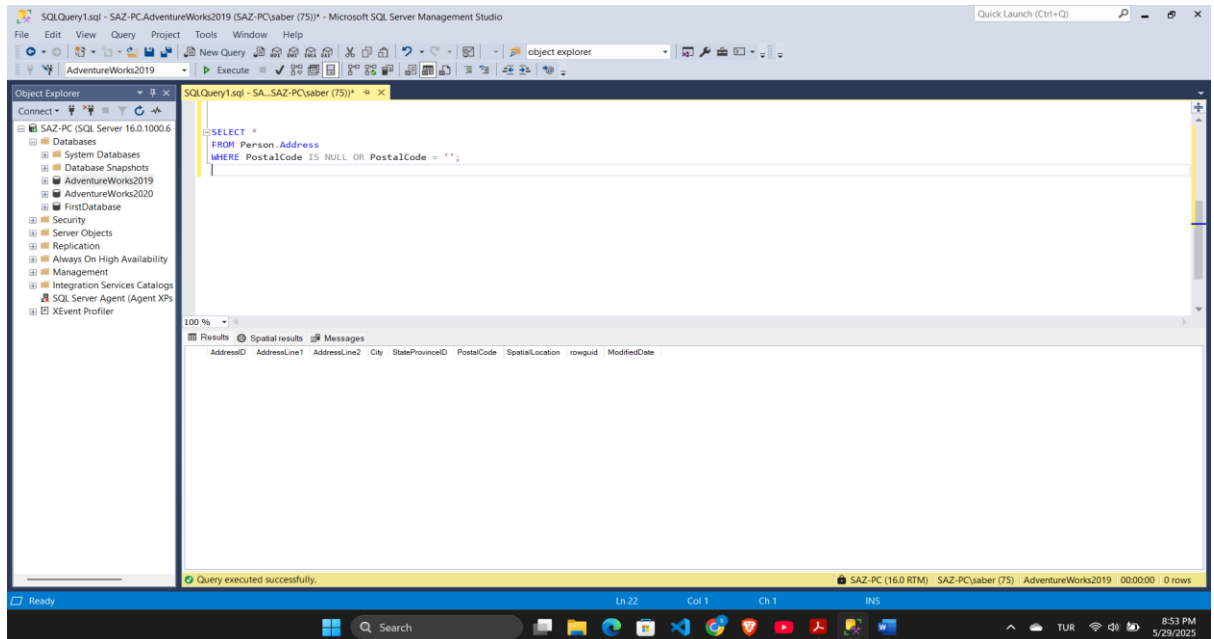
1. Extract (Veri Çekme)

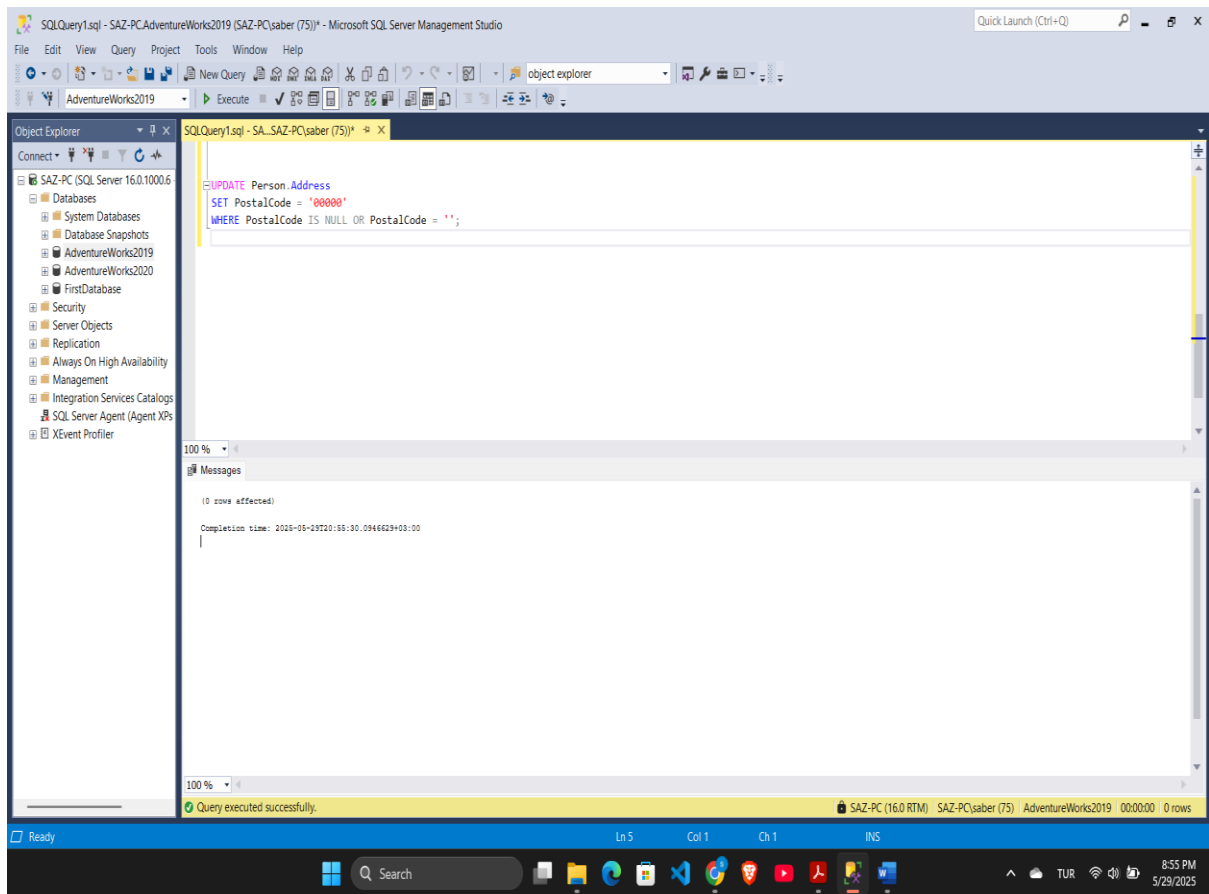
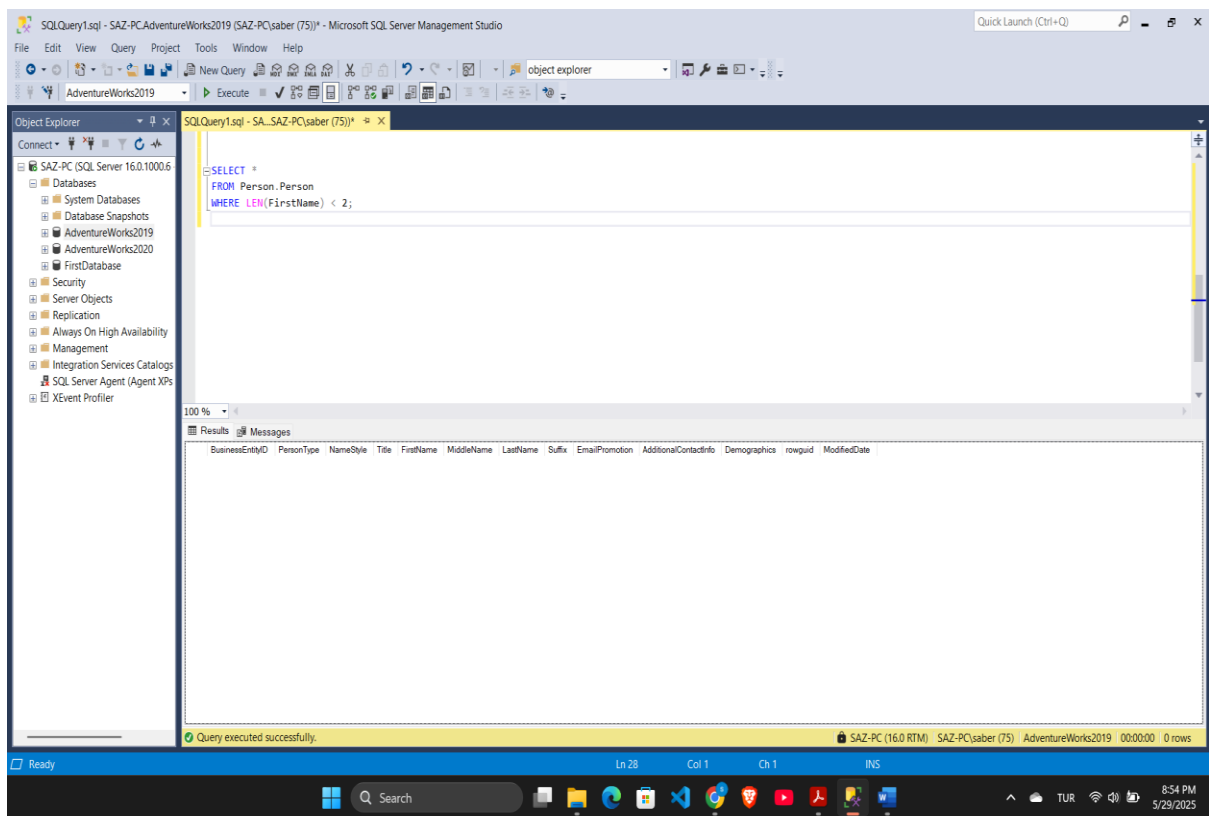
AdventureWorks2019 içerisinde analiz için Customer, SalesOrderHeader, Person, ve Address tabloları seçilmiştir. Bu tablolar arasında müşterilere ait demografik bilgiler, sipariş bilgileri ve adres detayları yer almaktadır.

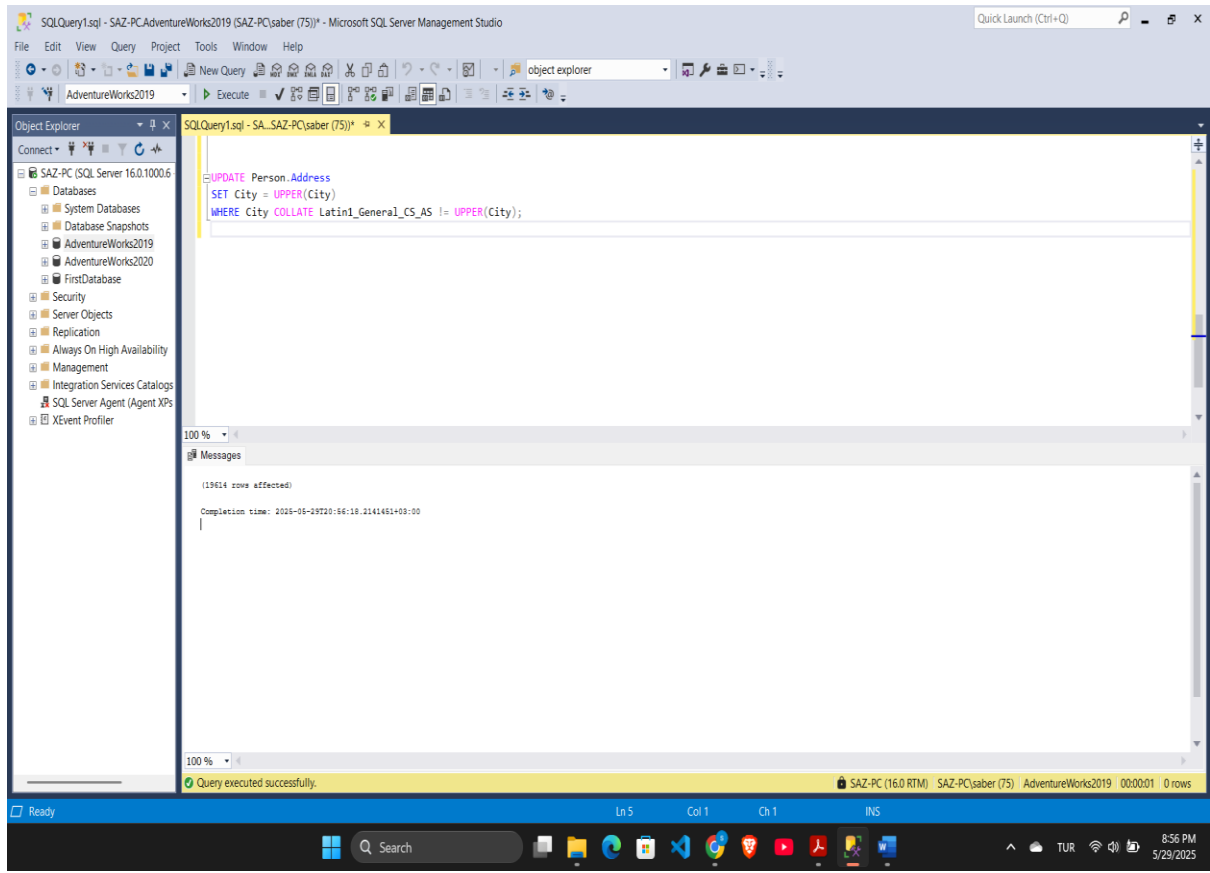


2. Transform (Veri Temizleme ve Dönüştürme)

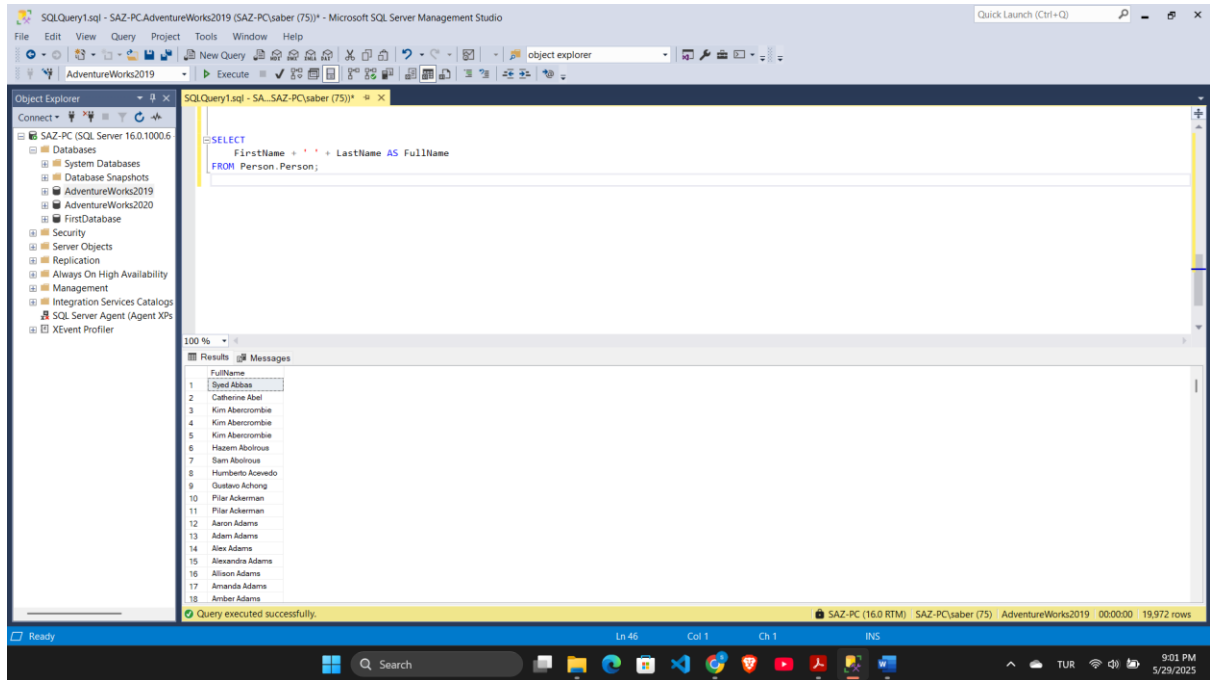
Veri Temizleme aşamasında eksik PostalCode, hatalı ya da kısa FirstName ve tutarsız şehir isimleri (örneğin küçük harfle başlayan şehirler) problemler analiz edilmiştir.

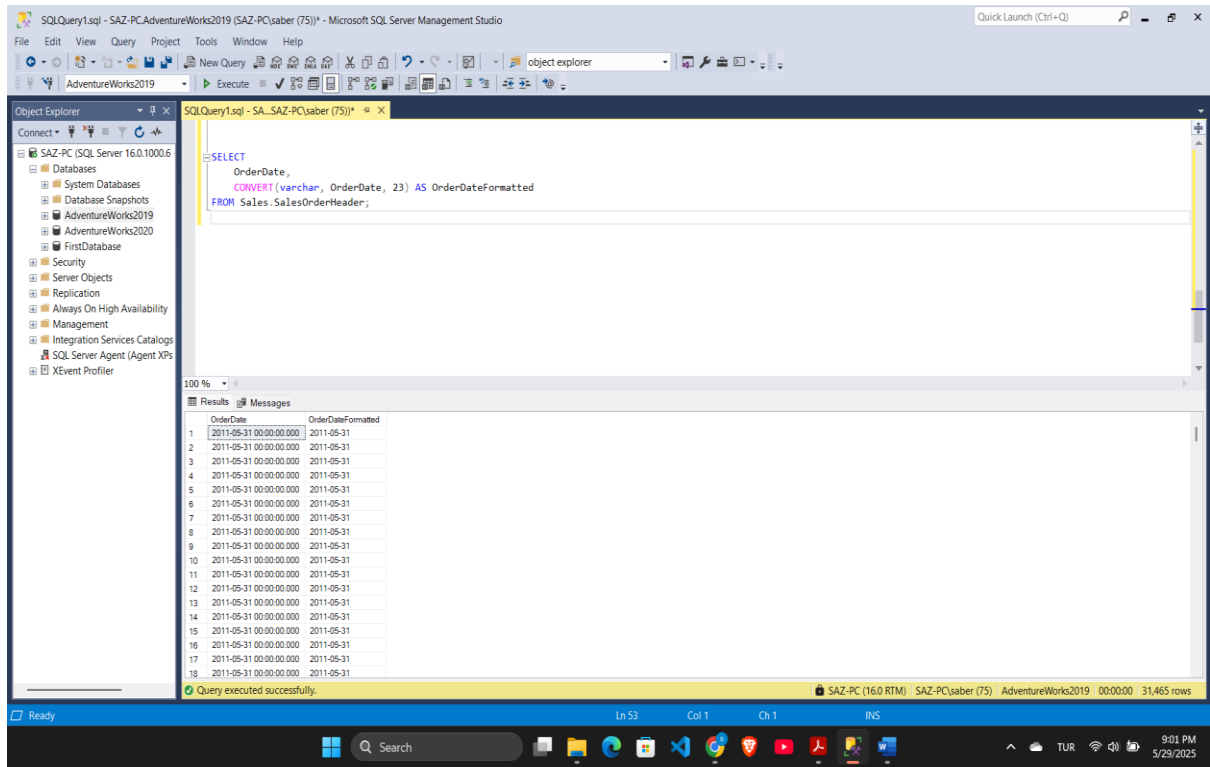






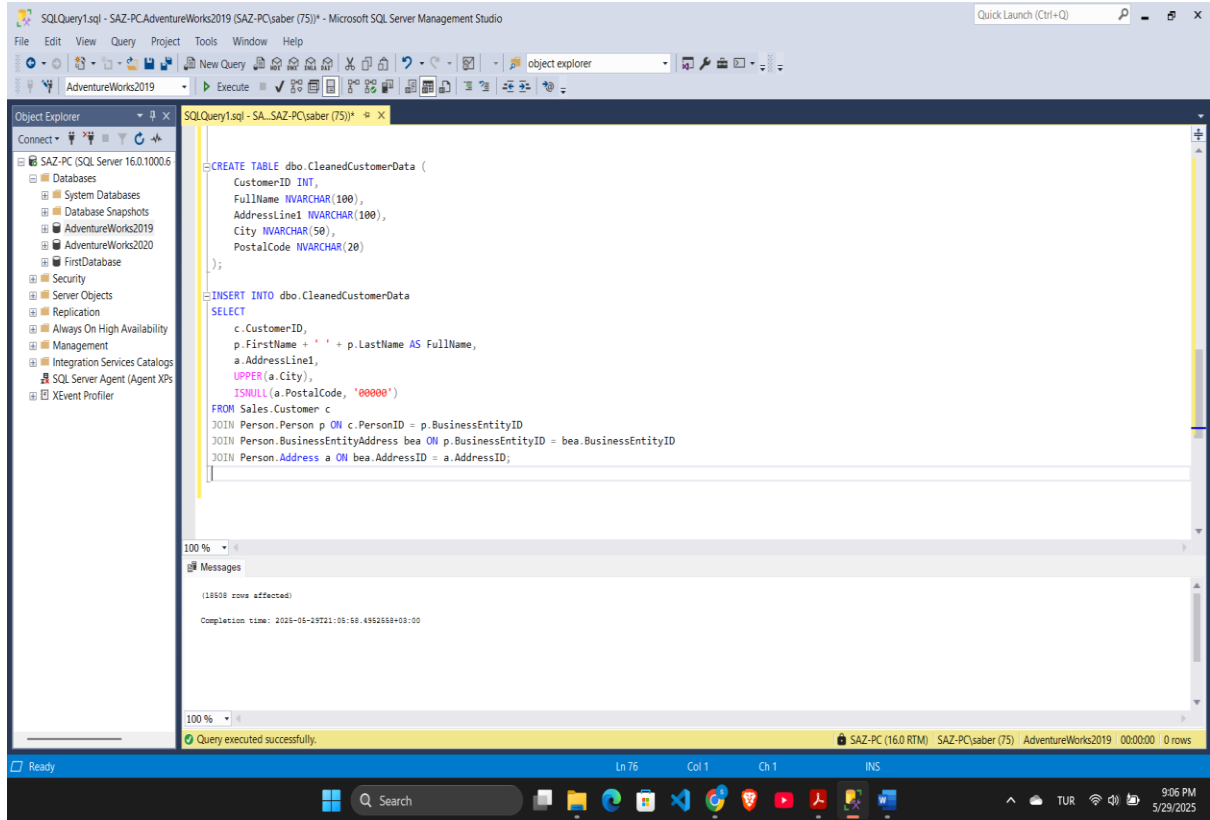
Veri dönüştürme aşamasında İsimleri birleştirme ve tarih formatlarını dönüştürme gibi işlemler yapılmıştır.





3. Load (Veri Yükleme)

Temizlenen ve dönüştürülen veriler, analiz için oluşturulan yeni bir tabloya aktarılmıştır.



6. Veri tabanı Yükseltme ve Sürüm Yönetimi

Veri tabanı Yükseltme Planı:

AdventureWorks veri tabanını 2019 sürümünden 2020 sürümüne yükseltme planı, mevcut veri tabanı yapısı analiz edilir, tam yedek alınır ve geçiş öncesi şema, kullanıcı izinleri gibi bilgiler belgelenir. Ardından, SQL Server hedef ortamda AdventureWorks2020 adlı boş bir veri tabanı oluşturulur ve AdventureWorks2019 yedeği bu veri tabanına restore edilir. Restore sonrası, yeni iş ihtiyaçlarına göre şema güncellemeleri yapılır (örneğin yeni sütun ekleme, veri tipi güncelleme) ve gerekirse veri dönüşümleri gerçekleştirilir.

Yükseltme sonrası test ortamında kapsamlı birim ve fonksiyon testleri yapılır; view'lar, trigger'lar, stored procedure'lar ve veri tutarlılığı kontrol edilir. Performans testleri ile sorgu süreleri ve indeks optimizasyonları değerlendirilir. Her aşamada geri dönüş (rollback) planı hazır tutulur ve yedekler güncel olarak saklanır. Bu strateji, kontrollü ve güvenli bir veri tabanı geçiş süreci sunar.

1. Mevcut veri tabanının yedeğini alma

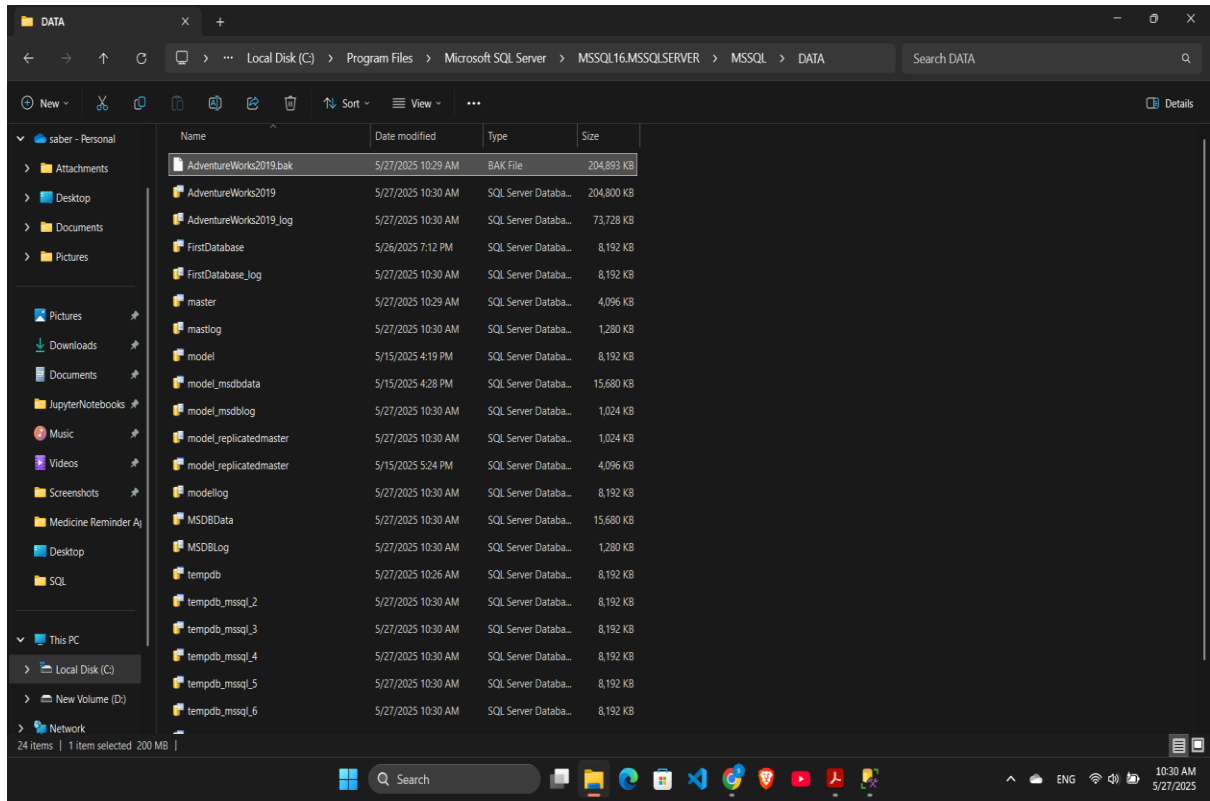
The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The main window shows a SQL query for backing up the AdventureWorks2019 database. The query is as follows:

```
-- Mevcut veritabanının yedeğini al
BACKUP DATABASE AdventureWorks2019
TO DISK = 'C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\AdventureWorks2019.bak'
WITH FORMAT,
MEDIANAME = 'AdventureWorks2019Backup',
NAME = 'Full Backup of AdventureWorks2019';
```

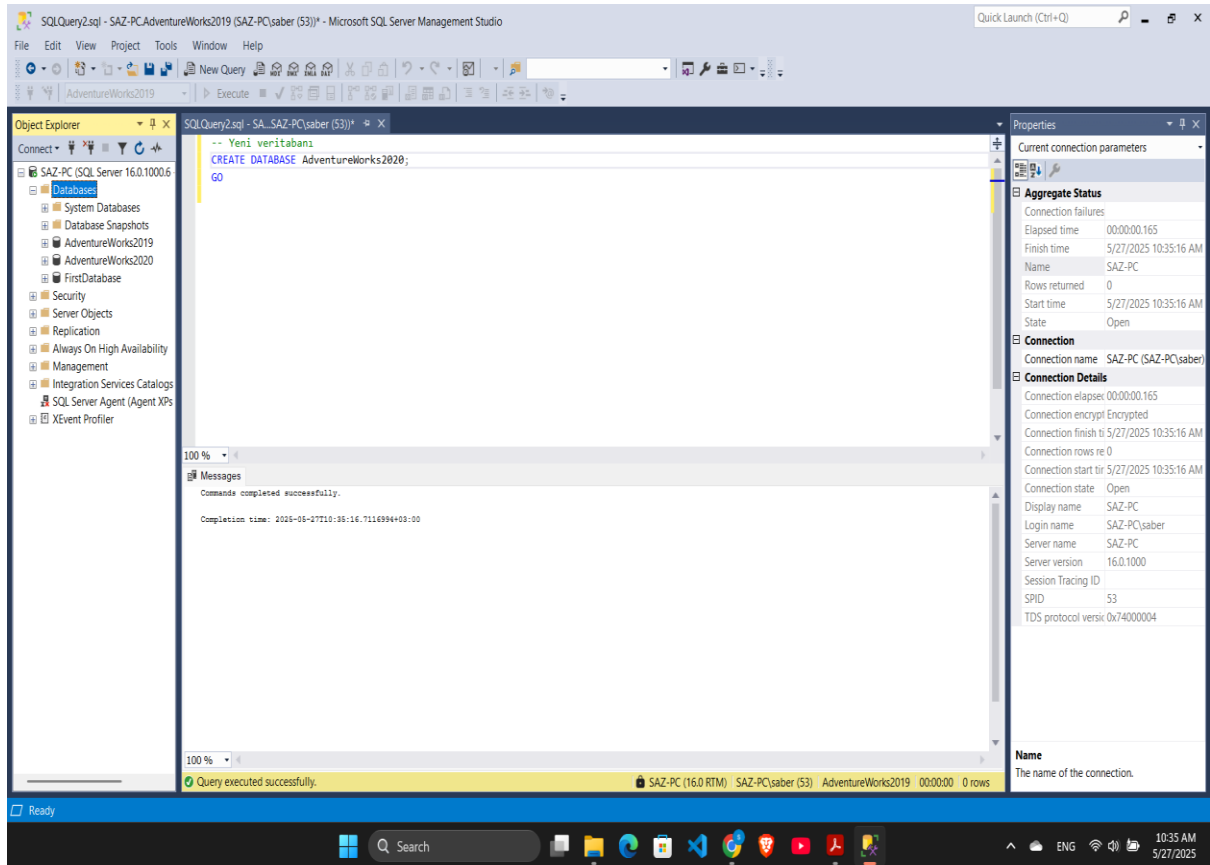
The query has been executed successfully, as indicated by the status bar at the bottom: "Query executed successfully." The Messages pane shows the following output:

```
Processed 26394 pages for database 'AdventureWorks2019', file 'AdventureWorks2019' on file 1.
Processed 1 pages for database 'AdventureWorks2019', file 'AdventureWorks2019_log' on file 1.
BACKUP DATABASE successfully processed 26395 pages in 7.009 seconds (27.998 MB/sec).
Completion time: 2025-05-27T10:29:28.5987229+03:00
```

The Properties pane on the right shows the connection details for the current connection (SAZ-PC (SAZ-PC\saber)). The connection state is Open, and the display name is SAZ-PC. The session tracing ID is 70, and the TDS protocol version is 0x74000004.



2. Yeni veri tabanı oluşturma



3. Eski veri tabanından yeni veri tabanına taşıma

The screenshot shows the Microsoft SQL Server Management Studio interface. The main window displays a SQL query for migrating data from AdventureWorks2019 to AdventureWorks2020. The query includes commands to use the master database, restore the AdventureWorks2020 database from a backup file, and then move the AdventureWorks2019 log file to the AdventureWorks2020 log file. The query is executed successfully, and the Messages pane shows the completion time and the number of pages processed.

```
USE master;
GO

-- 2. Hedef veritabanını tek kullanıcıya moduna al (bağlantıları sonlandırarak)
ALTER DATABASE AdventureWorks2020 SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
GO

-- verileri ve scriptleri AdventureWorks2020'ye aktarmak
RESTORE DATABASE AdventureWorks2020
FROM DISK = 'C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\AdventureWorks2019.bak'
WITH MOVE 'AdventureWorks2019' TO 'C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\AdventureWorks2020.mdf',
MOVE 'AdventureWorks2019_log' TO 'C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\AdventureWorks2020_log.ldf',
REPLACE;

ALTER DATABASE AdventureWorks2020 SET MULTI_USER;
GO
```

The Messages pane shows the following output:

```
Processed 26384 pages for database 'AdventureWorks2020', file 'AdventureWorks2019' on file 1.
Processed 1 pages for database 'AdventureWorks2020', file 'AdventureWorks2019_log' on file 1.
RESTORE DATABASE successfully processed 26385 pages in 0.272 seconds (769.096 MB/sec).
Completion time: 2025-05-27T11:02:20.2426761+03:00
```

The Properties pane on the right shows the connection details for the SAZ-PC (SAZ-PC\saber) connection.

4. Şema değişikliği

The screenshot shows the Microsoft SQL Server Management Studio interface. The main window displays a SQL query for making schema changes to the AdventureWorks2020 database. The query includes commands to add a new column to the Person table, add a new column to the Production.Product table, rename a column in the Production.Product table, drop a column in the Production.Product table, and create a new table named EmployeeLeave.

```
USE AdventureWorks2020;
GO

-- şema değişikliği:
-- yeni sütun ekleme
ALTER TABLE Person.Person
ADD SocialSecurityNumber NVARCHAR(11) NULL;

-- Ürün tablosuna üretim tarihi sütunu ekle
ALTER TABLE Production.Product
ADD ManufacturingDate DATE NULL;

EXEC sp_rename 'Production.ProductName', 'ProductName', 'COLUMN';

-- Product tablosundan eski bir sütunu sil
ALTER TABLE Production.Product
DROP COLUMN ProductLine;

-- Sütun veri tipini NVARCHAR(50) → NVARCHAR(100) olarak değiştir
ALTER TABLE Person.Person
ALTER COLUMN LastName NVARCHAR(100);

-- yeni tablo ekleme
CREATE TABLE EmployeeLeave (
    LeaveID INT PRIMARY KEY IDENTITY(1,1),
    BusinessEntityID INT,
    LeaveType NVARCHAR(50),
    StartDate DATE,
    EndDate DATE
);
```

The Messages pane shows the following output:

```
Commands completed successfully.
Completion time: 2025-05-27T12:39:23.2924992+03:00
```

The Properties pane on the right shows the connection details for the SAZ-PC (SAZ-PC\saber) connection.

5. Değişiklikleri izlemek için log tablosu ve DDL trigger oluşturma

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the Object Explorer with the AdventureWorks2020 database selected. The central pane shows a SQL query window with the following code:

```
USE AdventureWorks2020;
GO

-- Değişiklikleri izlemek için log tablosu oluşturun
CREATE TABLE SchemaChangeLog (
    ChangeID INT IDENTITY(1,1),
    EventData XML,
    ChangedBy SYSNAME,
    ChangeDate DATETIME DEFAULT GETDATE()
);
GO

-- Veritabanı seviyesinde DDL trigger oluşturun
CREATE TRIGGER trg_SchemaChangeLog
ON DATABASE
FOR DDL_DATABASE_LEVEL_EVENTS
AS
BEGIN
    INSERT INTO SchemaChangeLog(EventData, ChangedBy)
    VALUES (EVENTDATA(), SYSTEM_USER);
END;
GO
```

The right pane shows the Properties window for the connection. The 'Aggregate Status' tab is selected, showing connection details such as connection name, connection details, and connection state.

6. DDL trigger testi

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the Object Explorer with the AdventureWorks2020 database selected. The central pane shows a SQL query window with the following code:

```
-- Test verisi ekleme
INSERT INTO Person.Person (BusinessEntityID, PersonType, FirstName, LastName)
VALUES (21001, 'EM', 'Ali', 'Yılmaz');

-- Veri sorgulama testi
SELECT TOP 5 * FROM Person.Person ORDER BY ModifiedDate DESC;

-- DDL trigger testi: bir tablo ekleyip SchemaChangeLog'a kaydını kontrol et
CREATE TABLE TestTrigger (ID INT);
SELECT * FROM SchemaChangeLog ORDER BY ChangeDate DESC;
```

The right pane shows the Properties window for the connection. The 'Aggregate Status' tab is selected, showing connection details such as connection name, connection details, and connection state.

The bottom pane shows the Results window with the following data:

BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailPromotion	AdditionalContactInfo	Demographics
21001	EM	0	NULL	Ali	NULL	Yılmaz	NULL	1	NULL	<IndividualSurvey>http://schemas.mic...
307	SC	0	Ms.	Jay	NULL	Adams	NULL	1	<AdditionalContactInfo>http://schemas.mic...	<IndividualSurvey>http://schemas.mic...
309	SC	0	Ms.	Ronald	L	Adina	NULL	0	<AdditionalContactInfo>http://schemas.mic...	<IndividualSurvey>http://schemas.mic...
303	SC	0	Ms.	Margaret	J	Smith	NULL	0	<AdditionalContactInfo>http://schemas.mic...	<IndividualSurvey>http://schemas.mic...
305	SC	0	Ms.	Carla	J	Adams	NULL	0	<AdditionalContactInfo>http://schemas.mic...	<IndividualSurvey>http://schemas.mic...

SQLQuery3.sql - SAZ-PC\AdventureWorks2020 (SAZ-PC\saber (67)) - Microsoft SQL Server Management Studio

File Edit View Query Project Tools Window Help

AdventureWorks2020

Object Explorer

SAZ-PC (SQL Server 16.0.10006)

Databases

System Databases

Database Snapshots

AdventureWorks2019

AdventureWorks2020

Database Diagrams

Tables

Views

External Resources

Synonyms

Programmability

Service Broker

Storage

Security

FirstDatabase

Security

Server Objects

Replication

Always On High Availability

Management

Integration Services Catalogs

SQL Server Agent (Agent XPs)

XEvent Profiler

EventData1.xml

SQLQuery3.sql - SAZ-PC\saber (67))

SQLQuery2.sql - SAZ-PC\saber (75))

Properties

Current connection parameters

Aggregate Status

Connection failures

Elapsed time 00:00:00.063

Finish time 5/27/2025 12:54:58 PM

Name SAZ-PC

Rows returned 3

Start time 5/27/2025 12:54:58 PM

State Open

Connection

Connection name SAZ-PC (SAZ-PC\saber)

Connection Details

Connection elapsed: 00:00:00.063

Connection encrypt Encrypted

Connection finish 5/27/2025 12:54:58 PM

Connection rows re 3

Connection start 5/27/2025 12:54:58 PM

Connection state Open

Display name SAZ-PC

Login name SAZ-PC\saber

Server name SAZ-PC

Server version 16.0.10000

Session Tracing ID 67

SPID 67

TDS protocol versio 0x74000004

Name

The name of the connection.

Test verisi ekleme

INSERT INTO Person.Person (BusinessEntityID, PersonType, FirstName, LastName)

VALUES (21001, 'EM', 'Ali', 'Yilmaz');

Veri sorgulama testi

SELECT TOP 5 * FROM Person.Person ORDER BY ModifiedDate DESC;

DDL trigger testi: bir tablo ekleyip SchemaChangeLog'a kaydını kontrol et

CREATE TABLE TestTrigger (ID INT);

SELECT * FROM SchemaChangeLog ORDER BY ChangeDate DESC;

Results

Messages

ChangedBy ChangeDate

SAZ-PC\saber 2025-05-27 12:54:58.023

SAZ-PC\saber 2025-05-27 12:54:43.133

SAZ-PC\saber 2025-05-27 12:50:01.940

Query executed successfully.

SAZ-PC (16.0 RTM) SAZ-PC\saber (67) AdventureWorks2020 00:00:00 3 rows

Ready

Ln 30 Col 56 Ch 56 INS

Search

12:55 PM 5/27/2025

EventData2.xml - Microsoft SQL Server Management Studio

File Edit View Project Tools Window Help

AdventureWorks2020

Object Explorer

SAZ-PC (SQL Server 16.0.10006)

Databases

System Databases

Database Snapshots

AdventureWorks2019

AdventureWorks2020

Database Diagrams

Tables

Views

External Resources

Synonyms

Programmability

Service Broker

Storage

Security

FirstDatabase

Security

Server Objects

Replication

Always On High Availability

Management

Integration Services Catalogs

SQL Server Agent (Agent XPs)

XEvent Profiler

EventData2.xml

SQLQuery3.sql - SAZ-PC\saber (67))

SQLQuery2.sql - SAZ-PC\saber (75))

Properties

XML Document

Misc

Encoding Unicode (UTF-8)

Output

Schemas

Stylesheet

Encoding

Character encoding of the document.

<EVENT_INSTANCE>

<EventType>CREATE_TABLE</EventType>

<PostTime>2025-05-27T12:54:58.020</PostTime>

<SPID>67</SPID>

<ServerName>SAZ-PC</ServerName>

<LoginName>SAZ-PC\saber</LoginName>

<UserName>dbo</UserName>

<DatabaseName>AdventureWorks2020</DatabaseName>

<SchemaName>dbo</SchemaName>

<ObjectName>TestTrigger</ObjectName>

<ObjectType>TABLE</ObjectType>

<SqlCommand>

<SetOptions ANSI_NULLS="ON" ANSI_NULL_DEFAULT="ON" ANSI_PADDING="ON" QUOTED_IDENTIFIER="ON" ENCRYPTED="FALSE" />

<CommandText>CREATE TABLE TestTrigger (ID INT)</CommandText>

</SqlCommand>

</EVENT_INSTANCE>

Ready

Ln 1 Col 1 Ch 1 INS

Search

12:55 PM 5/27/2025

7. Geri dönüş (rollback) planı

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The central pane shows a SQL query window with the following script:

```
-- Geri dönüş planı: eski yedeği geri yükleme
USE master;
GO
RESTORE DATABASE AdventureWorks2019
FROM DISK = 'C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\AdventureWorks2019.bak'
WITH REPLACE;
```

The Messages pane below the query window shows the execution results:

```
Processed 26884 pages for database 'AdventureWorks2019', file 'AdventureWorks2019_1' on file 1.
Processed 1 pages for database 'AdventureWorks2019', file 'AdventureWorks2019_log' on file 1.
RESTORE DATABASE successfully processed 26885 pages in 0.321 seconds (817.800 MB/sec).
Completion time: 2025-05-27T12:59:06.7208449+03:00
```

The Properties pane on the right shows the connection details for the 'SAZ-PC (SAZ-PC\saber)' connection:

Aggregate Status	
Connection failures	
Elapsed time	00:00:00.879
Finish time	5/27/2025 12:59:06 PM
Name	SAZ-PC
Rows returned	0
Start time	5/27/2025 12:59:05 PM
State	Open

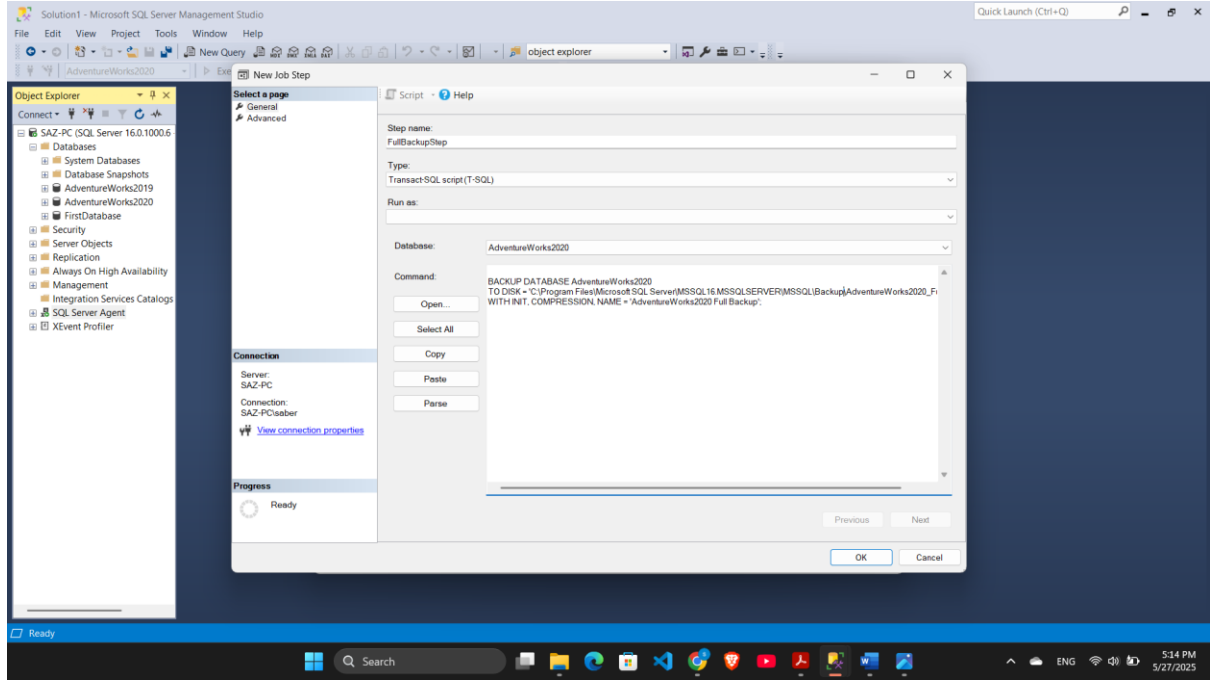
Connection	
Connection name	SAZ-PC (SAZ-PC\saber)
Connection Details	
Connection elapsed	00:00:00.879
Connection encrypt	Encrypted
Connection finish	5/27/2025 12:59:06 PM
Connection rows re	0
Connection start	5/27/2025 12:59:05 PM
Connection state	Open
Display name	SAZ-PC
Login name	SAZ-PC\saber
Server name	SAZ-PC
Server version	16.0.1000
Session Tracing ID	
SPID	65
TDS protocol versik	0x74000004

The status bar at the bottom indicates 'Query executed successfully.' and 'SAZ-PC (16.0 RT) (saber (65) master 00:00:00 0 rows)'.

7. Veritabanı Yedekleme ve Otomasyon Çalışması

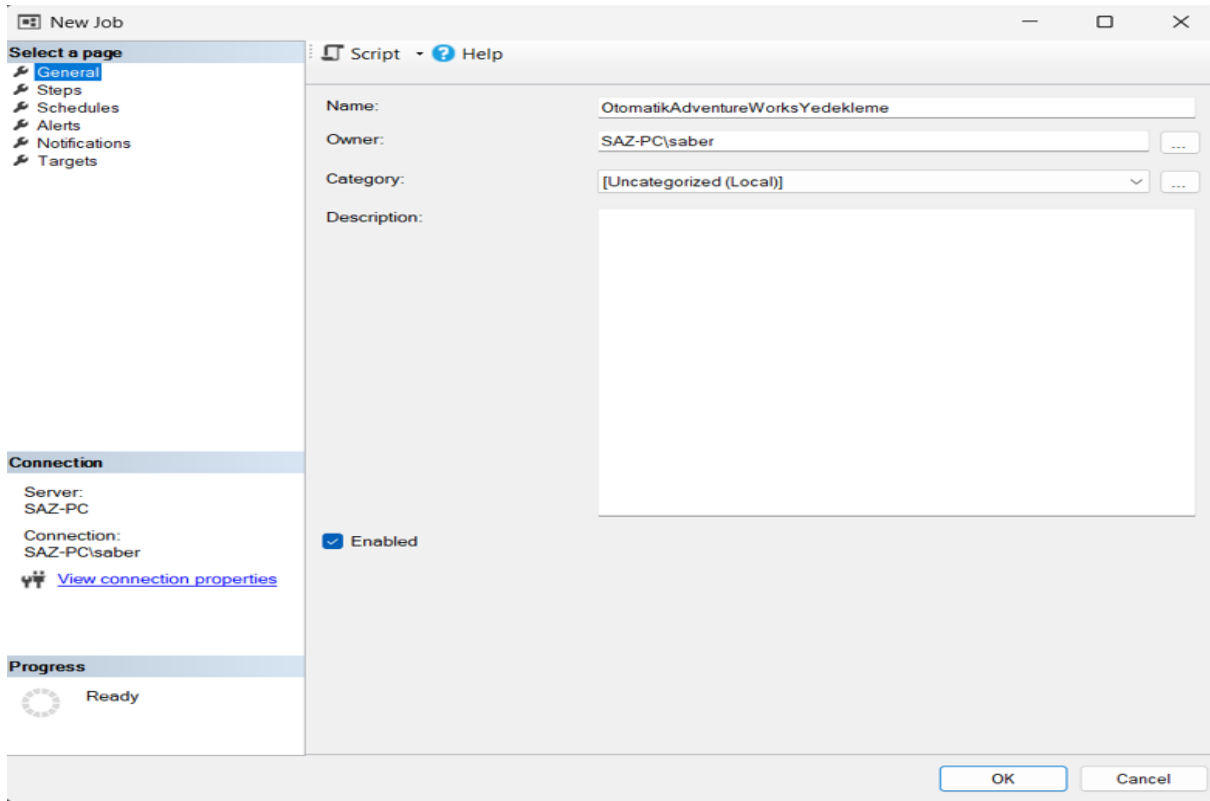
1. SQL Server Agent'ı Etkinleştirme

Otomatik görevlerin zamanlanarak çalıştırılabilmesi için ilk adım, SQL Server Agent servisinin etkinleştirilmesidir. Bu işlem için SQL Server Configuration Manager üzerinden "SQL Server Agent (MSSQLSERVER)" servisi bulunur ve başlatılır. Eğer servis çalışmıyorsa, zamanlanmış işler devreye giremez, bu nedenle bu adım sistemin temel yapı taşıdır.



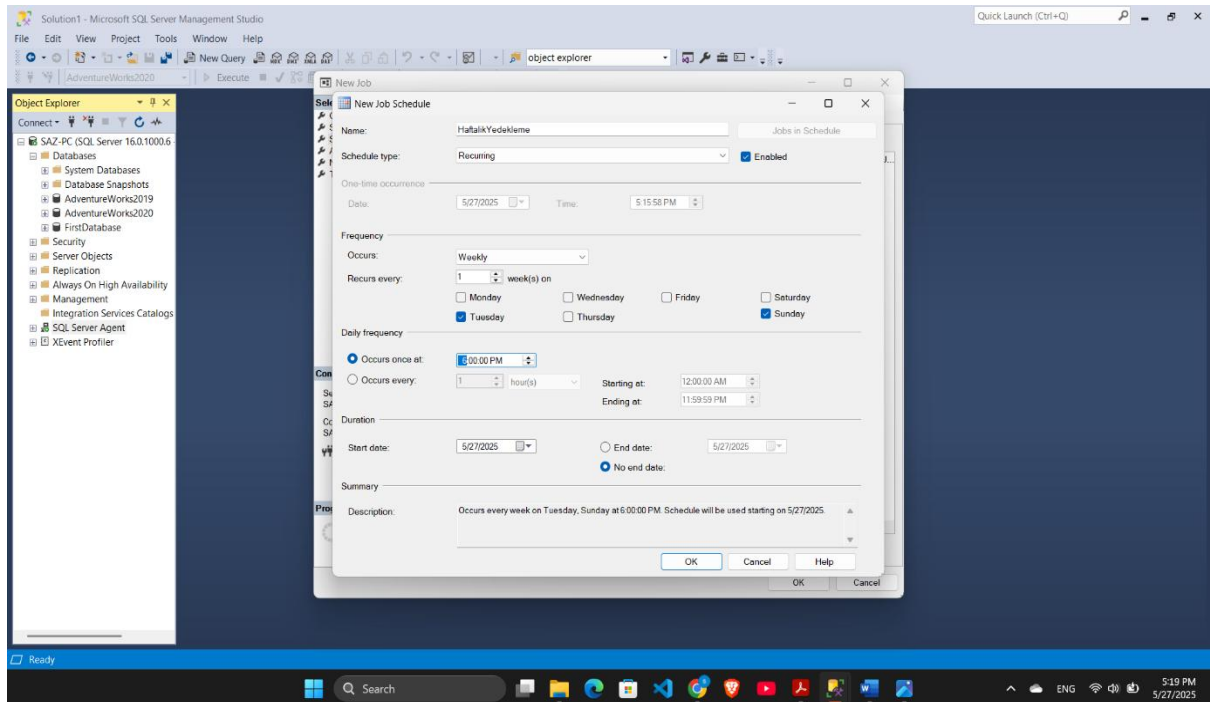
2. Otomatik Yedekleme Job'unun Oluşturulması

SQL Server Management Studio (SSMS) arayüzü kullanılarak yeni bir Job tanımlanmıştır. "OtomatikAdventureWorksYedekleme" adı verilen bu job, AdventureWorks2020 veritabanının tam yedeğini (full backup) alacak şekilde yapılandırılmıştır. Job içerisine eklenen "FullBackupStep" adındaki adım, BACKUP DATABASE komutu ile belirlenen dizine .bak uzantılı yedek dosyasını oluşturur. Bu işlem sırasında WITH INIT ve COMPRESSION seçenekleri kullanılarak eski yedeğin üzerine yazılması ve sıkıştırılması sağlanmıştır.



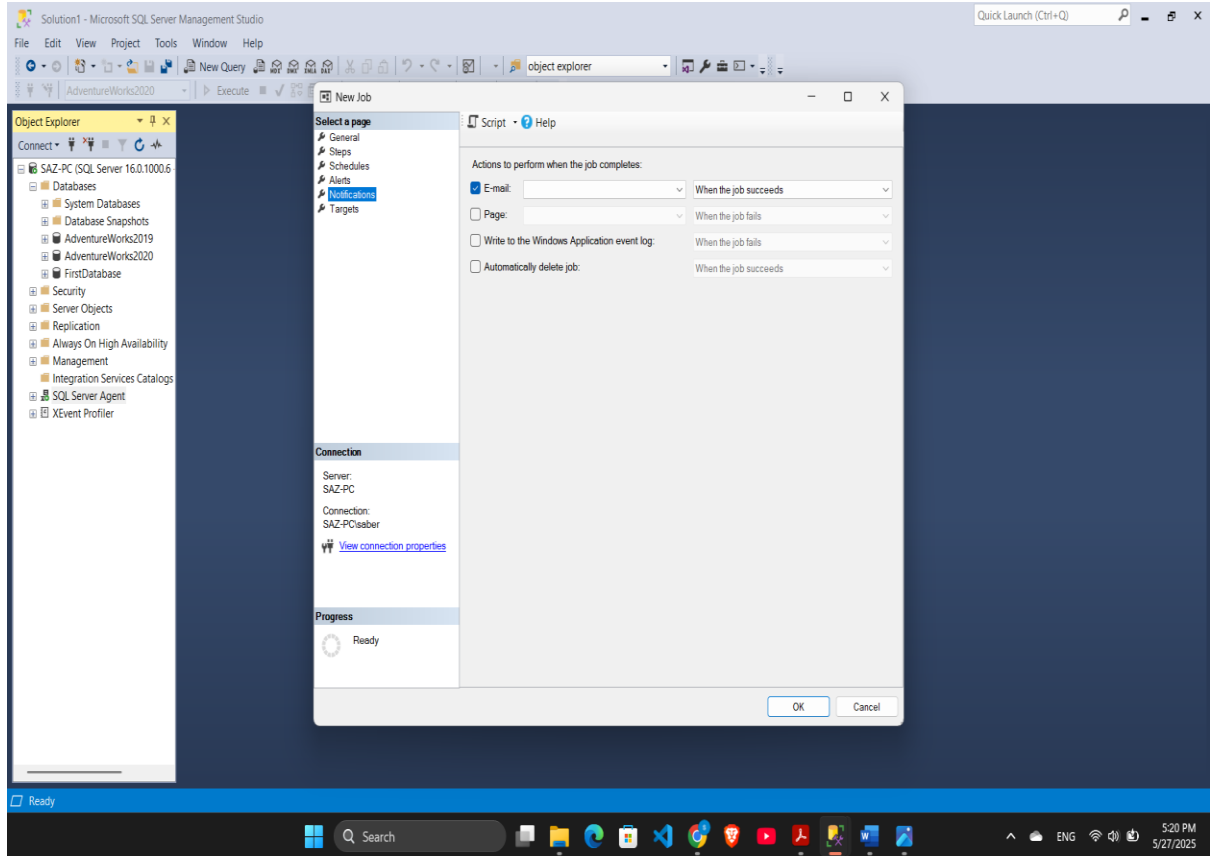
3. Yedekleme Görevinin Zamanlanması

Oluşturulan job'un belirli aralıklarla otomatik olarak çalışabilmesi için zamanlama ayarı yapılmıştır. "HaftalıkYedekleme" isimli zamanlama profili ile bu işin her hafta saat Salı günü 18:00'de çalışması tanımlanmıştır. Bu şekilde sistem yöneticisinin manuel olarak yedekleme başlatmasına gerek kalmadan süreç otomatik ilerleyecektir.



4. Job Başarılı Bildirimlerinin Ayarlanması

Yedekleme job'unun başarılı olması durumunda sistem yöneticisine (DBA) e-posta gönderilmesi sağlanmıştır. SSMS üzerinden Job ayarlarının “Notifications” sekmesinde, “When the job successes” seçeneği işaretlenmiş ve mail profili üzerinden ilgili operatöre bildirim gönderimi yapılandırılmıştır.

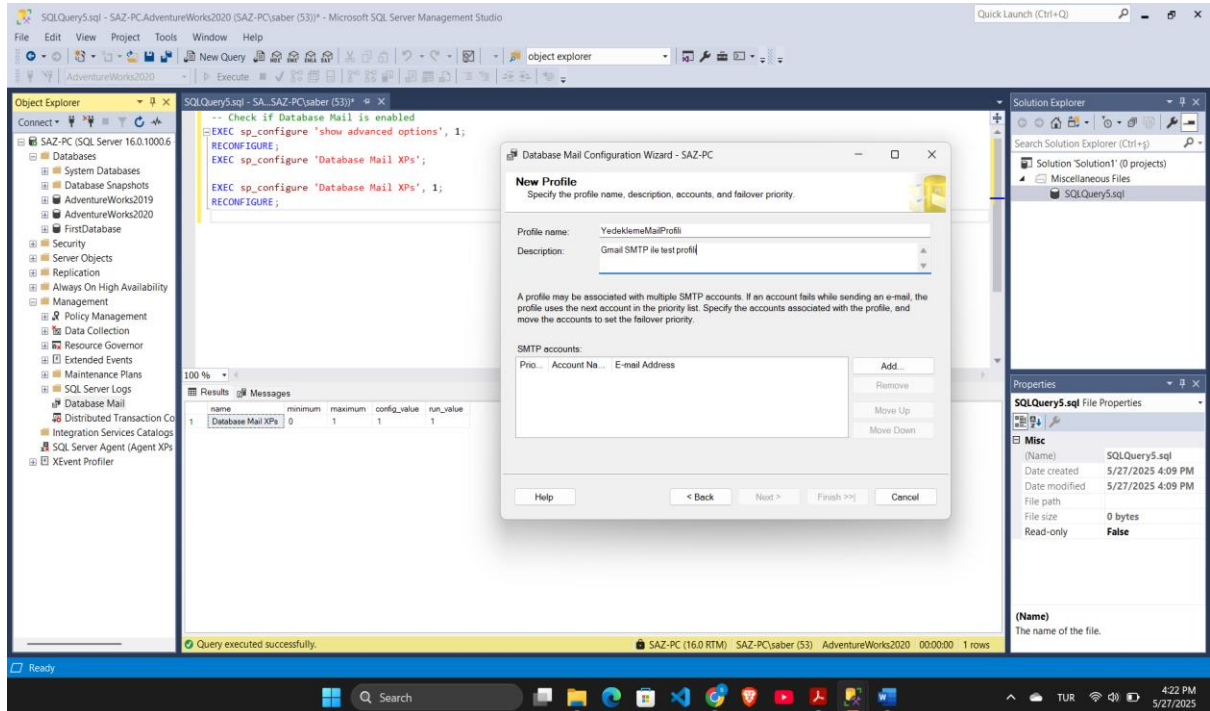
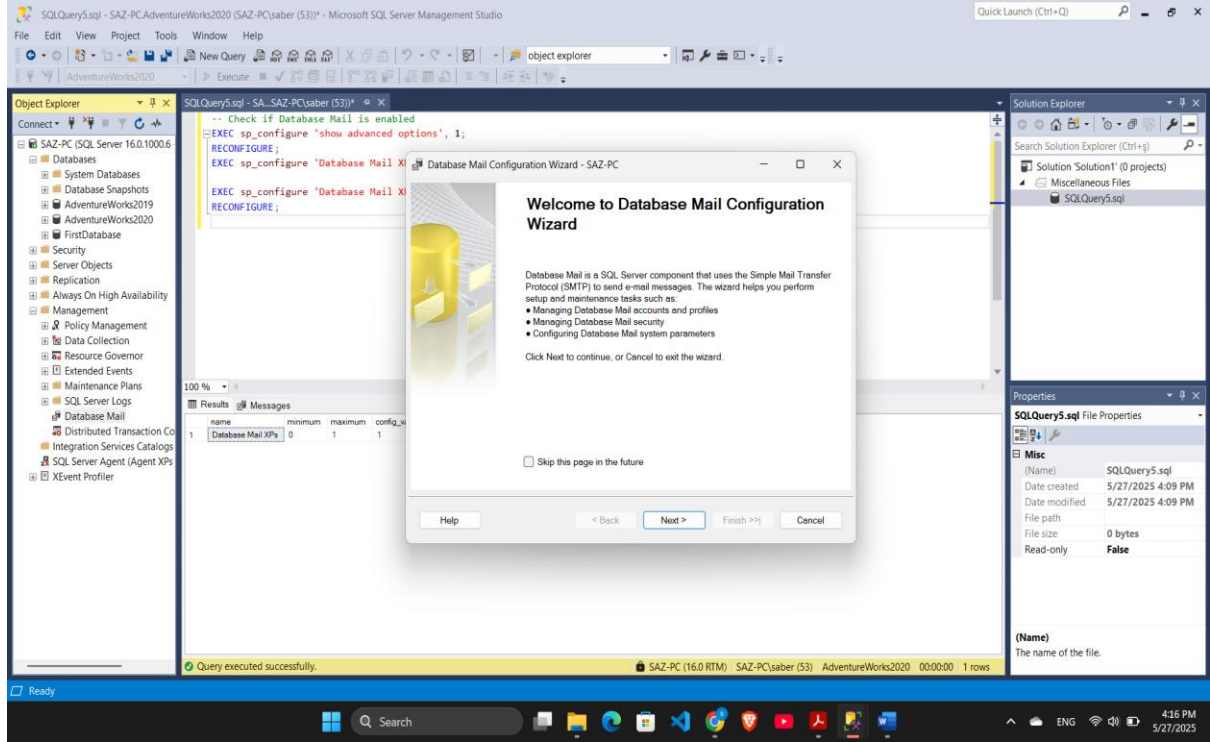


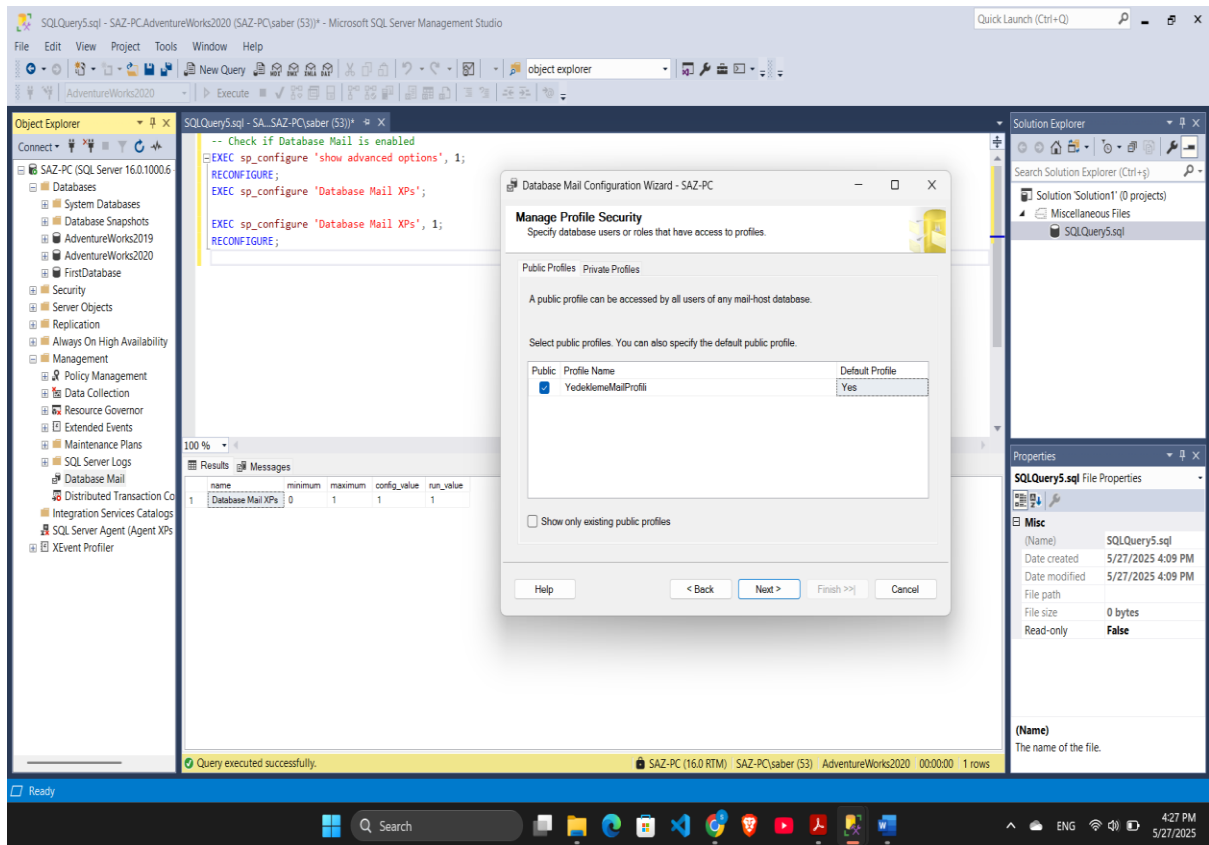
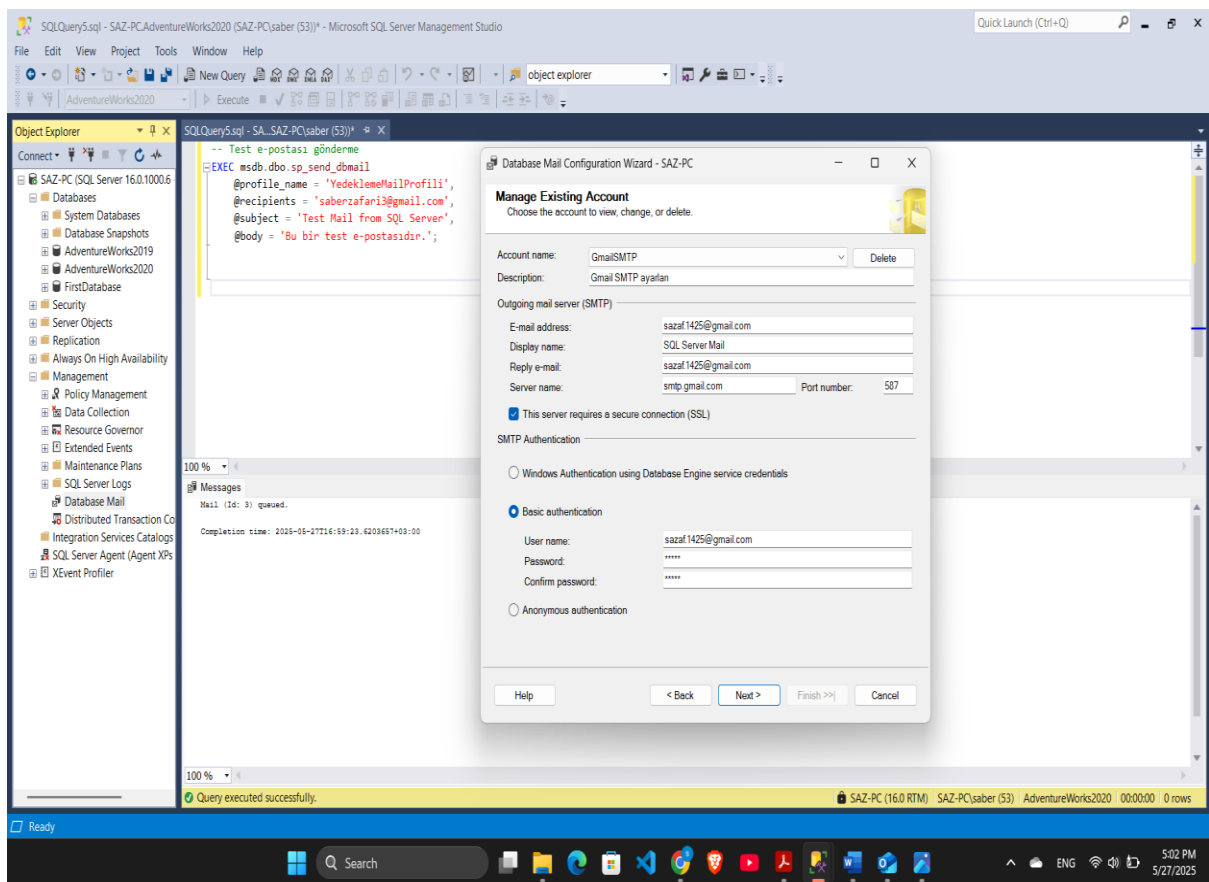
5. T-SQL Scripting ile yedekleme raporları oluşturma

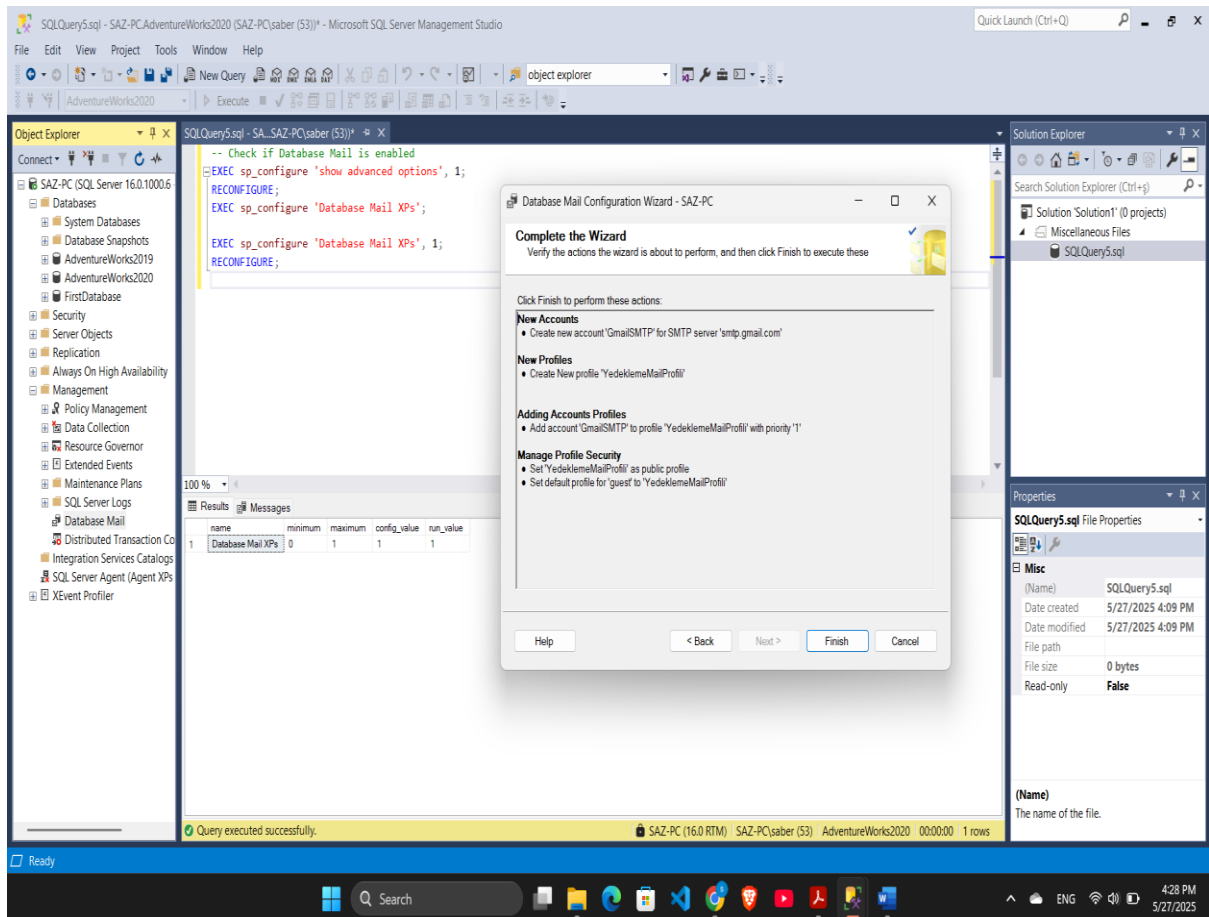
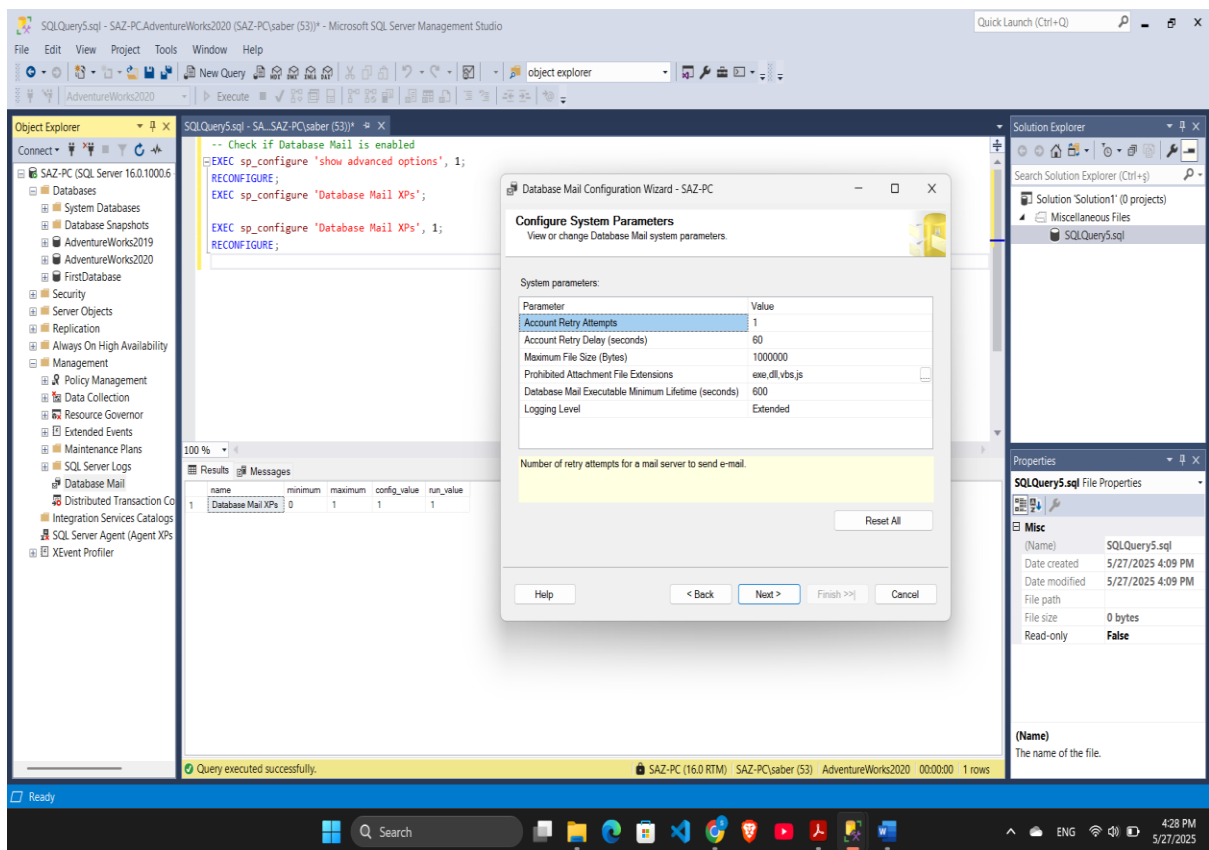
```
SELECT
    database_name,
    MAX(backup_finish_date) AS SonYedeklemeTarihi,
    DATEDIFF(HOUR, MAX(backup_finish_date), GETDATE()) AS SaatFarkı,
    CASE
        WHEN DATEDIFF(HOUR, MAX(backup_finish_date), GETDATE()) > 24 THEN 'UYARI: Yedek alınmadı!'
        ELSE 'OK'
    END AS Durum
FROM msdb.dbo.backupset
WHERE type = 'D'
GROUP BY database_name;
```

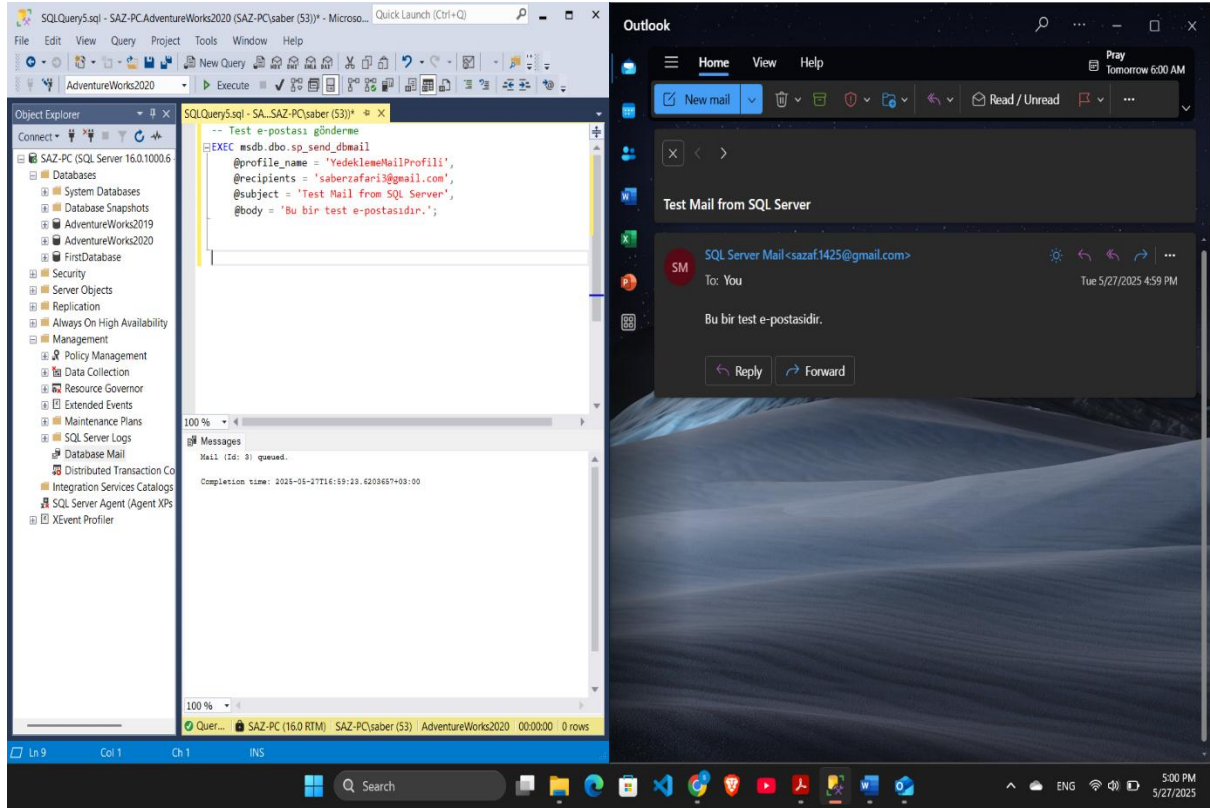

6. Database Mail Yapılandırması

Veri tabanı yöneticilerine e-posta gönderilmesini sağlayan Database Mail yapılandırması, SSMS üzerinden gerçekleştirilmiştir. Yeni bir mail profili oluşturularak, SMTP ayarları (örneğin Gmail veya Outlook) sisteme girilmiş ve bu profil SQL Server Agent ile ilişkilendirilmiştir. Bu yapılandırma ile SQL Server, sistem dışına otomatik e-posta gönderebilir hale getirilmiştir.









7. Yedekleme Hataları İçin SQL Alert Tanımı

Olası bir yedekleme hatasını tespit edip, ilgili yöneticiyi uyarmak amacıyla SQL Server üzerinde bir alert tanımı yapılmıştır. `sp_add_alert` prosedürü kullanılarak "Yedekleme Başarısız Alert" adıyla bir uyarı sistemi oluşturulmuş, bu sistem yalnızca 16 seviyesinde hata durumlarında çalışacak şekilde tanımlanmıştır. Bu uyarı, spesifik olarak "OtomatikAdventureWorksYedekleme" job'una bağlanarak yalnızca bu iş başarısız olduğunda devreye girmesi sağlanmıştır. Oluşturulan uyarı sisteminin bir kişiye e-posta gönderebilmesi için bir operatör tanımı yapılmıştır. `sp_add_operator` prosedürüyle "DBAOperator" adı altında bir operatör tanımlanmış ve bu operatöre ait e-posta adresi girilmiştir. Ardından, `sp_add_notification` prosedürü ile bu operatör daha önce oluşturulan uyarıya bağlanmıştır. Böylece herhangi bir yedekleme hatasında sistem otomatik olarak bu kullanıcıya e-posta göndererek bilgilendirme yapabilecektir.

